Mathematical Statistics with Applications

Student's Solutions Manual

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Chapter 1

Descriptive Statistics

EXERCISES 1.2

1.2.1. The suggested solutions:
For qualitative data we can have color, sex, race, Zip code and so on. For quantitative data we can have age, temperature, time, height, weight and so on. For cross section data we can have school funding for each department in 2000. For time series data we can have the crude oil price from 1995 to 2008.

1.2.3. The suggested questions can be

1. What types of data the amount is?
2. Are these Federal Agency get same amount of money? If not, why?
3. Which Federal Agency should get more money? Why?

The suggested inferences we can make is

1. These Federal Agency get different amount of money.
2. The differences of money between the Agencies are kind of big.

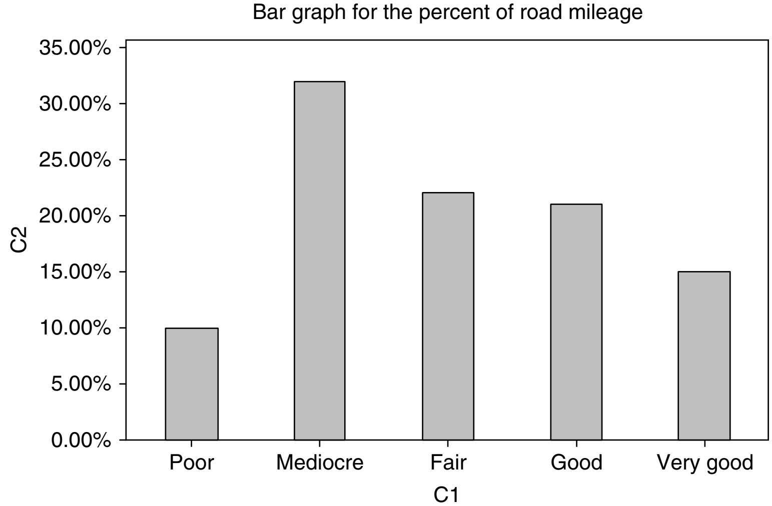
EXERCISES 1.3

1.3.1. For stratified sample, we can say suppose we decide to sample 100 college students from the population of 1000 (that is 10% of the population). We know these 1000 students come from three different major, Math, Computer Science and Social Science. We have Math 200, CS 400 and SS 400 students. Then we choose 10% of each of them Math 20, CS 40 and SS 40 by using random sampling within each major.
For cluster sample, we can say suppose we decide to sample some college students from the population of 2000. We know these 2000 students come from 20 different countries and we choose 3 out of the 20 countries by random sampling. Then we get all the individual information from each of the 3 countries.

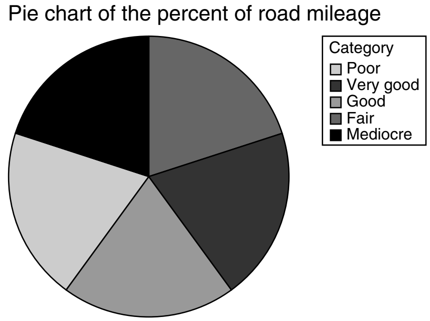
EXERCISES 1.4

1.4.1. By minitab

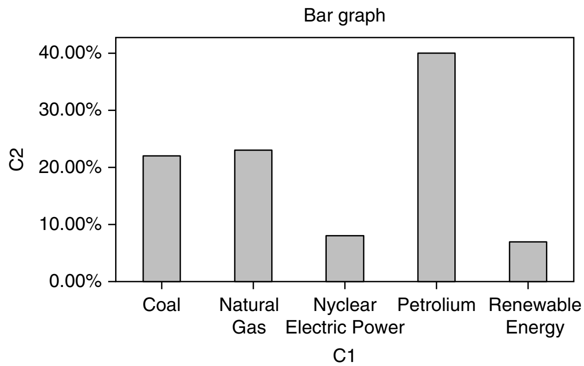
(a) Bar graph



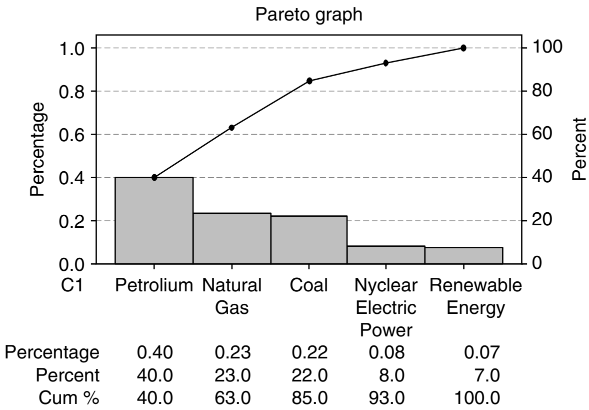
(b) Pie chart



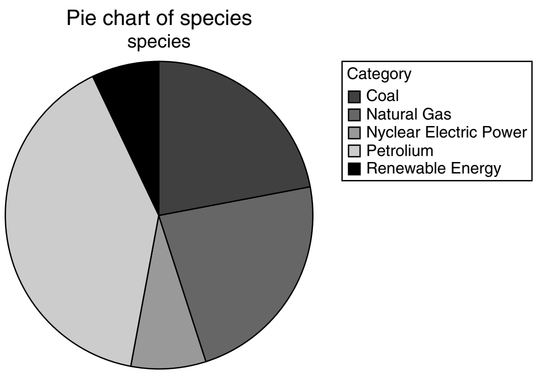
1.4.3. **(a)** Bar graph



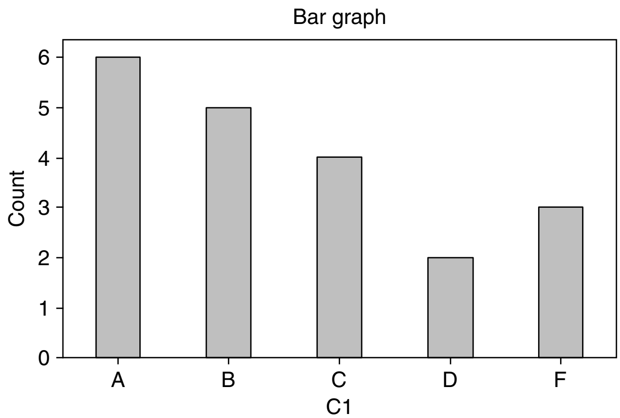
(b) Pareto chart



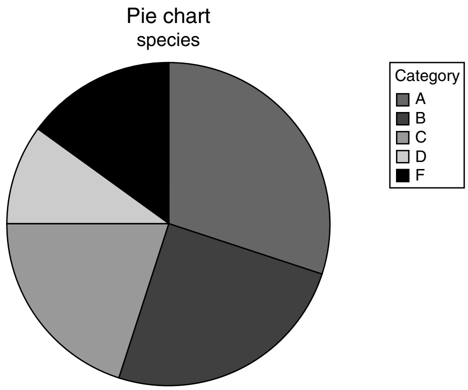
(c) Pie chart



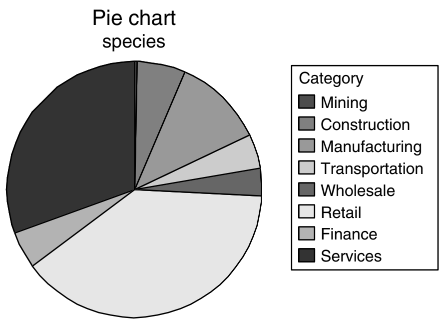
1.4.5. **(a)** Bar graph



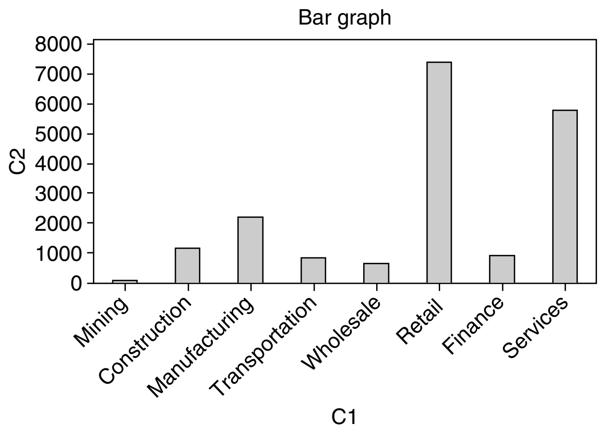
(b) Pie chart



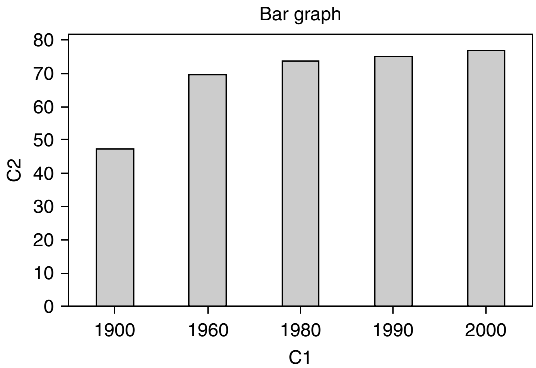
1.4.7. (a) Pie chart



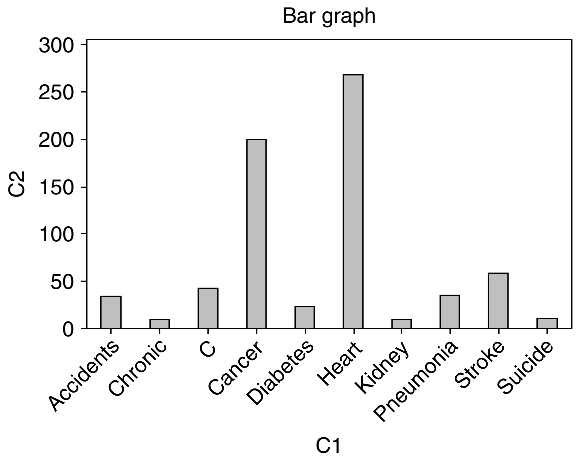
(b) Bar graph



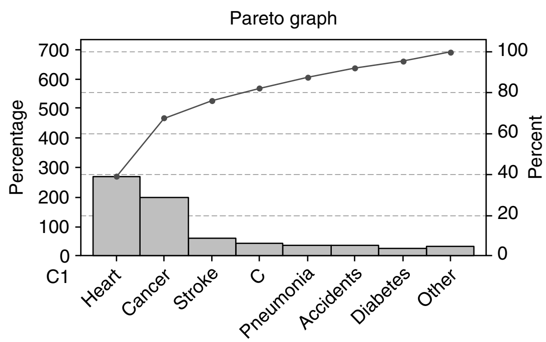
1.4.9. Bar chart



1.4.11. **(a)** Bar graph

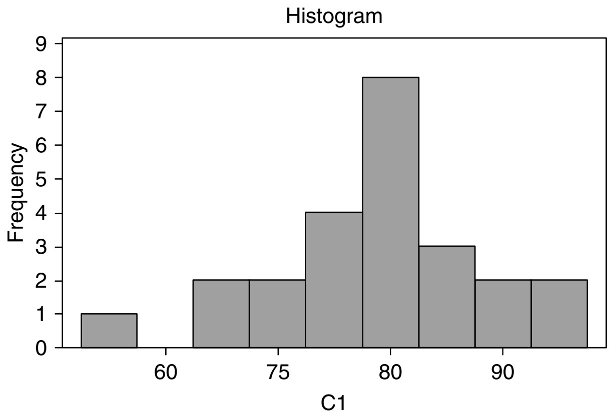


(b) Pareto graph



Percentage	268.0	119.4	58.5	42.3	35.1	34.5	23.9	30.2
Percent	38.7	28.8	8.5	6.1	5.1	5.0	3.5	4.4
Cum %	38.7	67.6	76.0	82.1	87.2	92.2	95.6	100.0

1.4.13.

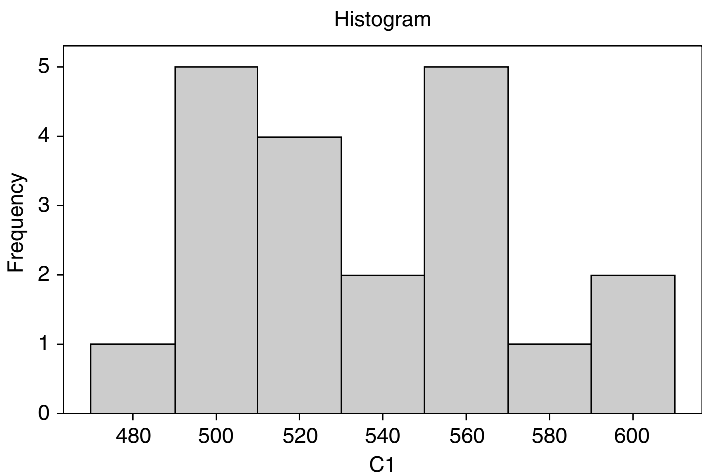


1.4.15. **(a)** Stem and leaf

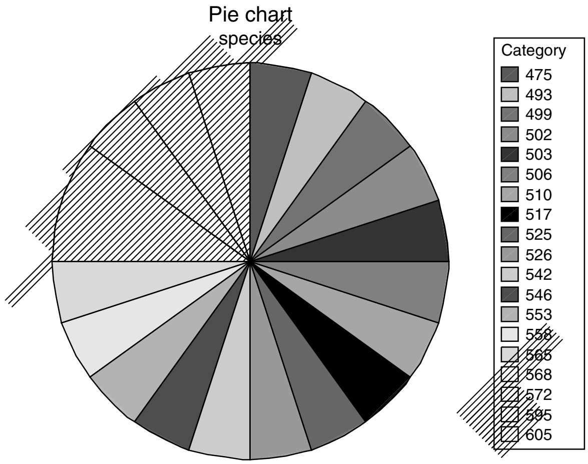
Stem-and-leaf of C1 N = 20
Leaf Unit = 10

1	4	7
3	4	99
8	5	00011
10	5	22
10	5	4455
6	5	6667
2	5	9
1	6	0

(b) Histogram



(c) Pie chart



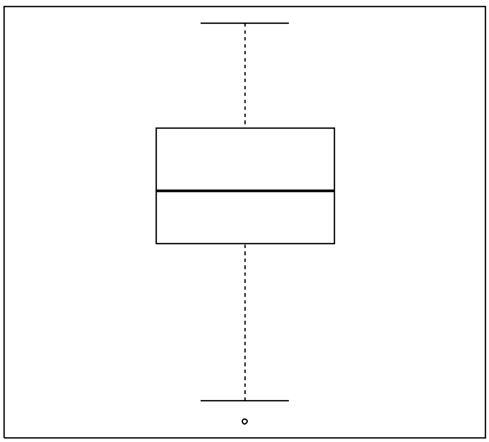
EXERCISES 1.5

1.5.1. Mean is 165.6667 and standard deviation is 63.15397

1.5.3. Data is 3,3,5,13 and standard deviation is 4.760952

1.5.5. **(a)** lower quantiles is 80, median is 95, upper quantiles is 115 and inter quantile range is 35. The lower limit of outliers is 27.5 and upper limit of outliers is 167.5.

(b) The box plot is



(c) Therefore there are no outliers.

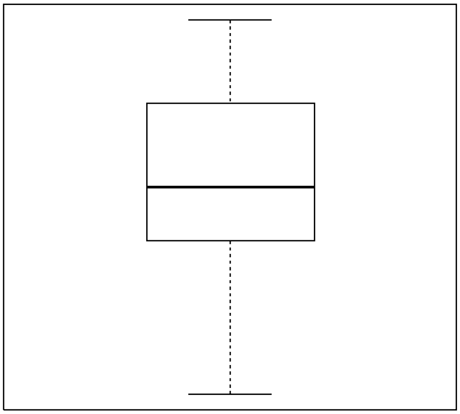
1.5.7.

$$\sum_{i=1}^{l} f_i(m_i - \bar{x}) = \sum_{i=1}^{l} f_i(m_i) - \sum_{i=1}^{l} \bar{x} = n\bar{x} - n\bar{x} = 0$$

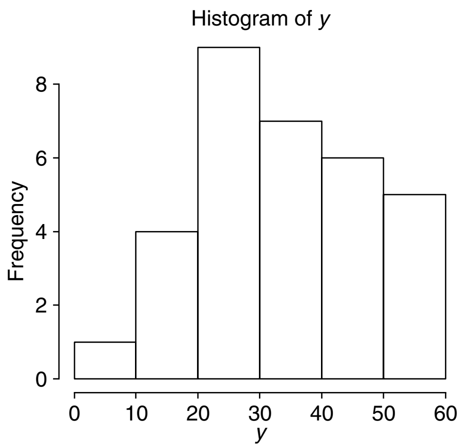
1.5.9. **(a)** Mean is 33.105, variance is 177.0430 and range is 48.19.

(b) Lower quantile is 24.9225, median is 32 and upper quantiles is 42.985. The inter quantile range is 18.0625. The lower limit of outliers is −2.17125 and upper limit of outliers is 70.07875. Therefore there are no outliers.

(c)



(d)



1.5.11. **(a)** Mean is 110, standard deviation is 83.4847.

(b) 68%, 95%, 99.7%.

1.5.13. **(a)** Mean is 3.7433, variance is 3.501 and standard deviation is 1.871323.

(b) Frequency table

Class	Interval	Frequency	mi	Mifi
1	0–1.6	4	.8	3.2
2	1.7–3.3	10	2.5	25
3	3.4–5	9	4.2	37.8
4	5.1–6.7	5	5.9	29.5
5	6.8–8.4	2	7.6	15.2

(c) By grouped data, Mean is 3.69, variance is 3.62 and standard deviation is 1.9. The results are similar to the none grouped data.

1.5.15.

$$L = 25$$
$$f_m = 139615$$
$$w = 4$$
$$F_b = 178859$$
$$n = 514661$$
$$\text{M} = L + \frac{w}{f_m}(.5n - F_b) = 27.24822.$$

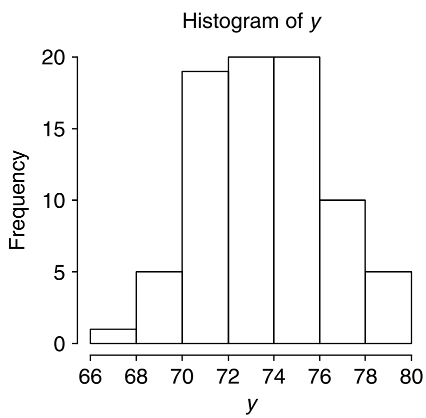
1.5.17. **(a)** Mean is 44.27, variance is 536.15 and standard deviation is 23.15.

(b)

$$L = 40$$
$$f_m = 59$$
$$w = 19$$
$$F_b = 69$$
$$n = 180$$
$$\text{M} = L + \frac{w}{f_m}(.5n - F_b) = 46.763.$$

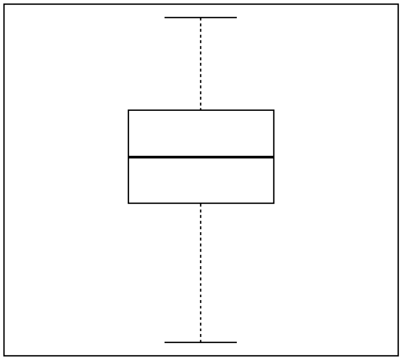
EXERCISES 1.8

1.8.1. **(a)**



(b) Mean is 74.0625, median is 74, variance is 7.223892 and standard deviation is 2.68773.

(c)



The lower limit of outliers is 66 and upper limit of outliers is 82. Therefore we have no outlier.

Chapter 2

Basic Concepts from Probability Theory

EXERCISES 2.2

2.2.1. **(a)** $S = \{(R, R, R), (R, R, L), (R, L, R), (L, R, R), (R, L, L), (L, R, L), (L, L, R), (L, L, L)\}$

(b) $P = \frac{7}{8}$

(c) $P = \frac{4}{8}$

(d) $P = \frac{3}{8}$

(e) $P = \frac{2}{8}$

2.2.3. **(a)** $P = \frac{5}{36}$

(b) $P = \frac{5}{36}$

(c) $P = \frac{4}{8}$

(d) $P = \frac{3}{8}$

2.2.5. $P\left(A \bigcup B\right) = \{(H, H), (H, T), (T, H)\}$.

2.2.7. **(a)** Probability space is
$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),$
$(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),$
$(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

(b) $P = \frac{6}{36}$

(c) $P = \frac{1}{36}$

2.2.9. **(a)** Probability space is $S = \{N, N, N, S, S\}$
N stands for normal and S stands for spoiled.

(b) $P = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20}$

(c) No more than one means no or just one.
$P = \frac{3}{5} \times \frac{2}{4} + 2 \times \frac{2}{5} \times \frac{3}{4} = .9$

2.2.11. $P = p + 2q$

2.2.13. **(a)** Since $A \subset B$ then let $A \cup A^c = B$, we know $P(A) + P(A^c) = P(B)$ by the axiom 3. Since $P(A^c) \geq 0$ by axiom 1 we know $P(A) + P(A^c) \geq P(A)$ then $P(A) \leq P(B)$.

(b) Let $C = A \cap B$, $A = A^a \cup C$ and $B = B^a \cup C$, by axiom 3 we know $P(A \cup B) = P(A^a) + P(B^a) + P(C)$.

Since $P(A) = P(A^a) + P(C)$ and $P(B) = P(B^a) + P(C)$ by axiom 3 again, we know $P(A^a) = P(A) - P(C)$ and $P(B^a) = P(B) - P(C)$. Plug them back in previous equation $P(A \cup B) = P(A^a) + P(B^a) + P(C)$ and we get the following equation $P(A \cup B) = P(A) + P(B) - P(C) = P(A) + P(B) - P(A \cap B)$. If $A \cap B = \phi$ then by axiom 1 we know $P(A \cap B) = 0$ and just plug in we complete the proof.

2.2.15. **(a)** From 2.2.13 we know $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and from axiom 2 we know $P(A \cup B) \leq 1$, we can see that $P(A) + P(B) - P(A \cap B) \leq 1$ and that complete the proof $P(A) + P(B) - 1 \leq P(A \cap B)$.

(b) From 2.2.13 we know $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$. From axiom 1 we know $P(A_1 \cap A_2) \geq 0$ it means $-P(A_1 \cap A_2) \leq 0$ and we can get the following inequality $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2) \leq P(A_1) + P(A_2)$.

2.2.17. **(a)** $P = .24 + .67 - .09 = .82$
(b) $P = 1 - .82 = .18$
(c) $P = 1 - .09 = .91$
(d) $P = 1 - .09 = .91$
(e) $P = 1 - .82 = .18$

2.2.19. **(a)** $P = .55$
(b) $P = .3$
(c) $P = .7$

2.2.21. **(a)** $P = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{1}{4} = .4$
(b) $P = \frac{3}{5} \times \frac{2}{4} + \frac{3}{5} \times \frac{2}{4} = .6$
(c) $P = 2 \times \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{1}{4} = .7$
(d) $P = \frac{3}{5} \times \frac{2}{4} = .3$

2.2.23. Without loss of generality let us assume A_n is increasing sequence then $A_1 \subset A_2 \subset \ldots \subset A_n \subset \ldots$. We know that if $A_1 \subset A_2 \subset \ldots \subset A_n \subset \ldots$ then $A_1 \cup A_2 \cup \ldots \cup A_n \cup \ldots = \bigcup_{i=1}^{\infty} A_i = \lim_{n\to\infty} A_n$. From the condition we know $\lim_{n\to\infty} A_n = \bigcup_{i=1}^{\infty} A_i$ and if we take probability on both sides then $\lim_{n\to\infty} P(A_n) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = P\left(\lim_{n\to\infty} A_n\right)$

EXERCISES 2.3

2.3.1. **(a)** 45
(b) 1

(c) 10
(d) 5400
(e) 2520

2.3.3. 1024

2.3.5. 53130

2.3.7. 155117520

2.3.9. 4^{40}

2.3.11. **(a)** $p = .4313 \times .44425 \times .46798 \times .5263 \times 1 = .04719$
(b) $p = .0001189$
(c) $p = .4313 \times .21419 \times .10344 \times 1 \times 1 = .009557$

2.3.13. 180

2.3.15. **(a)** $p = 1 - \frac{365 \times 364 \times \ldots (365 - 20 + 1)}{365^{20}} = .4114$
(b) $p = 1 - .2936 = .7063$
(c) If $n = 23$ then $p = .4927$

2.3.17. $p = 1 - .27778 - .16667 = .5556$

2.3.19. **(a)** 7776
(b) 3.954×10^{21}
(c) 5.36447×10^{28}
(d) 3.022285×10^{12}

2.3.21. The question is asking when the cell does the splitting to produce a child. There will be a cell with half of the chromosomes. According to this understanding we have
(a) 2^{23}
(b) $\frac{\binom{23}{9}}{2^{23}} = .097416$

EXERCISES 2.4

2.4.1. **(a)** .999
(b) $\frac{1}{3}$

2.4.3. **(a)** $P(A|B) + P(A^c|B) = \frac{P(A \cap B)}{P(B)} + \frac{P(A^c \cap B)}{P(B)} = \frac{P(B|A) \times P(A) + P(B|A^c) \times P(A^c)}{P(B)} = \frac{P(B)}{P(B)} = 1$
(b) **(i)** if $P(A|B) + P(A|B^c) = 1$ then we know $P(A|B^c) = 1 - P(A|B) = P(A^c|B)$ that means A and B are symmetric in probability. But it is not always true.

(ii) if $P(A|B) + P(A^c|B^c) = 1$ then we know $P(A^c|B^c) = 1 - P(A|B) = P(A^c|B)$
That means B and B^{c}'s conditional probability are same, which is same as A and B are independent. But that is not always true.

2.4.5. If A and B are independent then $P(A \cap B) = P(A) \times P(B)$

(i) $P(A^c \cap B) = P(B) - P(A \cap B) = P(B) - P(A) \times P(B) = (1 - P(A))(P(B)) = P(A^c) \times P(B)$ then we know A^c and B are independent.

(ii) According to (i) just switch A and B and we can prove it.

(iii) $P(A^c \cap B^c) = P(B^c) - P(A \cap B^c) = P(B^c) - P(A) \times P(B^c) = (1 - P(A))(P(B^c)) = P(A^c) \times P(B^c)$.

2.4.7. $P(E|F) = \frac{1}{13} = \frac{4}{52} = P(E)$ then E and F are independent

2.4.9. .1948

2.4.11. **(a)** $P = .031125$
(b) $P = .06$

2.4.13. .8

2.4.15. **(a)**
$$\begin{aligned} P(\text{a dime is selected}) &= \sum_{i=2}^{12} P(\text{a dime is selected}|\text{box } i \text{ is selected})\, P(\text{box } i \text{ is selected}) \\ &= \sum_{i=2}^{12} \tfrac{i}{12} P(\text{the sum of dies } = i) \\ &= \tfrac{2(1)+3(2)+4(3)+5(4)+6(5)+7(6)+8(5)+9(4)+10(3)+11(2)+12(1)}{12(36)} \\ &= .583333 \end{aligned}$$

(b) $P(\text{box 4 is selected}|\text{a penny is selected})$
$$= \frac{P(\text{box 4 is selected \& a penny is selected})}{P(\text{a penny is selected})} = \frac{(3/36)(8/12)}{1 - P(\text{a dime is selected})}$$
$$= .13333$$

2.4.17. $P = .6575$

2.4.19. $P = .60976$

2.4.21. **(a)** $P(\text{Accident rate}) = .25 \times .086 + .257 \times .044 + .347 \times .056 + .146 \times .098 = .066548$
(b) $P(\text{gourp4}|\text{Accident}) = \frac{.146 \times .098}{.0665} = .215$

2.4.23. $P = .16667$

2.4.25.
$$\begin{aligned} P(\text{Working}) &= P(B, C) + P(A, B, \text{not}C) + P(A, \text{not}B, C) \\ &= [1 - P(\text{not}B) - P(\text{not}C) - P(\text{not}B, \text{not}C)] + P(A)P(B|\text{not}C)P(\text{not}C) \\ &\quad + P(A)\,[P(\text{not}B) - P(\text{not}B, \text{not}C)] \\ &= [1 - 0.1 - 0.05 - 0.75 * 0.05] + 0.85 * 0.25 * 0.05 + 0.85 * [0.1 - 0.75 * 0.05] \\ &= 0.8875 + 0.010625 + 0.053125 = 0.95 \end{aligned}$$

2.4.27. **(a)** $P(\text{same type of blood}) = \frac{18}{40} \times \frac{17}{39} + \frac{16}{40} \times \frac{15}{39} + \frac{4}{40} \times \frac{3}{39} + \frac{2}{40} \times \frac{1}{39} = .358974$
(b) $P(\text{type is } O|\text{type is } B) = \frac{18}{39} = .4615$

2.4.29. Let E denote A ends up with all the money when he starts with i.

Let F denote A ends up with all the money when he starts with $N - i$. For A starts with $N - i$ means B starts with i because N is the total money A and B has so if we got $P(E)$ then $P(F) = 1 - P(E)$.
Let H denote the event that the first flip lands heads and p denote the probability to have H on the first flip.

$$P(E) = P(E|H)P(H) + P(E|H^C)P(H^C)$$

This probability represents that A gets a head and combined with the probability if B win the first coin.
Now we let $P(E) = P(E|H)p + P(E|H^C)(1 - p) = P_i$ and define this as the first round.
New, given that the first flip lands heads, the situation after the first bet is that A has $i + 1$ units and B has $N - (i + 1)$ units.
Since the successive flips are assumed to be independent with a common probability p of heads, it follows that, from that point on, A's probability of winning all the money is exactly the same as if the game were just starting with A having an initial fortune of $i + 1$ and B having an initial fortune of $N - (i + 1)$
Therefore, $P(E|H) = P_{i+1}$ and $P(E|H^C) = P_{i+1}$. Let q to be $1 - p$,

$$P_i = pP_{i+1} + qP_{i-1} \quad i = 1, 2, \ldots, N - 1$$

By applying the condition that $P_0 = 0$ and $P_N = 1$

$$P_i = 1P_i = (p + q)P_i = pP_{i+1} + qP_{i-1}$$

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}) \quad i = 1, 2, \ldots, N - 1$$

After plug in i
We got

$$P_2 - P_1 = \frac{q}{p}(P_1 - P_0) = \frac{q}{p}P_1$$

$$P_3 - P_2 = \frac{q}{p}(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1$$

$$P_N - P_{N-1} = \frac{q}{p}(P_{N-1} - P_{N-2}) = \left(\frac{q}{p}\right)^{N-1} P_1$$

If $\frac{q}{p} = 1$
Then $P_2 - P_1 = P_1$ and $P_2 = 2P_1$

$$P_3 = 3P_1$$

$$P_N = NP_1$$

$$P_N = 1$$

Which means $P_1 = \frac{1}{N}$
Therefore $P_i = \frac{i}{N}$

If $\frac{q}{p} \prec 1$, then $P_N - P_1 = 1 - P_1 = P_1 \dfrac{q}{p}\left(\dfrac{1-(\frac{q}{p})^{N-1}}{1-\frac{q}{p}}\right) = P_1\left(\dfrac{\frac{q}{p}-(\frac{q}{p})^N}{1-\frac{q}{p}}\right)$

$$1 = P_1\left(\frac{\frac{q}{p}-(\frac{q}{p})^N}{1-\frac{q}{p}}+1\right) = P_1\left(\frac{1-(\frac{q}{p})^N}{1-\frac{q}{p}}\right)$$

$$P_1 = \left(\frac{1-\frac{q}{p}}{1-(\frac{q}{p})^N}\right)$$

Add all equations, we got

$$P_N - P_1 = P_1\left(\frac{q}{p}+\left(\frac{q}{p}\right)^2+\cdots+\left(\frac{q}{p}\right)^{N-1}\right)$$

Add first $i-1$ of all equations, we got

$$P_i - P_1 = P_1\left(\frac{q}{p}+\left(\frac{q}{p}\right)^2+\cdots+\left(\frac{q}{p}\right)^{i-1}\right) = P_1\frac{q}{p}\left(\frac{1-\left(\frac{q}{p}\right)^{i-1}}{1-\frac{q}{p}}\right)$$

$$P_i = \left(\frac{1-\frac{q}{p}}{1-\left(\frac{q}{p}\right)^N}\right)\left(\frac{1-\left(\frac{q}{p}\right)^i}{1-\frac{q}{p}}\right) = \left(\frac{1-\left(\frac{q}{p}\right)^i}{1-\left(\frac{q}{p}\right)^N}\right)$$

Then if we start with $N-i$, just replace p by q and i by $N-i$.

$$Q_i = \left(\frac{1-\left(\frac{p}{q}\right)^{N-i}}{1-\left(\frac{p}{q}\right)^N}\right) \text{ if } \frac{p}{q} \prec 1$$

$$Q_i = \frac{N-i}{N} \text{ if } \frac{p}{q} = 1$$

$$P_i + Q_i = 1$$

EXERCISES 2.5

2.5.1. **(a)** $c = e^{-\lambda}$
(b) $P = e^{-\lambda}$
(c) $P = 1 - e^{-\lambda} - \lambda e^{-\lambda} - \frac{\lambda^2 e^{-\lambda}}{2}$

2.5.3. $F(x) = \begin{cases} 0, & \text{where} \quad x \prec -5 \\ .2, & \text{where} \quad -5 \leq x \prec 0 \\ .3, & \text{where} \quad 0 \leq x \prec 3 \\ .7, & \text{where} \quad 3 \leq x \prec 6 \\ 1, & \text{where} \quad 6 \leq 6 \end{cases}$

2.5.5. $p(x) = \begin{cases} 0, & \text{where} \quad -\infty \prec x \leq -1 \\ .2, & \text{where} \quad -1 \prec x \leq 3 \\ .6, & \text{where} \quad 3 \prec x \leq 9 \\ .2, & \text{where} \quad x \geq 9 \end{cases}$

2.5.7. **(a)** $c = \frac{1}{9}$

(b) $P = .7037$

(c) $F(x) = \begin{cases} 0, & \text{where} \quad x \prec 0 \\ \frac{1}{27}x^3, & \text{where} \quad 0 \leq x \leq 3 \\ 1, & \text{where} \quad x \succ 3 \end{cases}$

2.5.9. $f(x) = \begin{cases} 0, & \text{where} \quad x \leq 0 \\ \frac{2x}{(1+x)^2}, & \text{where} \quad x \succ 0 \end{cases}$

2.5.11. $p = .7013 - .55809 = .1432$

$p = .7364$

2.5.13. $f(t) = \begin{cases} 0, & \text{where} \quad t \prec 0 \\ \frac{\beta(t-\gamma)^{\beta-1}}{\alpha} e^{-\frac{(t-\gamma)^\beta}{\alpha}}, & \text{where} \quad 0 \leq t \end{cases}$

EXERCISES 2.6

2.6.1. $m(t) = \frac{1}{6}\left(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)$

$\text{Var}(x) = 2.9167$

2.6.3. **(a)** $E(Y) = 3.6$

$E(Y^2) = 17.2$

$E(Y^3) = 95.3$

$\text{VAR}(Y) = 4.24$

(b) $M_y(t) = e^{-t} \times .1 + .05 + e^{2t} \times .25 + e^{5t} \times .4 + e^{6t} \times .2$

2.6.5. $E(X) = \sum_x xP(X = x) = \sum_{n=1}^{\infty} 2^n P(X = 2^n) = \sum_{n=1}^{\infty} 2^n \frac{1}{2^n} = \sum_{n=1}^{\infty} 1 = \infty$

2.6.7. $a = \frac{1}{2}$

$b = 1$

2.6.9. **(a)** $E(c) = \sum cf(x) = c$

(b) $E(cg(x)) = \sum cg(x)f(x) = c\sum g(x)f(x) = cE(g(x))$

(c) $E\left(\sum_i g_i(x)\right) = \sum\sum_i g_i(x) f_i(x) = \sum_i\sum g(x) f(x) = \sum_i E(g_i(x))$

(d) $V(ax+b) = E((ax+b) - E(ax+b))^2 = E(ax+b-aE(x)-b)^2 = E(a(x-E(x))^2$
$= a^2 V(x)$

Plug in $b = 0$ get another one.

2.6.11. $E(X) = c \times 1 + 0 = c$
$V(X) = Ex^2 - (E(x))^2 = c^2 - c^2 = 0$
CDF is $F(X) = \varepsilon(x-c)$ where ε is indicator function

2.6.13. $M_x(t) = e^{\lambda(e^t-1)}$
$E(X) = M_x^{'}(0) = \lambda e^t e^{\lambda(e^t-1)}|_{t=0} = \lambda$
$E(X^2) = M_x^{''}(0) = \lambda e^t e^{\lambda(e^t-1)} + (\lambda e^t)^2 e^{\lambda(e^t-1)}|_{t=0} = \lambda + \lambda^2$
$V(X) = E(X^2) - (E(x))^2 = \lambda + \lambda^2 - \lambda^2 = \lambda$

2.6.15. (a) $p = \frac{1}{x+1}$ let $q = 1 - p = \frac{x}{x+1}$ where x start from 0 to infinity means number of failures before the 1st success. Therefore the total number of trails is $x+1$.

$E(x) = \sum_{x=0}^{\infty} xp(1-p)^x = p\sum_{x=0}^{\infty} xp(q)^x q\frac{1}{q} = qp\sum_{x=0}^{\infty}\binom{x+1}{x}(q)^x = qpp^{-2} = \frac{q}{p}$ by negative binomial

Another way to prove is

$$E(x) == \sum_{x=0}^{\infty} xp(1-p)^x = p\sum_{x=0}^{\infty} x(q)^{x-1}q = pq\sum_{x=0}^{\infty}\frac{d(q)^x}{dq}$$

$$= pq\frac{d\left(\sum_{x=0}^{\infty}(q)^x\right)}{dq} = pq\frac{d\left(\frac{1}{1-q}\right)}{dq} = pq\frac{1}{(1-q)^2} = \frac{pq}{p^2} = \frac{q}{p}$$

$$E(x^2) = \sum_{x=0}^{\infty} x^2 p(1-p)^x = pq\sum_{x=0}^{\infty} x^2(q)^{x-1} = pq\sum_{x=0}^{\infty}\frac{d(xq^x)}{dq}$$

$$= pq\frac{d\left(\sum_{x=0}^{\infty}(xq^x)\right)}{dq} = pq\frac{d\left(\frac{1}{p}\sum_{x=0}^{\infty}(xpq^x)\right)}{dq} = pq\frac{d\left(\frac{1}{1-q}E(x)\right)}{dq}$$

$$= pq\frac{d\left(\frac{1}{1-q}\frac{q}{p}\right)}{dq} = pq\frac{d\left(\frac{q}{(1-q)^2}\right)}{dq} = pq\frac{1-2q+q^2+2q-2q^2}{(1-q)^4} = \frac{pq(1-q^2)}{p^4} = \frac{pq-pq^3}{p^4}$$

$$V(x) = E(x^2) - (E(x))^2 = \frac{pq-pq^3}{p^4} - \frac{q^2}{p^2} = \frac{pq-pq^3-p^2q^2}{p^4} = \frac{pq-pq^2(q+p)}{p^4}$$

$$= \frac{pq(1-q)}{p^4} = \frac{p^2q}{p^4} = \frac{q}{p^2}$$

(b) $M_x(t) = \sum_{x=0}^{\infty} e^{xt} p(1-p)^x = p \sum_{x=0}^{\infty} (e^t q)^x = \frac{p}{1-e^t q} = \frac{p}{1-(1-p)e^t}$
When $\left|e^t q\right| \prec 1$ and that is $e^t q \prec 1$ take ln both sides give us $\ln(e^t) \prec \ln\left(\frac{1}{1-p}\right)$
That is when $t \prec -\ln(1-p)$.

2.6.17. $E(x) = \int_0^1 x(x^2)dx + \int_1^2 x\left(\frac{6x-2x^2-3}{2}\right)dx + \int_2^3 x\frac{(x-3)^2}{2}dx = \frac{1}{8} + 1 + \frac{3}{8} = 1.5$

2.6.19. $M_x(t) = \int_0^{\infty} e^{xt}\frac{1}{2}e^{-x}dx + \int_{-\infty}^{0} e^{xt}\frac{1}{2}e^{x}dx = \frac{-1}{(t+1)(t-1)}$

2.6.21. $M_y(t) = \int_0^{\infty} e^{ty}\alpha e^{-\alpha y}dy = \alpha\int_0^{\infty} e^{ty-\alpha y}dy = \alpha\frac{1}{(t-\alpha)}\int_0^{\infty} e^{(t-\alpha)y}d(t-\alpha)y = \alpha\frac{1}{(t-\alpha)}e^{(t-\alpha)y}|_0^{\infty} =$
$\alpha\frac{-1}{(t-\alpha)} = \frac{\alpha}{(\alpha-t)} t \prec \alpha$ and $0 \prec \alpha$
Since $M_y(t) = M_x(t) = \frac{\alpha}{(\alpha-t)}$ and MGF uniquely define the PDF therefore we know that x has the same distribution as y and

$$g(x) = \begin{cases} \alpha e^{-\alpha y}, & 0 \prec \alpha \text{ and } 0 \prec x \\ 0, & \text{otherwise} \end{cases}$$

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Chapter 3

Additional Topics in Probability

EXERCISES 3.2

3.2.1. **(a)** $$P(X = 7) = \binom{10}{7}(0.5)^7(0.5)^{10-7} = 120(0.5)^7(0.5)^3 = 0.117$$

(b) $$P(X \le 7) = 1 - P(8) - P(9) - P(10) = 1 - 0.044 - 0.010 - 0.001 = 0.945$$

(c) $$P(X > 0) = 1 - P(0) = 1 - 0.001 = 0.999$$

(d) $$E(X) = 10(0.5) = 5$$
$$\mathrm{Var}(X) = 10(0.5)(1 - 0.5) = 2.5$$

3.2.3. **(a)** $P(Z > 1.645) = 0.05,$ so $z_0 = 1.645.$

(b) $P(Z < 1.175) = 0.88,$ so $z_0 = 1.175.$

(c) $P(Z < -1.28) = 0.10,$ so $z_0 = -1.28.$

(d) $P(Z > -1.645) = 0.95,$ so $z_0 = -1.645.$

3.2.5. **(a)** $$P(X \le 20) = P\left(Z \le \frac{20-10}{5}\right) = P(Z \le 2) = 0.9772$$

(b) $$P(X > 5) = P\left(Z > \frac{5-10}{5}\right) = P(Z > -1) = 0.8413$$

(c) $$P(12 \le X \le 15) = P\left(\frac{12-10}{5} \le Z \le \frac{15-10}{5}\right) = P(0.4 \le Z \le 1) = P(Z \le 1) - P(Z < 0.4) = 0.8413 - 0.6554 = 0.1859$$

(d) $P(|X - 12| \le 15) = P(-15 \le X - 12 \le 15) = P(-3 \le X \le 27)$

$$= P\left(\frac{-3-10}{5} \le Z \le \frac{27-10}{5}\right)$$
$$= P(-2.6 \le Z \le 3.4) = P(Z \le 3.4) - P(Z < -2.6)$$
$$= 0.9997 - 0.0047 = 0.9950$$

3.2.7. Let X = the number of people satisfied with their health coverage, then $n = 15$ and $p = 0.7$.

(a) $P(X = 10) = \binom{15}{10}(0.7)^{10}(0.3)^{15-10}$

$$= 3003(0.7)^{10}(0.3)^5$$
$$= 0.206$$

There is a 20.6% chance that exactly 10 people are satisfied with their health coverage.

(b) $P(X \le 10) = 1 - P(X = 11) - P(X = 12) - P(X = 13) - P(X = 14) - P(X = 15)$

$$= 0.515$$

There is a 51.5% chance that no more than 10 people are satisfied with their health coverage.

(c) $E(X) = np = 15(0.7) = 10.5$

3.2.9. Let X = the number of defective tubes in a certain box of 400, then $n = 400$ and $p = 3/100 = 0.03$.

(a) $P(X = r) = \binom{400}{r}(0.03)^r(0.97)^{400-r}$

(b) $P(X \ge k) = \sum_{i=k}^{400}\binom{400}{i}(0.03)^i(0.97)^{400-i}$

(c) $P(X \le 1) = P(X = 0) + p(X = 1) = 0.0000684$

(d) Part (c) shows that the probability that at most one defective is 0.0000684, which is very small.

3.2.11. $p(x) = \frac{e^{-\lambda}\lambda^x}{x!} \ge 0$ since $\lambda > 0$ and $x \ge 0$.

Since each $p(x) \ge 0$, then

$$p(x) \le \sum_x p(x) = \sum_x \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\sum_x \frac{\lambda^x}{x!} = e^{-\lambda}\left(1 + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \cdots\right)$$

$= e^{-\lambda}(e^{\lambda}) = 1$ here we apply Taylor's expansion on e^{λ}.

This shows that $p(x) \le 0$ and $\sum_x p(x) = 1$.

3.2.13. The probability density function is given by

$$f(x) = \begin{cases} \dfrac{1}{10}, & 0 \le x \le 10 \\ 0, & \text{otherwise} \end{cases}$$

Hence,

$$P(5 \le X \le 9) = \int_5^9 \frac{1}{10} dx = 0.4.$$

Hence, there is a 40% chance that a piece chosen at random will be suitable for kitchen use.

3.2.15. The probability density function is given by

$$f(x) = \begin{cases} \frac{1}{100}, & 0 \le x \le 100 \\ 0, & \text{otherwise} \end{cases}$$

(a) $P(60 \le X \le 80) = \int_{60}^{80} \frac{1}{100} dx = 0.2.$

(b) $P(X > 90) = \int_{90}^{100} \frac{1}{100} dx = 0.1.$

(c) There is a 20% chance that the efficiency is between 60 and 80 units; there is 10% chance that the efficiency is greater than 90 units.

3.2.17. Let X = the failure time of the component. And X follows exponential distribution with rate 0.05. Then the p.d.f. of X is given by

$$f(x) = 0.05e^{-0.05x}, \ x > 0.$$

Hence,

$$R(10) = 1 - F(10) = 1 - \int_0^{10} 0.05e^{-0.05x} dx = 1 - \left(1 - e^{-0.5}\right) = e^{-0.5} = 0.607.$$

3.2.19. The uniform probability density function is given by

$$f(x) = 1, \quad 0 \le x \le 1.$$

Hence,

$$P(0.5 \le X \le 0.65 | X \le 0.75) = \frac{P(0.5 \le X \le 0.65 \text{ and } X \le 0.75)}{P(X \le 0.75)}$$

$$= \frac{P(0.5 \le X \le 0.65)}{P(X \le 0.75)} = \frac{\int_{0.5}^{0.65} 1dx}{\int_0^{0.75} 1dx} = \frac{0.15}{0.75} = 0.2$$

3.2.21. First, find z_0 such that $P(Z > z_0) = 0.15$.

$P(Z > 1.036) = 0.15$, so $z_0 = 1.036$.

$x_0 = 72 + 1.036 \cdot 6 = 78.22$

The minimum score that a student has to get to an "A" grade is 78.22.

3.2.23. $P(1.9 \le X \le 2.02) = P\left(\frac{1.9 - 1.96}{0.04} \le Z \le \frac{2.02 - 1.96}{0.04}\right) = P(-1.5 \le Z \le 1.5) = 0.866$

$P(X < 1.9 \text{ or } X > 2.02) = 1 - P(1.9 \le X \le 2.02) = 0.134$

13.4% of the balls manufactured by the company are defective.

3.2.25. **(a)** $P(X > 125) = P\left(Z > \frac{125 - 115}{10}\right) = P(Z > 1) = 0.16$

(b) $P(X < 95) = P\left(Z < \frac{95 - 115}{10}\right) = P(Z < -2) = 0.023$

(c) First, find z_0 such that $P(Z < z_0) = 0.95$.
$P(Z < 1.645) = 0.95$, so $z_0 = 1.645$.
$x_0 = 115 + 1.645 \cdot 10 = 131.45$

(d) There is a 16% chance that a child chosen at random will have a systolic pressure greater than 125 mm Hg. There is a 2.3% chance that a child will have a systolic pressure less than 95 mm Hg. 95% of this population have a systolic blood pressure below 131.45.

3.2.27. First find z_1, z_2 and z_3 such that

$$P(Z > z_1) = 0.2,\ P(Z > z_2) = 0.5 \text{ and } P(Z > z_3) = 0.8$$

Using standard normal table, we can find that $z_1 = -0.842$, $z_2 = 0$ and $z_3 = 0.842$.
Then
$y_1 = 0 + (-0.842) \cdot 0.65 = -0.5473 \Rightarrow x_1 = \exp(y_1) = 0.58$, similarly we can obtain $x_2 = 1$ and $x_3 = 1.73$.
For the probability of surviving 0.2, 0.5 and 0.8 the experimenter should choose doses 0.58, 1 and 1.73, respectively.

3.2.29. **(a)**
$$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx} \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} dx$$
$$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty x^{\alpha-1} \exp\left(-\frac{1-\beta t}{\beta} x\right) dx$$
$$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty \left(\frac{\beta}{1-\beta t} u\right)^{\alpha-1} e^{-u} \frac{\beta}{1-\beta t} du \text{ by letting } u = \frac{1-\beta t}{\beta} x \text{ with } 1 - \beta t > 0$$
$$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \left(\frac{\beta}{1-\beta t}\right)^\alpha \int_0^\infty u^{\alpha-1} \exp(-u) du$$
note that the integrand is the kernel density of $\Gamma(\alpha, 1)$
$$= \frac{1}{\Gamma(\alpha)\beta^\alpha} \left(\frac{\beta}{1-\beta t}\right)^\alpha \Gamma(\alpha) \cdot 1^\alpha$$
$$= (1 - \beta t)^{-\alpha} \text{ when } t < \tfrac{1}{\beta}.$$

(b) $E(X) = M'_X(0) = -\alpha(1-\beta t)^{-\alpha-1}(-\beta)\big|_{t=0} = \alpha\beta$, and

$$E(X^2) = M_X^{(2)}(0) = \frac{d}{dt}[-\alpha(1-\beta t)^{-\alpha-1}(-\beta)]\big|_{t=0}$$

$$= \alpha(\alpha+1)(1-\beta t)^{-\alpha-2}(-\beta)^2\big|_{t=0} = \alpha(\alpha+1)\beta^2$$

Then $\text{Var}(X) = E(X^2) - E(X)^2 = \alpha(\alpha+1)\beta^2 - (\alpha\beta)^2 = \alpha\beta^2$.

3.2.31. **(a)** First consider the following product

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^\infty u^{\alpha-1}e^{-u}du \int_0^\infty v^{\beta-1}e^{-v}dv$$

$$= \int_0^\infty x^{2(\alpha-1)}e^{-x^2}2xdx \int_0^\infty y^{2(\beta-1)}e^{-y^2}2ydv \text{ by letting } u = x^2 \text{ and } v = y^2$$

$$= \left(2\int_0^\infty |x|^{2\alpha-1}\, e^{-x^2}dx\right)\left(2\int_0^\infty |y|^{2\beta-1}\, e^{-y^2}dy\right) \text{ noting that the integrands are even functions}$$

$$= \left(\int_{-\infty}^\infty |x|^{2\alpha-1}\, e^{-x^2}dx\right)\left(\int_{-\infty}^\infty |y|^{2\beta-1}\, e^{-y^2}dy\right)$$

$$= \int_{-\infty}^\infty\int_{-\infty}^\infty |x|^{2\alpha-1}\,|y|^{2\beta-1}\, e^{-(x^2+y^2)}dxdy$$

Transforming to polar coordinates with $x = r\cos\theta$ and $y = r\sin\theta$

$$\Gamma(\alpha)\Gamma(\beta) = \int_0^{2\pi}\int_0^\infty |r\cos\theta|^{2\alpha-1}\,|r\sin\theta|^{2\beta-1}\, e^{-r^2}rdrd\theta$$

$$= \int_0^\infty r^{2\alpha+2\beta-2}e^{-r^2}rdr \int_0^{2\pi} \left|(\cos\theta)^{2\alpha-1}(\sin\theta)^{2\beta-1}\right| d\theta$$

$$= \left(\frac{1}{2}\int_0^\infty s^{\alpha+\beta-1}e^{-s}ds\right)\left(4\int_0^{\pi/2}(\cos\theta)^{2\alpha-1}(\sin\theta)^{2\beta-1}d\theta\right) \text{ by letting } s = r^2$$

$$= \Gamma(\alpha+\beta)2\int_0^{\pi/2}(\cos\theta)^{2\alpha-1}(\sin\theta)^{2\beta-1}d\theta$$

$$= \Gamma(\alpha+\beta)2\int_0^1 t^{\alpha-1/2}(1-t)^{\beta-1/2}\frac{1}{2\sqrt{t(1-t)}}dt \text{ by letting } t = \cos^2\theta$$

$$= \Gamma(\alpha+\beta)\int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt$$
$$= \Gamma(\alpha+\beta)B(\alpha,\beta)$$

Hence, we have shown that

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

(b) $$E(X) = \frac{1}{B(\alpha,\beta)}\int_0^1 x\cdot x^{\alpha-1}(1-x)^{\beta-1}dx = \frac{B(\alpha+1,\beta)}{B(\alpha,\beta)}\int_0^1 \frac{x^{(\alpha+1)-1}(1-x)^{\beta-1}}{B(\alpha+1,\beta)}dx$$
$$= \frac{B(\alpha+1,\beta)}{B(\alpha,\beta)}\cdot 1 = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$$
$$= \frac{\alpha\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta)\Gamma(\alpha+\beta)}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\alpha}{\alpha+\beta}, \text{ and}$$

$$E(X^2) = \frac{1}{B(\alpha,\beta)}\int_0^1 x^2\cdot x^{\alpha-1}(1-x)^{\beta-1}dx = \frac{B(\alpha+2,\beta)}{B(\alpha,\beta)}\int_0^1 \frac{x^{(\alpha+2)-1}(1-x)^{\beta-1}}{B(\alpha+2,\beta)}dx$$
$$= \frac{B(\alpha+2,\beta)}{B(\alpha,\beta)}\cdot 1 = \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$$
$$= \frac{\alpha(\alpha+1)\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta)(\alpha+\beta+1)\Gamma(\alpha+\beta)}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}.$$

Then $\mathrm{Var}(X) = E(X^2) - E(X)^2 = \dfrac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 = \dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$

3.2.33. In this case, the number of breakdowns per month can be assumed to have Poisson distribution with mean 3.

(a) $P(X=1) = \frac{e^{-3}3^1}{1!} = 0.1494$. There is a 14.94% chance that there will be just one network breakdown during December.

(b) $P(X \geq 4) = 1 - P(0) - P(1) - P(2) - P(3) = 0.3528$. There is a 35.28% chance that there will be at least 4 network breakdowns during December.

(c) $P(X \leq 7) = \sum_{x=0}^{7} \frac{e^{-3}3^x}{x!} = 0.9881$. There is a 98.81% chance that there will be at most 7 network breakdowns during December.

3.2.35. **(a)** $$P(1 < X < 4) = \int_1^4 \frac{1}{\Gamma(2)\cdot 1^2}x^{2-1}e^{-x/1}dx = \int_1^4 xe^{-x}dx$$
$$= \left(-xe^{-x}\right)\Big|_1^4 - \int_1^4 (-e^{-x})dx \text{ by integration by parts}$$
$$= 0.6442$$

The probability that an acid solution made by this procedure will satisfactorily etch a tray is 0.6442.

(b) $P(1 < X < 4) = \int\limits_1^4 \frac{1}{\Gamma(1) \cdot 2^1} x^{1-1} e^{-x/2} dx = \frac{1}{2} \int\limits_1^4 e^{-x/2} dx = \left(-e^{-x/2} \right)\Big|_1^4 = 0.4712.$

The probability that an acid solution made by this procedure will satisfactorily etch a tray is 0.4712.

EXERCISES 3.3

3.3.1. **(a)** The joint probability function is

$$P(X = x, Y = y) = \frac{\binom{8}{x}\binom{6}{y}\binom{10}{4-x-y}}{\binom{24}{4}},$$

where $0 \le x \le 4,\ 0 \le y \le 4$, and $0 \le x + y \le 4$.

(b) $P(X = 3, Y = 0) = \dfrac{\binom{8}{3}\binom{6}{0}\binom{10}{4-3-0}}{\binom{24}{4}} = 0.053.$

(c) $P(X < 3, Y = 1) = \sum\limits_{x=0}^{2} P(X = x, Y = 1) = \sum\limits_{x=0}^{2} \dfrac{\binom{8}{x}\binom{6}{1}\binom{10}{4-x-1}}{\binom{24}{4}} = 0.429.$

(d)

	y					
x	0	1	2	3	4	Sum
0	0.020	0.068	0.064	0.019	0.001	0.172
1	0.090	0.203	0.113	0.015		0.421
2	0.119	0.158	0.040			0.317
3	0.053	0.032				0.085
4	0.007					0.007
Sum	0.289	0.461	0.217	0.034	0.001	1.00

3.3.3. $\int\limits_{-1}^{1}\int\limits_{-1}^{1} f(x, y)dxdy = \int\limits_{-1}^{1}\int\limits_{-1}^{1} c(1-x)(1-y)dxdy = c\left(\int\limits_{-1}^{1} (1-x)dx\right)\left(\int\limits_{-1}^{1} (1-y)dy\right)$

$= c\left(x - \frac{x^2}{2}\Big|_{-1}^{1} \right)\left(y - \frac{y^2}{2}\Big|_{-1}^{1} \right) = 4c$

Thus, if $c = 1/4$, then $\int_{-1}^{1}\int_{-1}^{1} f(x,y)dxdy = 1$. And we also see that $f(x,y) \geq 0$ for all x and y. Hence, $f(x,y)$ is a joint probability density function.

3.3.5. By definition, the marginal pdf of X is given by the row sums, and the marginal pdf of Y is obtained by the column sums. Hence,

x_i	-1	3	5	otherwise
$f_X(x_i)$	0.6	0.3	0.1	0

y_i	-2	0	1	4	otherwise
$f_Y(y_i)$	0.4	0.3	0.1	0.2	0

3.3.7. From **Exercise 3.3.5** we can calculate the following.

$$P(X=-1|Y=0) = \frac{P(X=-1, Y=0)}{f_Y(0)} = \frac{0.1}{0.3} = 0.33.$$

3.3.9. **(a)** The marginal of X is

$$f_X(x) = \int_x^2 f(x,y)dy = \int_x^2 \frac{8}{9}xydy = \frac{4}{9}(4x - x^3),\ 1 \leq x \leq 2.$$

(b) $$P(1.5 < X < 1.75, Y > 1) = \int_{1.5}^{1.75}\left(\int_x^2 \frac{8}{9}xydy\right)dx = \int_{1.5}^{1.75} \frac{4}{9}\left(4x - x^3\right)dx$$

$$= \frac{4}{9}\left(2x^2 - \frac{x^4}{4}\right)\Bigg|_{1.5}^{1.75} = 0.2426.$$

3.3.11. Using the joint density in **Exercise 3.3.9** we can obtain the joint mgf of (X, Y) as

$$M_{(X,Y)}(t_1,t_2) = E(e^{t_1X+t_2Y}) = \int_1^2\int_x^2 e^{t_1x+t_2y}\frac{8}{9}xydydx$$

$$= \int_1^2 \frac{8}{9}xe^{t_1x}\left(\int_x^2 e^{t_2y}ydy\right)dx = \int_1^2 \frac{8}{9}xe^{t_1x}\left(K - \frac{x}{t_2}e^{t_2x} + \frac{1}{t_2^2}e^{t_2x}\right)dx$$

$$\text{where } K = \frac{e^{2t_2}}{t_2^2}(2t_2 - 1)$$

$$= \frac{8}{9}K\int_1^2 xe^{t_1x}dx - \frac{8}{9t_2}\int_1^2 x^2e^{(t_1+t_2)x}dx + \frac{8}{9t_2^2}\int_1^2 xe^{(t_1+t_2)x}dx$$

$$= \frac{8}{9}K\left(\frac{x}{t_1}e^{t_1x} - \frac{1}{t_1^2}e^{t_1x}\right)\Bigg|_1^2$$

$$- \frac{8}{9t_2}\left(\frac{x^2}{t_1+t_2}e^{(t_1+t_2)x} - \frac{2x}{(t_1+t_2)^2}e^{(t_1+t_2)x} + \frac{2}{(t_1+t_2)^3}e^{(t_1+t_2)x}\right)\Bigg|_1^2$$

$$+ \frac{8}{9t_2^2}\left(\frac{x}{t_1+t_2}e^{(t_1+t_2)x} - \frac{1}{(t_1+t_2)^2}e^{(t_1+t_2)x}\right)\Bigg|_1^2$$

After simplification we then have

$$M_{(X,Y)}(t_1,t_2) = \frac{t_1+3t_2-t_1^2-3t_2^2-4t_1t_2+t_1^2t_2+2t_1t_2^2+t_2^3}{t_2^2(t_1+t_2)^3}e^{t_1+t_2} + \frac{(2t_2-1)(1-t_1)}{t_1^2t_2^2}e^{t_1+2t_2}$$
$$+ \left(\frac{-t_1-3t_2+2t_1^2+6t_2^2+8t_1t_2-4t_1^2t_2+-8t_1t_2^2-4t_2^3}{t_2^2(t_1+t_2)^3} + \frac{(2t_2-1)(2t_1-1)}{t_1^2t_2^2}\right)e^{2t_1+2t_2}$$

3.3.13. **(a)** $f_X(x) = \sum_{y=0}^{n} f(x,y) = \sum_{y=0}^{n}\left[\frac{6xy}{n(n+1)(2n+1)}\right]^2 = \frac{36x^2}{[n(n+1)(2n+1)]^2}\sum_{y=0}^{n} y^2$

$$= \frac{6x^2}{n(n+1)(2n+1)}, x = 1, 2, \ldots, n.$$

$$f_Y(y) = \sum_{x=0}^{n} f(x,y) = \sum_{x=0}^{n}\left[\frac{6xy}{n(n+1)(2n+1)}\right]^2 = \frac{36y^2}{[n(n+1)(2n+1)]^2}\sum_{x=0}^{n} x^2$$

$$= \frac{6y^2}{n(n+1)(2n+1)}, y = 1, 2, \ldots, n.$$

Given $y = 1, 2, \ldots, n$, we have

$$f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{\left[\frac{6xy}{n(n+1)(2n+1)}\right]^2}{\frac{6y^2}{n(n+1)(2n+1)}} = \frac{6x^2}{n(n+1)(2n+1)}, x = 1, 2, \ldots, n.$$

(b) Given $x = 1, 2, \ldots, n$, we have

$$f(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{\left[\frac{6xy}{n(n+1)(2n+1)}\right]^2}{\frac{6x^2}{n(n+1)(2n+1)}} = \frac{6y^2}{n(n+1)(2n+1)}, y = 1, 2, \ldots, n.$$

3.3.15. **(a)** $E(XY) = \sum_{x,y} xy \cdot f(x,y) = \sum_{x=1}^{3}\sum_{y=1}^{3} xy \cdot f(x,y) = \frac{35}{12}.$

(b) $E(X) = \sum_{x,y} x \cdot f(x,y) = \sum_{x=1}^{3}\sum_{y=1}^{3} x \cdot f(x,y) = \frac{5}{3}$, and

$$E(Y) = \sum_{x,y} y \cdot f(x,y) = \sum_{x=1}^{3}\sum_{y=1}^{3} y \cdot f(x,y) = \frac{11}{6}.$$

Then, $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = \frac{35}{12} - \frac{5}{3}\cdot\frac{11}{6} = -\frac{5}{36}$.

(c) $\mathrm{Var}(X) = \sum_{x,y}[x - E(X)]^2 \cdot f(x,y) = \sum_{x=1}^{3}\sum_{y=1}^{3}\left(x - \frac{5}{3}\right)^2 \cdot f(x,y) = \frac{5}{9}$, and

$\mathrm{Var}(Y) = \sum_{x,y}[y - E(Y)]^2 \cdot f(x,y) = \sum_{x=1}^{3}\sum_{y=1}^{3}\left(y - \frac{11}{6}\right)^2 \cdot f(x,y) = \frac{23}{36}$.

Then, $\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} = \frac{-5/36}{\sqrt{(5/9)(23/36)}} = -0.233$.

3.3.17. Assume that a and c are nonzero.

$\mathrm{Cov}(U,V) = \mathrm{Cov}(aX+b, cY+d) = ac\mathrm{Cov}(X,Y)$,

$\mathrm{Var}(U) = \mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$, and $\mathrm{Var}(V) = \mathrm{Var}(cY+d) = c^2\mathrm{Var}(Y)$.

Then, $\rho_{UV} = \frac{\mathrm{Cov}(U,V)}{\sqrt{\mathrm{Var}(U)\mathrm{Var}(V)}} = \frac{ac\mathrm{Cov}(X,Y)}{\sqrt{a^2\mathrm{Var}(X)c^2\mathrm{Var}(Y)}} = \frac{ac}{\sqrt{(ac)^2}}\rho_{XY}$

$$= \frac{ac}{|ac|}\rho_{XY} = \begin{cases}\rho_{XY}, & \text{if } \ ac > 0\\ -\rho_{XY}, & \text{otherwise}\end{cases}.$$

3.3.19. We fist state the famous Cauchy–Schwarz inequality:
$|E(XY)| \le \sqrt{E(X^2)E(Y^2)}$ and the equality holds if and only if there exists some constant α and β, not both zero, such that $P(\alpha\,|X|^2 = \beta\,|Y|^2) = 1$.
Now, consider

$$|\rho_{XY}| = 1 \Leftrightarrow \left|\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}\right| = 1 \Leftrightarrow |\mathrm{Cov}(X,Y)| = \sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$$

$$\Leftrightarrow |E(X-\mu_X)(Y-\mu_Y)| = \sqrt{E(X-\mu_X)^2E(Y-\mu_Y)^2}$$

By the Cauchy–Schwarz inequality we have

$$P\left(\alpha\,|X-\mu_X|^2 = \beta\,|Y-\mu_Y|^2\right) = 1$$

$\Leftrightarrow P(X - \mu_X = K(Y-\mu_Y)) = 1$ for some constant K

$\Leftrightarrow P(X = aY + b) = 1$ for some constants a and b.

3.3.21. **(a)** First, we compute the marginal densities.

$$f_X(x) = \int_x^\infty f(x,y)dy = \int_x^\infty e^{-y}dy = e^{-x},\ x \ge 0, \text{ and}$$

$$f_Y(y) = \int_0^y f(x,y)dx = \int_0^y e^{-y}dx = ye^{-y},\ y \ge 0.$$

For given $y \geq 0$, we have the conditional density as

$$f(x|Y=y) = \frac{f(x,y)}{f_Y(y)} = \frac{e^{-y}}{\frac{1}{y}e^{-y}} = \frac{1}{y}, \quad 0 \leq x \leq y.$$

Then, $(X|Y=y)$ follows *Uniform*$(0,y)$. Thus, $E(X|Y=y) = \frac{y}{2}$.

(b) $E(XY) = \int_y \int_x xy \cdot f(x,y)dxdy = \int_0^\infty \left(\int_0^y xye^{-y}dx\right)dy = \int_0^\infty \frac{1}{2}y^3e^{-y}dy = 3,$

$E(X) = \int_x x \cdot f_X(x)dx = \int_0^\infty xe^{-x}dx = 1$, and

$E(Y) = \int_y y \cdot f_Y(y)dy = \int_0^\infty y^2e^{-y}dy = 2.$

Then, $\text{Cov}(X,Y) = E(XY) - E(X)E(Y) = 3 - 1 \cdot 2 = 1.$

(c) To check for independence of X and Y

$$f_X(1)f_Y(1) = e^{-2} \neq e^{-1} = f(1,1).$$

Hence, X and Y are not independent.

3.3.23. Let $\sigma^2 = \text{Var}(X) = \text{Var}(Y)$. Since X and Y are independent, we have $\text{Cov}(X,Y) = E(XY) - E(X)E(Y) = E(X)E(Y) - E(X)(Y) = 0$. Then, $\text{Cov}(X, aX+Y) = a\text{Cov}(X,X) + \text{Cov}(X,Y) = a\text{Var}(X) = a\sigma^2$, and $\text{Var}(aX+Y) = a^2\text{Var}(X) + \text{Var}(Y) = (a^2+1)\sigma^2$. Thus, $\rho_{X,aX+Y} = \frac{\text{Cov}(X,aX+Y)}{\sqrt{\text{Var}(X)\text{Var}(aX+Y)}} = \frac{a\sigma^2}{\sqrt{\sigma^2(a^2+1)\sigma^2}} \frac{a}{\sqrt{a^2+1}}$.

EXERCISES 3.4

3.4.1. The pdf of X is $f_X(x) = \frac{1}{a}$ if $0 < x < a$ and zero otherwise.

$$F_Y(y) = P(Y < y) = P(cX + d < y) = P\left(X < \frac{y-d}{c}\right) = \int_{-\infty}^{\frac{y-d}{c}} f_X(x)dx$$

$$= \begin{cases} 1, & \text{if } \frac{y-d}{c} \geq a \\ \frac{y-d}{ac}, & \text{if } 0 < \frac{y-d}{c} < a \\ 0, & \text{otherwise} \end{cases} = \begin{cases} 1, & \text{if } y \geq ac+d \\ \frac{y-d}{ac}, & \text{if } d < y < ac+d \\ 0, & \text{otherwise} \end{cases}$$

Then, $f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{1}{ac}$ if $d < y < ac + d$ and zero otherwise.

3.4.3. Let $U = XY$ and $V = Y$.

Then $X = U/V$ and $Y = V$, and

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{v} & -\frac{u}{v^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{v}.$$

Then the joint pdf of U and V is given by

$$f_{U,V}(u,v) = f_{X,Y}\left(\frac{u}{v}, v\right) \cdot |J| = f_{X,Y}\left(\frac{u}{v}, v\right)\left|\frac{1}{v}\right|$$

Then the pdf of U is given by

$$f_U(u) = \int_v f_{U,V}(u,v)dv = \int_{-\infty}^{\infty} f_{X,Y}\left(\frac{u}{v}, v\right)\left|\frac{1}{v}\right| dv.$$

3.4.5. The joint pdf of (X, Y) is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_1\sigma_2} e^{-\left(\frac{x^2}{2\sigma_1^2}+\frac{y^2}{2\sigma_2^2}\right)}, \ -\infty < x < \infty, \ -\infty < y < \infty; \ \sigma_1, \ \sigma_2 > 0.$$

We can easily show that the marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{x^2}{\sigma_1^2}}, \ -\infty < x < \infty, \text{ and}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{y^2}{\sigma_2^2}}, \ -\infty < y < \infty.$$

This implies that $X \sim N\left(0, \sigma_1^2\right)$ and $Y \sim N\left(0, \sigma_2^2\right)$.
Also, notice that $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all x and y, thus X and Y are independent.
By the definition of Chi-Square distribution and the independency of X and Y, we know that $\frac{1}{\sigma_1^2}X^2 + \frac{1}{\sigma_2^2}Y^2 \sim \chi^2(2)$. Therefore, the pdf of $U = \frac{1}{\sigma_1^2}X^2 + \frac{1}{\sigma_2^2}Y^2$ is

$$f_U(u) = \frac{1}{2}e^{-\frac{u}{2}}, \quad u > 0.$$

3.4.7. Let $U = X + Y$ and $V = Y$.
Then $X = U - V$ and $Y = V$, and

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1.$$

Then the joint pdf of U and V is given by

$$f_{U,V}(u,v) = f_{X,Y}(u-v, v) \cdot |J| = f_{X,Y}(u-v, v).$$

Thus, the pdf of U is given by $f_U(u) = \int_{-\infty}^{\infty} f_{X,Y}(u-v, v)dv$.

3.4.9. **(a)** Here let $g(x) = \frac{x-\mu}{\sigma}$, and hence, $g^{-1}(z) = \sigma z + \mu$. Thus, $\frac{d}{dz}g^{-1}(z) = \sigma$. Also, $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$, $-\infty < x < \infty$. Therefore, the pdf of Z is $f_Z(z) = f_X(g^{-1}(z))$ $\left|\frac{d}{dz}g^{-1}(z)\right| = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}$, $-\infty < z < \infty$, which is the pdf of $N(0,1)$.

(b) The cdf of U is given by

$$F_U(u) = P(U \le u) = P\left(\frac{(X-\mu)^2}{\sigma^2} \le u\right) = P\left(-\sqrt{u} \le \frac{X-\mu}{\sigma} \le \sqrt{u}\right) = P\left(-\sqrt{u} \le Z \le \sqrt{u}\right)$$

$$= \int_{-\sqrt{u}}^{\sqrt{u}} \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}\,dz = 2\int_{0}^{\sqrt{u}} \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}\,dz, \text{ since the integrand is an even function.}$$

Hence, the pdf of U is $f_U(u) = \frac{d}{du}F_U(u) = \frac{2}{\sqrt{2\pi}}e^{-\frac{u}{2}}\frac{1}{2\sqrt{u}} = \frac{1}{\sqrt{2\pi}}u^{-\frac{1}{2}}e^{-\frac{u}{2}}$, $u > 0$ and zero otherwise, which is the pdf of $\chi^2(1)$.

3.4.11. Since the support of the pdf of V is $v > 0$, then $g(v) = \frac{1}{2}mv^2$ is a one-to-one function on the support. Hence, $g^{-1}(y) = \sqrt{\frac{2y}{m}}$. Thus, $\frac{d}{de}g^{-1}(y) = \frac{1}{2}\sqrt{\frac{m}{2y}\frac{2}{m}}$. Therefore, the pdf of E is given by

$$f(y) = f_V(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| = c\frac{2y}{m}e^{-\beta\frac{2y}{m}}\frac{1}{\sqrt{2my}} = \frac{c\sqrt{2y}}{\sqrt{m^3}}e^{-\frac{2\beta\cdot y}{m}}, \quad y > 0.$$

3.4.13. Let $U = \sqrt{X^2+Y^2}$ and $V = \tan^{-1}\left(\frac{Y}{X}\right)$. Here U is considered to be the radius and V is the angle. Hence, this is a polar transformation and hence is one-to-one.
Then $X = U\cos V$ and $Y = U\sin V$, and

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \cos v & -u\sin v \\ \sin v & u\cos v \end{vmatrix} = u\cos^2 v + u\sin^2 v = u.$$

Then the joint pdf of U and V is given by

$$f_{U,V}(u,v) = f_{X,Y}(uv, v)\cdot|J| = \frac{1}{2\pi\sigma^2}\exp\left\{-\frac{1}{2\sigma^2}\left(u^2\cos^2 v + u^2\sin^2 v\right)\right\}u.$$

$$= \frac{u}{2\pi\sigma^2}e^{-\frac{u^2}{2\sigma^2}}, \; u > 0,\; 0 \le v < 2\pi.$$

3.4.15. The joint pdf of (X, Y) is

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) = \frac{1}{4}e^{-\frac{x+y}{2}}, \; x, y > 0.$$

Apply the result in **Exercise 3.4.14** with $\beta = 2$. We have the joint pdf of $U = \frac{X-Y}{2}$ and $V = Y$ as $f_{U,V}(u,v) = \frac{1}{2}e^{-(u+v)}$, $v > -2u$, $v > 0$. Thus, the pdf of U is given by

$$f_U(u) = \int_v f_{U,V}(u,v)dv$$

$$= \begin{cases} \displaystyle\int_0^\infty \frac{1}{2}e^{-(u+v)}dv = \frac{1}{2}e^{-u}, & u \ge 0 \\ \displaystyle\int_{-2u}^\infty \frac{1}{2}e^{-(u+v)}dv = \frac{1}{2}e^{u}, & u < 0. \end{cases}$$

EXERCISES 3.5

3.5.1. **(a)** Note that X follows $\beta(5,5)$, then apply the result in **Exercise 3.2.31** we have $\mu = 1/2$ and $\sigma^2 = 1/44$. From the Chebyshev's theorem

$$P(\mu - K\sigma < X < \mu + K\sigma) \ge 1 - \frac{1}{K^2}.$$

Equating $\mu - K\sigma$ to 0.2 and $\mu + K\sigma$ to 0.8 with $\mu = 1/2$ and $\sigma = \sqrt{1/44} = 0.15$, we obtain $K = 2$. Hence, $P(0.2 < X < 0.8) \ge 1 - \dfrac{1}{2^2} = 0.75$.

(b) $P(0.2 < X < 0.8) = \int_{0.2}^{0.8} 630x^4(1-x)^4 dx = 0.961$.

3.5.3. $\sigma^2 = E(X-\mu)^2 = \sum_x (x-\mu)^2 f(x)$

$$= \sum_{x\le\mu-K\sigma} (x-\mu)^2 f(x) + \sum_{\mu-K\sigma<x<\mu+K\sigma} (x-\mu)^2 f(x) + \sum_{x\ge\mu+K\sigma} (x-\mu)^2 f(x)$$

$$\ge \sum_{x\le\mu-K\sigma} (x-\mu)^2 f(x) + \sum_{x\ge\mu+K\sigma} (x-\mu)^2 f(x)$$

Note that $(x-\mu)^2 \ge K^2\sigma^2$ for $x \le \mu - K\sigma$ or $x \ge \mu + K\sigma$. Then the above inequality can be written as

$$\sigma^2 \ge K^2\sigma^2 \left[\sum_{x\le\mu-K\sigma} f(x) + \sum_{x\ge\mu+K\sigma} f(x) \right]$$

$$= K^2\sigma^2 \left[P(X \le \mu - K\sigma) + P(X \ge \mu + K\sigma) \right]$$

$$= K^2\sigma^2 P(|X-\mu| \ge K\sigma)$$

This implies that $P(|X-\mu| \ge K\sigma) \le \frac{1}{K^2}$ or $P(|X-\mu| < K\sigma) \ge 1 - \frac{1}{K^2}$.

3.5.5. Apply Chebyshev's theorem we have

$$P\left(\left|\frac{X_n}{n}-p\right|<0.1\right)=P(|X_n-np|<0.1n)\ge 1-\frac{E(X_n-np)^2}{(0.1n)^2}$$
$$=1-\frac{\mathrm{Var}(X_n)}{0.01n^2}=1-\frac{np(1-p)}{0.01n^2}$$
$$=1-100\frac{p(1-p)}{n}\ge 1-\frac{100}{n}\cdot\frac{1}{4},\text{ since } p(1-p)\le\frac{1}{4}.$$

This implies that we want to find n such that

$$1-\frac{100}{n}\cdot\frac{1}{4}=0.9\Rightarrow\frac{25}{n}=0.1\Rightarrow n=250.$$

3.5.7. Let $X_1,\ldots,X_n$ denote each toss of coin with value 1 if head occurs and 0 otherwise. Then, $X_1,\ldots,X_n$ are independent variables which follow *Bernoulli* distribution with $p=1/2$. Thus, $E(X_i)=1/2$, and $\mathrm{Var}(X_i)=1/4$. For any $\varepsilon>0$, from the law of large numbers we have

$$P\left(\left|\frac{S_n}{n}-\frac{1}{2}\right|<\varepsilon\right)\to 1 \text{ as } n\to\infty,\text{ i.e. } \frac{S_n}{n} \text{ will be near to } \frac{1}{2} \text{ for large } n.$$

If the coin is not fair, then the fraction of heads, $\frac{S_n}{n}$, will be near to the true probability of getting head for large n.

3.5.9. Note that $E(X_i)=0$, and $\mathrm{Var}(X_i)=i$. Hence, $X_1,\ldots,X_n$ are not identically distributed. Thus, the conditions of the law of large numbers stated in the text are not satisfied. There is a weaker version of the weak law of large numbers which requires only $E(X_i)=\mu$, and $\mathrm{Var}(\overline{X})\to 0$ as $n\to\infty$. However, in this case

$$\mathrm{Var}(\overline{X})=\mathrm{Var}\left(\frac{\sum_{i=1}^{n}X_i}{n}\right)=\frac{\sum_{i=1}^{n}\mathrm{Var}(X_i)}{n^2}=\frac{\sum_{i=1}^{n}i}{n^2}=\frac{\frac{n(n+1)}{2}}{n^2}\to\frac{1}{2}\text{ as } n\to\infty.$$

Therefore, the conditions of the weaker version are not satisfied, either.

3.5.11. First note that $E(X_i)=2$ and $\mathrm{Var}(X_i)=2$. From the CLT, $\dfrac{\overline{X}_{100}-2}{2/\sqrt{100}}$ follows approximately $N(0,1)$. Hence, we have

$$P(\overline{X}_{100}>2)=P\left(\frac{\overline{X}_{100}-2}{2/\sqrt{100}}>\frac{2-2}{2/\sqrt{100}}\right)\approx P(Z>0)=0.5,\text{ where } Z\sim N(0,1).$$

3.5.13. First note that $E(X_i)=1/2$ and $\mathrm{Var}(X_i)=1/12$. Then, by CLT we know that $Z_n=\frac{S_n-n/2}{\sqrt{n/12}}=\frac{\overline{X}-1/2}{\sqrt{1/12}/\sqrt{n}}$ approximately follows $N(0,1)$ for large n.

3.5.15. From the Chebyshev's theorem

$$P(\mu - K\sigma < X < \mu + K\sigma) \geq 1 - \frac{1}{K^2}.$$

Equating $\mu - K\sigma$ to 104 and $\mu + K\sigma$ to 140 with $\mu = 122$ and $\sigma = 2$, we obtain $K = 9$. Hence,

$$P(104 < X < 140) \geq 1 - \frac{1}{9^2} = 0.988.$$

3.5.17. Let $X_i = 1$ if the ith person in the sample is color blind and 0 otherwise. Then each X_i follows *Bernoulli* distribution with estimated probability 0.02, and $E(X_i) = 0.02$ and $\text{Var}(X_i) = 0.0196$. Let $S_n = \sum_{i=1}^{n} X_i$.
We want $P(S_n \geq 1) = 0.99$. By the CLT, $\frac{S_n/n - 0.02}{\sqrt{0.0196/n}}$ follows approximately $N(0, 1)$. Then,

$$0.99 = P(S_n \geq 1) \approx P\left(Z \geq \frac{1/n - 0.02}{\sqrt{0.0196/n}}\right).$$

Using the normal table, $\frac{1/n - 0.02}{\sqrt{0.0196/n}} = -2.33$. Solving this equation, we have $n = 359.05$. Thus, the sample size must be at least 360.

3.5.19. Let $X_1, \ldots, X_{100}$ be iid with $\mu = 1$ and $\sigma^2 = 0.04$. Let $\overline{X} = \frac{1}{100}\sum_{i=1}^{100} X_i$.
By the CLT, $\frac{\overline{X} - 1}{\sqrt{0.04/100}}$ follows approximately $N(0, 1)$. Then,

$$P(0.99 \leq \overline{X} \leq 1) \approx P\left(\frac{0.99 - 1}{\sqrt{0.04/100}} \leq Z \leq \frac{1 - 1}{\sqrt{0.04/100}}\right) = P(-0.5 \leq Z \leq 0) = 0.1915.$$

3.5.21. Let $X_i = 1$ if the ith dropper in the sample is defective and 0 otherwise. Then each X_i follows *Bernoulli* distribution with estimated probability $\frac{500}{10000} = 0.05$, and $E(X_i) = 0.05$ and $\text{Var}(X_i) = 0.0475$. Let $S_{125} = \sum_{i=1}^{125} X_i$.From the CLT, $\frac{S_{125} - 125 \cdot 0.05}{\sqrt{0.0475 \cdot 125}}$ follows approximately $N(0, 1)$. Hence, we have

$$P(S_{125} \leq 2) \approx P\left(Z \leq \frac{2 - 125 \cdot 0.05}{\sqrt{0.0475 \cdot 125}}\right) = P(Z \leq -1.744) = 0.041.$$

Chapter 4

Sampling Distributions

EXERCISES 4.1

4.1.1. **(a)** There are $\binom{5}{3} = 10$ equally likely possible samples of size 3, so the probability for each is $1/10$ without replacement:

	$\overline{X}$	M	S
$(-2, -1, 0)$	-1	-1	1
$(-2, -1, 1)$	$-2/3$	-1	$2\frac{1}{3}$
$(-2, -1, 2)$	$-1/3$	-1	$4\frac{1}{3}$
$(-2, 0, 1)$	$-1/3$	-0	$2\frac{1}{3}$
$(-2, 0, 2)$	0	0	4
$(-2, 1, 2)$	$1/3$	1	$4\frac{1}{3}$
$(-1, 0, 1)$	0	0	1
$(-1, 0, 2)$	$1/3$	0	$2\frac{1}{3}$
$(-1, 1, 2)$	$2/3$	1	$2\frac{1}{3}$
$(0, 1, 2)$	1	1	1

(i)

$\overline{X}$	-1	$-2/3$	$-1/3$	0	$1/3$	$2/3$	1
$p(\overline{X})$	$1/10$	$1/10$	$2/10$	$2/10$	$2/10$	$1/10$	$1/10$

(ii)

M	-1	0	1
$p(M)$	$3/10$	$4/10$	$3/10$

(iii)

S	1	$\sqrt{7/3}$	2	$\sqrt{13/3}$
$p(S)$	$3/10$	$4/10$	$1/10$	$2/10$

(iv)
$$E(\overline{X}) = (-1)\left(\tfrac{1}{10}\right) + \left(-\tfrac{2}{3}\right)\left(\tfrac{1}{10}\right) + \left(-\tfrac{1}{3}\right)\left(\tfrac{1}{10}\right) + 0\left(\tfrac{1}{10}\right) + \left(\tfrac{1}{3}\right)\left(\tfrac{2}{10}\right) + \left(\tfrac{2}{3}\right)\left(\tfrac{1}{10}\right) + (1)\left(\tfrac{1}{10}\right) = 0$$
$$E(\overline{X}^2) = (-1)^2\left(\tfrac{1}{10}\right) + \left(-\tfrac{2}{3}\right)^2\left(\tfrac{1}{10}\right) + \left(-\tfrac{1}{3}\right)^2\left(\tfrac{1}{10}\right) + 0^2\left(\tfrac{2}{10}\right) + \left(\tfrac{1}{3}\right)^2\left(\tfrac{2}{10}\right) + \left(\tfrac{2}{3}\right)^2\left(\tfrac{1}{10}\right) + 1^2\left(\tfrac{1}{10}\right) = \tfrac{1}{3}$$
$$\mathrm{Var}(\overline{X}) = E(\overline{X}^2) - (E\overline{X})^2 = \frac{1}{3} - 0^2 = \frac{1}{3}$$

(b) We can get $5^3 = 125$ samples of size 3 with replacement

4.1.3. Population: $\{1, 2, 3\}$. $p(x) = \tfrac{1}{3}$, for x in $\{1, 2, 3\}$

(a) $\mu = \frac{1}{N}\sum_{i=1}^{N} c_i = 2,\ \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(c_i - \mu)^2 = \tfrac{2}{3}$

(b)

Sample	$\overline{X}$	Sample	$\overline{X}$	Sample	$\overline{X}$
(1, 1, 1)	1	(2, 1, 1)	1⅓	(3, 1, 1)	1⅔
(1, 1, 2)	1⅓	(2, 1, 2)	1⅔	(3, 1, 2)	2
(1, 1, 3)	1⅔	(2, 1, 3)	2	(3, 1, 3)	2⅓
(1, 2, 1)	1⅓	(2, 2, 1)	1⅔	(3, 2, 1)	2
(1, 2, 2)	1⅔	(2, 2, 2)	2	(3, 2, 2)	2⅓
(1, 2, 3)	2	(2, 2, 3)	2⅓	(3, 2, 3)	2⅔
(1, 3, 1)	1⅔	(2, 3, 1)	2	(3, 3, 1)	2⅓
(1, 3, 2)	2	(2, 3, 2)	2⅓	(3, 3, 2)	2⅔
(1, 3, 3)	2⅓	(2, 3, 3)	2⅔	(3, 3, 3)	3

$\overline{X}$	1	1⅓	1⅔	2	2⅓	2⅔	3
$p(\overline{X})$	1/27	1/9	2/9	7/27	2/9	1/9	1/27

(c) $E(\overline{X}) = \sum_{\overline{X}} \bar{x} \cdot p(\bar{x}) = 2,\ E\left(\overline{X}^2\right) = \sum_{\overline{X}} \bar{x}^2 \cdot p(\bar{x}) = 4\tfrac{2}{9}$, then $\mathrm{Var}(\overline{X}) = \tfrac{2}{9}$

4.1.5. Since $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - n\bar{x}^2$, we have $E(S')^2 = \frac{1}{n}\sum_i EX_i^2 - \frac{nE\overline{X}^2}{n}$

Assuming the sampling from a population with mean μ and variance σ^2, we have

$$E(S')^2 = \frac{1}{n}n\left(\sigma^2 + \mu^2\right) - \left(\frac{\sigma^2}{n} + \mu^2\right)$$

$$= \sigma^2 - \frac{\sigma^2}{n}$$
$$= \left(\frac{n-1}{n}\right)\sigma^2 < \sigma^2 = E(S^2)$$

4.1.7. Let X be the weight of sugar $X \sim N(\mu = 5\,\text{lb}, \sigma = 0.2\,\text{lb})$
Then $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the mean weight, where $n = 15$.
By Corollary 4.2.2, $E(\overline{X}) = \mu$, and $\text{Var}(\overline{X}) = \frac{\sigma^2}{n}$. Then $\overline{X} \sim N(5, {}^{0.2^2}\!/_{15})$, and $\frac{\overline{X}-5}{\sqrt{0.2^2/15}} = Z \sim N(0, 1^2)$. Therefore, the probability requested is $P(-0.2 < \overline{X} - 5 < 0.2) = P(-\sqrt{15} < Z < \sqrt{15}) = 0.9999$

4.1.9. Let X be the height. $X \sim N(\mu = 66, \sigma^2 = 2^2)$, and $\frac{\overline{X}-66}{2/\sqrt{26}} = Z \sim N(0, 1^2)$. Then, $P(\overline{X} > 170) = P(Z > 2\sqrt{26}) = 0$

4.1.11. Let X be the time. $X \sim N(\mu = 95, \sigma^2 = 10^2)$. Then $\frac{X-95}{10} = Z \sim N(0, 1^2)$. Therefore, $P(X < 85) = P(Z < 1) = 0.8413$, or 84.13% of measurement times will fall below 85 seconds.

4.1.13. According the information, $\mu = 215$ and $\sigma = 35$.
(a) If $n = 55$, we can assume $X \sim N(\mu, \sigma)$, then $P(\overline{X} > 230) = P\left(Z > \frac{230-215}{35/\sqrt{55}}\right) = 0.0007$
(b) If $n = 200$, we can assume $X \sim N(\mu, \sigma)$, then $P(\overline{X} > 230) = P\left(Z > \frac{230-215}{35/\sqrt{200}}\right) = 0$
(c) If $n = 35$, we can assume $X \sim N(\mu, \sigma)$, then $P(\overline{X} > 230) = P\left(Z > \frac{230-215}{35/\sqrt{35}}\right) = 0.0056$
(d) Increasing the sample size, decrement the probability

4.1.15. Let T be the temperature.
Since $n = 60$, we assume $T \sim N(98.6, 0.95^2)$. Then $\overline{T} \sim N(98.6, {}^{0.95^2}\!/_{60})$. Therefore, $P(\overline{T} \geq 99.1) = 0$

EXERCISES 4.2

4.2.1. We have that $Y \sim \chi^2_{(15)}$
(a) We can see, for example in a table, that $P(Y \leq 6.26) = 0.025$. Then $y_0 = 6.26$
(b) Choosing upper and lower tail area to 0.025, and since $P(Y \leq 27.5) = 0.975$, and $P(Y \leq 6.26) = 0.025$, then $P(a < Y < b) = 0.95$, then $b = \chi^2_{0.975,15} = 27.5$, $a = \chi^2_{0.025,15} = 6.26$
(c) $P(Y \geq 22.307) = 1 - P(Y < 22.3) = 0.10$

4.2.3. If $X \sim \Gamma(k, \theta)$, then $X \sim \chi^2_{(v)} \equiv \Gamma(k = v/2, \theta = 2)$. In our case $T \sim \Gamma(1, 2)$, then $\overline{T} = \frac{1}{n}\sum_{i=1}^{n} T \sim \Gamma(n, 2/n)$
(a) With $n = 3$, $\overline{T} \sim \Gamma(3, 2/3)$
(b) $P(\overline{T} > 2) = 0.4232$

4.2.5. Since $X_1, X_2, \ldots, X_5$ are i.i.d. $N(55, 223)$, then $Y = \sum_{i=1}^{5} \frac{(X_i-55)^2}{223} \sim \chi^2_{(5)}$

(a) Since $Z = Y - \frac{n(\overline{X}-55)^2}{223}$, and $\frac{n(\overline{X}-55)^2}{223} \sim \chi^2_{(1)}$, Z is Chi-square distributed with 4 degrees of freedom and Y is Chi-square distributed with 5 degrees of freedom

(b) Yes

(c) (i) $P(0.62 \leq Y \leq 0.76) = 0.0075$ (ii) $P(0.77 \leq Z \leq 0.95) = 0.0251$

4.2.7. Since the random sample comes from a normal distribution, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$. Setting the upper and lower tail area equal to 0.05, even this is not the only choices, and using a Chi-square table with $n-1 = 14$ degrees of freedom, we have $\frac{(n-1)b}{\sigma^2} = \chi^2_{0.95,14} = 23.68$, and $\frac{(n-1)a}{\sigma^2} = \chi^2_{0.05,14} = 6.57$. Then, with $\sigma = 1.41$, $b = 3.36$, and $a = 0.93$

4.2.9. Since $T \sim t_8$

(a) $P(T \leq 2.896) = 0.99$

(b) $P(T \leq -1.860) = 0.05$

(c) Since t-distribution is symmetric, we find a such that $P(T > a) = \frac{0.01}{2}$. Then $a = 3.35$

4.2.11. According with the information, $\mu = 11.4$, $n = 20$, $\overline{y} = 11.5$, and $s = 2$, then $t = \frac{\overline{y}-\mu}{s/\sqrt{n}} = 0.224$. The degrees of freedom are $n-1 = 19$, so the critic value is 1.328 at $\alpha = 0.05$-level. Then, the data tend to agree with the psychologist claims.

4.2.13. If $X \sim \chi^2_{(v)}$, then $X \sim \Gamma\left(\alpha = \frac{v}{2}, \beta = 2\right)$, then $E(X) = \left(\frac{v}{2}\right)(2) = v$ and $\text{Var}(X) = \left(\frac{v}{2}\right)(2)^2 = 2v$

4.2.15. If $X_1, X_2, \ldots, X_n$ is from $N(\mu, \sigma^2)$

then, by Theorem 4.2.8, $\frac{(n-1)S^2}{\sigma^2}$ is from $\chi^2_{(n-1)}$

then, by Exercise 4.2.13, $\text{Var}\left[\frac{(n-1)S^2}{\sigma^2}\right] = 2(n-1)$

Since $\text{Var}(aX) = a^2\text{Var}(X)$, $\frac{(n-1)^2}{\sigma^4}\text{Var}(S^2) = 2(n-1)$

Simplifying after multiplying by $\frac{\sigma^4}{(n-1)^2}$, we obtain $\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$

4.2.17. If X and Y are independent random variables from an exponential distribution with common parameter $\theta = 1$, then using 4.2.16 with $n = 1$, $2X \sim \chi^2_{(2)}$ and $2Y \sim \chi^2_{(2)}$ then $\frac{X}{Y} = \frac{2X}{2Y} \sim F(2, 2)$

4.2.19. If $X \sim F(9, 12)$

(a) $P(X \leq 3.87) = 0.9838$

(b) $P('X \leq 0.196) = 0.01006$

(c) $F_{0.975}(9, 12) = 0.025$ then $F_{0.975} = 3.4358$.
$0.025 = P(X < F_{0.975}) = P\left(\frac{1}{X} > \frac{1}{F_{0.975}}\right)$, where $\frac{1}{X} \sim F(12, 9)$
Then $\frac{1}{F_{0.975}} = 3.8682$ and $F_{0.975} = 0.258518$. Thus, $a = 0.2585$, $b = 3.4358$

4.2.21. If $X \sim F(n_1, n_2)$ the PDF is given by

$$f(x) = \begin{cases} \dfrac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\Gamma(n_2/2)}\left(\dfrac{n_1}{n_2}\right)\left(\dfrac{n_1}{n_2}x\right)^{\frac{n_1}{2}-1}\left(1+\dfrac{n_1}{n_2}x\right)^{-(n_1+n_2)/2}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

Then

$$EX = \int_0^\infty x\frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\Gamma(n_2/2)}\left(\frac{n_1}{n_2}\right)\left(\frac{n_1}{n_2}x\right)^{\frac{n_1}{2}-1}\left(1+\frac{n_1}{n_2}x\right)^{-(n_1+n_2)/2} dx$$
$$= \frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\Gamma(n_2/2)}\left(\frac{n_1}{n_2}\right)\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}-1}\int_0^\infty x^{\frac{n_1}{2}}\left(1+\frac{n_1}{n_2}x\right)^{-(n_1+n_2)/2} dx$$

Let $y = 1-\left(1+\frac{n_1}{n_2}x\right)^{-1}$ then $x = \left(\frac{n_1}{n_2}\right)^{-1} y(1-y)^{-1}$ and $dx = \left(\frac{n_1}{n_2}\right)^{-1}(1-y)^{-2}dy$ and $\lim\limits_{x\to\infty}\left\{1-\left(1+\frac{n_1}{n_2}x\right)^{-1}\right\} = 1$, then

$EX = \frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\Gamma(n_2/2)}\left(\frac{n_1}{n_2}\right)^{\frac{n_2}{2}}\left(\frac{n_2}{n_1}\right)^{\frac{n_2}{2}+1}\int_0^1 y^{\frac{n_2}{2}}(1-y)^{n_1-2}\,dy$, which converges for $n_1 > 2$.

For $\alpha > 0, \beta > 0 \int_0^1 y^{\alpha-1}(1-y)^{\beta-1}dy = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$, where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}dx$ with the property $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$.

$$\text{Then } EX = \frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\Gamma(n_2/2)}\left(\frac{n_1}{n_2}\right)^{\frac{n_2}{2}}\left(\frac{n_2}{n_1}\right)^{\frac{n_2}{2}+1}\frac{\Gamma(n_1/2+1)\Gamma(n_2/2-1)}{\Gamma[(n_1+n_2)/2]}$$
$$= \frac{n_2}{n_1}\frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)(n_2/2-1)\Gamma(n_2/2-1)}\left(\frac{n_1}{2}\right)\frac{\Gamma(n_1/2)\Gamma(n_2/2-1)}{\Gamma[(n_1+n_2)/2]}$$
$$= \frac{n_2}{n_2-2}, n_2 > 2$$

Similarly,

$$EX^2 = \frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)\Gamma(n_2/2)}\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}\left(\frac{n_2}{n_1}\right)^{\frac{n_2}{2}+2}\int_0^1 y^{\frac{n_1}{2}}(1-y)^{\frac{n_2}{2}-3}dy, \text{ which converges for } n_2 > 4$$
$$= \frac{\Gamma[(n_1+n_2)/2]}{\Gamma(n_1/2)(n_2/2-1)(n_2/2-2)\Gamma(n_2/2-2)}\left(\frac{n_2}{n_1}\right)^2\frac{(n_1/2+1)(n_1/2)\Gamma(n_1/2)\Gamma(n_2/2-2)}{\Gamma[(n_1+n_2)/2]}$$
$$= \left(\frac{n_2}{n_1}\right)^2\frac{n_1(n_1+2)}{(n_2-2)(n_2-4)}, \ n_2 > 4.$$

Now, $\text{Var}(X) = EX^2 - (EX)^2$. Therefore,

$$EX = \frac{n_2}{n_2-2}, n_2 > 2 \quad \text{and} \quad \text{Var}X = \frac{n_2^2(2n_1+2n_2-4)}{n_1(n_2-2)^2(n_2-4)}$$

4.2.23. If $X_1, X_2, \ldots, X_{n_1}$ is a random sample from a normal population with mean μ_1 and variance σ^2 and if $Y_1, Y_2, \ldots, Y_{n_2}$ is a random sample from an independent normal population with mean μ_2 and variance σ^2, then $\overline{X} \sim N\left(\mu_1, {}^{\sigma^2}\!/_{n_1}\right), \overline{Y} \sim N\left(\mu_2, {}^{\sigma^2}\!/_{n_2}\right)$, $\dfrac{(n_1-1)S_1^2}{\sigma^2} \sim \chi^2_{(n_1-1)}$, and $\dfrac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{(n_2-1)}$.

Then $\overline{X} - \overline{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}\right)$ and $\frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{(n_1+n_2-2)}$

then $\frac{\overline{X}-\overline{Y}-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma^2}{n_1}+\frac{\sigma^2}{n_2}}} \sim N(0, 1^2)$ and $\frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{(n_1+n_2-2)}$

Then, since the samples are independent, we have by definition that

$$\frac{\dfrac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\dfrac{\sigma^2}{n_1} + \dfrac{\sigma^2}{n_2}}}}{\sqrt{\left[\dfrac{(n_1-1)S_1^2}{\sigma^2} + \dfrac{(n_2-1)S_2^2}{\sigma^2}\right] \Big/ [n_1 + n_2 - 2]}} \sim T_{(n_1+n_2-2)}$$

This after simplification becomes:

$$\frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\dfrac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}} \sim T_{(n_1+n_2-2)} \qquad \text{Q.E.D.}$$

4.2.25. If $X \sim \chi^2_{(v)}$ with $v > 0$, then the pdf of X is given by

$$f(x) = \begin{cases} \dfrac{1}{\Gamma(v/2)2^{v/2}} e^{-x/2} x^{v/2-1}, & 0 < x < \infty \\ 0, & x \le 0 \end{cases}$$

Then, by definition of MGF,

$$M_X(t) = \frac{1}{\Gamma(v/2)2^{v/2}} \int_0^\infty e^{x(t-1/2)} x^{v/2-1} dx = \left(\frac{1}{1-2t}\right)^{v/2} \int_0^\infty \frac{e^{-w} w^{v/2-1}}{\Gamma(v/2)} dw$$

$$= (1-2t)^{-v/2}, \quad t < \frac{1}{2}$$

Since $\Gamma(v/2) = \int_0^\infty e^{-w} w^{v/2-1} dw$ and $M_X'(t) = v(1-2t)^{-v/2-1}$, then $EX = M_X'(t)\,|_{t=0} = v$
$M_X''(t) = v(v+2)(1-2t)^{-v/2-2}$, then $EX^2 = M_X''(t)\,|_{t=0} = v^2 + 2v$

Therefore, $\operatorname{Var} X = EX^2 - (EX)^2 = 2v$

4.2.27. Let X be a random variable with PDF

$$f(x) = \begin{cases} \dfrac{2}{\pi(1+x^2)}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

X does not follows T_1 distribution (i.e. X is not $t(1)$ distributed), even $X^2 \sim F(1,1)$. Therefore, $X^2 \sim F(1,n)$ does not (necessarily) imply $X \sim t(n)$

EXERCISES 4.3

4.3.1. $f(x) = \frac{1}{10}e^{-x/10}$, $x > 0$, then the cumulative distribution of τ_1 is

$$F_{\tau_1}(t) = \int_0^t \frac{1}{10}e^{-x/10}dx = -e^{-x/10}\Big|_0^t = 1 - e^{-t/10}.$$

Let Y represent the life length of the system, then $Y = \min(\tau_1, \tau_2)$ and $F_Y(y) = 1 - [1 - F_{\tau_i}(y)]^2$, then the pdf of Y is of the form $f_Y(y) = 2f_{\tau_i}(y)[1 - F_{\tau_i}(y)]$ and is given by

$$f_y(y) = \begin{cases} \dfrac{1}{5}e^{-y/5}, & 0 < y < \infty \\ 0, & \text{otherwise} \end{cases}$$

4.3.3. X_1, X_2 take values $0, 1$; X_3 take values $1, 2, 3$, and $Y_1 = \min\{X_1, X_2, X_3\}$
Since the values of X_1, X_2 are less or equal to the values for X_3, Y_1 take values $0, 1$
Since the values of X_3 are greater than the values for X_1, X_2, then $Y_3 = \max\{X_1, X_2, X_3\}$ take values $1, 2, 3$
Since $Y_1 \le Y_2 \le Y_3$, Y_2 take values $0, 1$

4.3.5. Let $X_1, X_2, \ldots, X_n$ be a random sample from exponential distribution with mean θ, then the common pdf is given by $f(x) = \frac{1}{\theta}e^{-\frac{x}{\theta}}$, if $x > 0$
Using Theorem 4.3.2, the pdf of the k-th order statistic is given by

$$f_k(y) = f_{Y_k}(y) = \frac{n!}{(k-1)!(n-k)!}f(y)(F(y))^{k-1}(1-F(y))^{n-k}, \text{ where } F(y) = 1 - e^{-\frac{y}{\theta}}$$

then $f_k(y) = \frac{n!}{(k-1)!(n-k)!}f(y)\left[1 - e^{-\frac{y}{\theta}}\right]^{k-1}\left[e^{-\frac{y}{\theta}}\right]^{n-k}$

Then, the pdf of Y_1 is $f_1(y) = nf(y)\left[e^{-\frac{y}{\theta}}\right]^{n-1} = \frac{n}{\theta}e^{-\frac{ny}{\theta}}$, which is the pdf of an exponential distribution with mean $\frac{n}{\theta}$, and the pdf of Y_n is

$$\begin{aligned} f_n(y) &= nf(y)[F(y)]^{n-1} \\ &= \frac{n}{\theta}e^{-\frac{y}{\theta}}\left[1 - e^{-\frac{ny}{\theta}}\right]^{n-1} \end{aligned}$$

4.3.7. $X_1, \ldots, X_n$ a random sample are i.i.d with pdf $f(x) = \frac{1}{2}, 0 \le x \le 2$
then $F(x) = \int_0^x \frac{1}{2} dx = \frac{x}{2}$, if $0 \le x \le 2$

$$\text{then } F(x) = \begin{cases} 1, & x > 2 \\ \dfrac{x}{2}, & 0 \le x \le 2 \\ 0, & x < 0 \end{cases}$$

Then, using Theorem 4.3.3 the joint pdf of Y_1 and Y_n is given by

$$\begin{aligned} f_{Y_1,Y_n}(x,y) &= \frac{n!}{(1-1)!(n-1-1)!(n-n)!}\left[\frac{x}{2}\right]^{1-1}\left[\frac{1}{2}y - \frac{1}{2}x\right]^{n-1-1} \cdot \left[1 - \frac{y}{2}\right]^{n-n} \cdot \frac{1}{2} \cdot \frac{1}{2} \\ &= \frac{n(n-1)}{2^n}(y-x)^{n-2}, \text{ if } 0 \le x < y \le 2 \text{ and } f_{Y_1,Y_n}(x,y) = 0, \text{ otherwise.} \end{aligned}$$

Now, let $R = Y_n - Y_1$ and $Z = Y_n$, and consider the functions $r = y_n - y_1, z = y_n$, then their inverses are $y_1 = z - r, y_n = z$.
Then the corresponding Jacobian of the one-to-one transformation is

$$J = \begin{vmatrix} \dfrac{\partial y_1}{\partial r} & \dfrac{\partial y_1}{\partial z} \\ \dfrac{\partial y_n}{\partial r} & \dfrac{\partial y_n}{\partial z} \end{vmatrix} = \begin{vmatrix} -1 & 1 \\ 0 & 1 \end{vmatrix} = -1$$

Then the joint pdf of R and Z is

$$\begin{aligned} g(r,z) &= |-1| f_{Y_1,Y_n}(z-r, z) \\ &= \frac{n(n-1)}{2^n} r^{n-2}, \text{ if } 0 \le r \le z \le 2, \text{ and } g(w,z) = 0, \text{ otherwise.} \end{aligned}$$

Then, the pdf of the range $R = Y_n - Y_1$ is

$$f_R(r) = \int_r^2 g(r,z)dz = \frac{n(n-1)(2-r)r}{2^n}, \text{ if } 0 \le r \le 2 \text{ and } f_R(r) = 0, \text{ otherwise.}$$

4.3.9. $X_1, \ldots, X_n$ a random sample from $N(10, 4)$

$$P(Y_n > 10) = 1 - P(Y_n \le 10)$$

The CDF $F_n(y)$ of Y_n is $[F(y)]^n$, where $F(y)$ is the cdf of X evaluated in y

Then, $P(Y_n \le y) = F_n(y) = [F(y)]^n = [P(X \le y)]^n$

and $P(X \le y) = P\left(Z \le \frac{y-10}{2}\right)$

Then $P(Y_n \le y) = \left[P\left(Z \le \frac{y-10}{2}\right)\right]^n$

Then $P(Y_n > y) = 1 - P(Y_n < y)$
$= 1 - \left[P\left(Z \le \frac{y-10}{2}\right)^n\right]$

Therefore $P(Y_n > 10) = 1 - [P(Z \leq 0)]^n$
$= 1 - (0.5)^n$

4.3.11. $X_1, \ldots, X_n$ is a random sample from Beta$(x = 2, \beta = 3)$
The joint pdf of Y_1 and Y_n, according Theorem 4.3.3, is given by

$$f_{Y_1,Y_n}(x, y) = \frac{n!}{(1-1)!(n-1-1)!(n-n)!}[F(x)]^{1-1}[F(y) - F(x)]^{n-1-1}[1 - F(y)]^{n-n} f(x) f(y)$$

$$= n(n-1)[F(y) - F(x)]^{n-2} f(x) f(y), \quad \text{if} \quad x < y$$

Since $X_i \sim$ Beta$(X = 2_1 \beta = 3)$ for $i = 1, 2, \ldots, n$, the pdf is

$$f(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \chi^{\alpha-1}(1 - x)^{\beta-1}, \quad x \in [0, 1]$$

and, the DF is

$$F(x) = \begin{cases} 0, & x \leq 0 \\ \dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x t^{\alpha-1}(1-t)^{\beta-1}dt, & 0 \leq x \leq 1 \\ 1, & x \geq 1 \end{cases}$$

In our case,

$$f(x) = \frac{\Gamma(5)}{\Gamma(2)\Gamma(3)} x^{2-1}(1-x)^{3-1} = \frac{4!}{1!2!} x(1-x)^2 = 12x(1-x)^2, \quad \text{if} \quad 0 \leq x \leq 1$$

and

$$F(x) = \int_0^x 12t(1-t)^2 dt = 12\left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right], \quad \text{if} \quad 0 \leq x \leq 1$$

Then, the joint pdf $f_{Y_1,Y_n}(x, y)$, using the 4.3.3, is given by

$$f_{Y_1,Y_n}(x, y) = n(n-1)\left[12\left(\frac{y^2}{2} - \frac{2y^3}{3} + \frac{y^4}{4}\right) - 10\left(\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right)\right]^{n-2} 12x(1-x)^2 12y(1-y)^2$$

$$= 12^n n(n-1)\left[\frac{1}{2}\left(y^2 - x^2\right) - \frac{2}{3}\left(y^3 - x^3\right) + \frac{1}{4}\left(y^4 - x^4\right)\right]^{n-2} xy(1-x)^2(1-y)^2,$$

if $0 \leq x \leq y \leq 1$, and $f_{Y_1,Y_n}(x, y) = 0$, otherwise

EXERCISES 4.4

4.4.1. $X_1, X_2, \ldots, X_n$, where $n = 150$, $\mu = 8$, $\sigma^2 = 4$, then $\mu_{\overline{x}} = 8$ and $\sigma_{\overline{x}}^2 = 4$
By Theorem 4.4.1: $\lim_{n\to\infty} P\left(Z \leq \dfrac{\overline{X} - \mu_{\overline{x}}}{\sigma_{\overline{x}}}\right) = \frac{1}{\sqrt{2k}} \int_{\infty}^{z} e^{-\mu^2/2} du$
then $P(7.5 < \overline{X} < 10) = P\left(\frac{75-8}{2} < Z < \frac{10-8}{2}\right) = 0.44$

4.4.3. Let T be the time spent by a customer coming to certain gas station to fill up gas
Suppose $T_1, T_2, \ldots, T_n$ are independent random variables, with $\mu_t = 3$ minutes, $\sigma_t^2 = 1.5$ minutes, and $n = 75$
Then $Y = \sum_{i=1}^{n} T_i$ is the total time spent by the n customers
Since, $Y = 3$ hours $= 180$ minutes, $P(Y < 180) = P\left(\overline{Y} < \frac{180}{n}\right) = P(\overline{Y} < 2.4)$
and by Theorem 4.4.1, $\overline{Y} \sim N\left(3, \dfrac{1.5}{75}\right)$
For practical purposes $n = 60$ is large enough
Therefore, $P(Y < 180) = P\left(Z < \dfrac{2.4 - 3}{\sqrt{0.02}}\right) = 0$, where $Z \sim N(0, 1^2)$
There is 0% chance that the total time spent by the customers is less than 3 hours.

4.4.5. 1250 students took it, and $\mu = 69\%$, $\sigma = 5.4\%$, and $n = 60$ students (is large enough).
Then, $\mu_{\overline{x}} = 69$ and $\sigma_{\overline{x}} = \frac{5.4}{\sqrt{60}} = 0.6997137$
Then $P(\overline{X} \leq 75.08) = P\left(Z \leq \frac{75.08-69}{0.697137}\right) = 1$
There is almost 100% chance that the average score is less than 75.08%.

4.4.7. $X_i \sim N(\mu_1, \sigma_1^2)$ for $i \in \{1, 2, \ldots, n\}$, then $\overline{X} \sim N(\mu_1, \sigma_1^2/n)$
$Y_j \sim N(\mu_2, \sigma_2^2)$ for $j \in \{1, 2, \ldots, m\}$, then $\overline{Y} \sim N(\mu_2, \sigma_2^2/m)$
Therefore,

$$\overline{X} - \overline{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}\right)$$

4.4.9. $X \sim$ Binomial $(n = 20, p = 0.2)$

$$P(X \leq 10) = \sum_{x=0}^{10} \binom{20}{x} 0.2^x (1 - 0.2)^{20-x} = 0.9944$$

Using normal approximation:

$$P(X \leq 10) = P\left(Z \leq \frac{10 - np + 0.5}{\sqrt{np(1-p)}}\right) = 0.99986$$

4.4.11. $q = 6\%$ of person making reservations will not show up each day
Rental company reserves for $n = 215$ persons
200 automobiles available.
$p = 0.94$ is the probability of the person making reservation will show up each day
Let X be the number of the persons making reservation will show up.
Then $X \sim$ Binomial$(n = 215, p = 0.94)$
The probability requested is $P(X \leq 200) = P\left(Z \leq \frac{200 - np + 0.5}{\sqrt{np(1-p)}}\right) = 0.3228$

4.4.13. SIDS occurs between the ages 28 days and one year

Rate of death due to SIDS is 0.0013 per year.

Randon sample of 5000 infants between the ages 28 day and are your

Let X be the number of SIDS related deaths

$p = 0.00103,\ n = 5000$

Then $X \sim \ \mathrm{Bin}(n = 5000, p = 0.00103)$

The probability requested is

$$P(X > 10) \approx P\left(Z > \frac{10 - np - 0.5}{\sqrt{np(1-p)}}\right) = 0.0274$$

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Chapter 5

Point Estimation

EXERCISES 5.2

5.2.1. The pdf of a geometric distribution is:

$f(x) = p(1-p)^{x-1}$ for $x = 1, 2$...........also, $\mu = \frac{1}{p}$.

By the methods of moments $E(X) = \overline{x} = \frac{1}{p}$

$\therefore \hat{p} = \overline{x}$

(b) For the given data

$\overline{x} = \frac{\sum x_i}{n} = \frac{2+4+\cdots+22+12}{18} = 16.11$

$\therefore \hat{p} = 16.11$

5.2.3. $\sum X_i \sim U(\theta-1, \theta+1)$ By definition:

$$f(x) = \begin{cases} \frac{1}{\theta+1-\theta+1}, & \text{if } \theta-1 < x < \theta+1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) By definition

$E(x) = \overline{x} = \frac{\theta-1+\theta+1}{2}$

$\therefore \overline{x} = \theta$

This implies the moment of estimator for $\theta = \overline{x}$ (b) $\hat{\theta} = \overline{x}$

$\frac{11.72+12.81+12.09+13.47+12.37}{4}$

$\hat{\theta} = 53.18$

5.2.5. The pdf of the exponential function is given as:

$$f(x) = \begin{cases} e^{-(x-\theta)}, & \text{if } x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

5.2.9. Similar as above

5.2.11. We have

$E(X) = \mu = \overline{x}$

Variance$(\sigma^2) = \frac{1}{n}\sum\left(X_i - \overline{X}\right)^2$

$\sigma^2 = \frac{1}{n}\left[\sum x_i^2 - \overline{x}^2\right]$

$\therefore \hat{\sigma} = \sqrt{\frac{1}{n}\left[\sum X_i{}^2 - \left(\frac{\sum X_i}{n}\right)^2\right]}$

The method of moments estimator for σ is given by $T(X_1, \ldots\ldots. X_n) = \sqrt{\frac{1}{n}\left[\sum X_i{}^2 - \left(\frac{\sum X_i}{n}\right)^2\right]}$

5.2.13. The method of moments estimator for $\mu = T(X_1, \ldots\ldots. X_n) = E(X) = \overline{X}$

$\therefore T(X_1, \ldots\ldots. X_n) = \overline{X}$

Since μ and σ^2 both are unknown. $\sigma^2 = \frac{1}{n}\sum\left(X_i - \overline{X}\right)^2$

This implies $\sigma^2 = \frac{(n-1).1}{n.(n-1)}\sum\left(X_i - \overline{X}\right)^2$,

$\therefore \sigma^2 = \frac{n-1}{n}s^2$

Let $s'^2 = \frac{n-1}{n}s^2$

$\therefore$ method of moments estimator for σ^2 is given by:

$s'^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$

EXERCISES 5.3

5.3.1. $f(x) = \binom{n}{x}p^x(1-p)^{n-x}$

$$L(p, x_1, x_2, \ldots. x_n) = \log\left\{\prod_{i=1}^{n}\binom{n}{x_i}\right\} + \sum X_i \log p + \left(n - \sum X_i\right)\log(1-p)$$

$$\frac{\partial}{\partial p}\log L(p, X_1, X_2, \ldots. X_n) = \frac{\sum X_i}{p} + \frac{n - \sum X_i}{1-p}(-1)$$

$$\frac{\partial}{\partial p}\log L(p, X_1, X_2, \ldots. X_n) = \frac{\sum X_i}{p} - \frac{n - \sum X_i}{1-p}$$

For maximum likelihood estimator of p

$$\frac{\partial}{\partial p}\log L(p, X_1, X_2, \ldots. X_n) = 0$$

$$\frac{\sum X_i}{p} - \frac{n - \sum X_i}{1-p} = 0$$

$(1-p)\sum X_i - p\left(n - \sum X_i\right) = 0$. This implies $\sum X_i = p\left\{\sum X_i + n - \sum X_i\right\}$

$p = \frac{\sum X_i}{n}$,

$\therefore p = \overline{X}$.

Hence, MLE of $p = \hat{p} = \overline{X}$.

By invariance property $\hat{q} = 1 - \hat{p}$ is MLE of q.

5.3.3. $f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$ implies $L(\theta) = \frac{1}{\theta^n} e^{\frac{-\sum X_i}{\theta}}$

$\ln L(\theta) = -n \ln \theta - \frac{\sum X_i}{\theta}$

Now taking the derivatives with respect to α and β and setting both equal to zero, we have

$\frac{\partial}{\partial \theta} \ln L = \frac{-n}{\theta} + \frac{\sum X_i}{\theta^2} = 0$

$-n\theta + \sum X_i = 0.$

$\therefore \hat{\theta} = \overline{X}$

From the given data:

MLE of θ is given by $\theta = \overline{X} = \frac{1+2+\cdots\cdots+7+2}{14}. \therefore \hat{\theta} = 6.07.$

5.3.5. Here pdf of X is given by

$$f(x) = \begin{cases} \frac{2x}{\alpha^2} e^{\frac{-x^2}{\alpha^3}} & \text{if, } x > 0 \\ 0, & \text{otherwise} \end{cases} \tag{5.1}$$

$L(\alpha, X_1, \ldots\ldots.. X_n) = \frac{2}{(\alpha^2)^2} \prod_{i=1}^n X_i e^{-\frac{x^2}{\alpha^2}}$

$\ln L = \ln 2 - 2 \ln \alpha + \sum_{i=1}^n \ln X_i - \sum_{i=1}^n \frac{{X_i}^2}{\alpha^2}$

$\frac{\partial}{\partial \alpha} \ln L = -\frac{2n}{\alpha} + \frac{2}{\alpha^3} \sum_{i=1}^n X_i^2$

$\frac{\partial}{\partial \alpha} = 0$ implies,

$-\frac{2n}{\alpha} + \frac{2}{\alpha^3} \sum_{i=1}^n X_i^2 = 0$

$-n\alpha^2 + \sum X^2 = 0$

$\alpha^2 = \frac{\sum X_i^2}{n}$

$\therefore \hat{\alpha} = \sqrt{\frac{\sum X_i^2}{n}}.$

5.3.7.

$$f(x) = \begin{cases} \frac{\alpha}{\beta^\alpha} x^{\alpha-1} e^{\left(-\frac{x}{\beta}\right)^\alpha} & \text{if, } x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

$L(\alpha, \beta, X) = \frac{\alpha^n}{\beta^{n\alpha}} \prod_{i=1} n X_i^{\alpha-1} e^{\left(\frac{\sum X_i}{\beta}\right)^\alpha}$

$L(\alpha, \beta, x) = n \ln \alpha - \alpha \ln \beta + (\alpha - 1) \sum \ln X_i - \left(\frac{\sum X_i}{\beta}\right)^2$

$\frac{\partial}{\partial \alpha} L(\alpha, \beta, x) = \frac{n}{\alpha} + \sum_{i=1}^n \ln X_i - n \ln \beta - \left(\frac{\sum X_i}{\beta}\right)^\alpha \ln \frac{\sum X_i}{\beta}$

$\frac{\partial}{\partial \beta} L(\alpha, \beta, x) = -\frac{\alpha n}{\beta} - \left(\sum X_i\right)^\alpha (a) \beta^{-\alpha-1}$

$\therefore \frac{\partial}{\partial \beta} L(\alpha, \beta, x) = -\frac{\alpha n}{\beta} + \alpha \beta^{(-\alpha-1)} \left(\sum X_i\right)^\alpha$

For maximum likelihood estimator of α: $\frac{\partial}{\partial\alpha}\ln L = 0$. This implies;

$\frac{n}{\alpha} + \sum_{i=1}^{n} \ln X_i - n\ln\beta - \left(\frac{\sum X_i}{\beta}\right)^{\alpha} \ln\frac{\sum X_i}{\beta} = 0$

similarly, $-\frac{\alpha n}{\beta} + \alpha\beta^{(-\alpha-1)}\left(\sum X_i\right)^{\alpha} = 0$

$-\frac{\alpha n}{\beta} + \alpha\beta^{(-\alpha-1)}\left(\sum X_i\right)^{\alpha}$

$\frac{\alpha n}{\beta} - \frac{\alpha}{\beta}\left(\frac{\sum X_i}{\beta}\right)^{\alpha} = 0$ solving for β we get $\beta = \left\{\frac{-\sum X_i}{\alpha n}\right\}^{\frac{\alpha}{\alpha+1}}$

Hence,

$$\frac{n}{2} + \sum X_i - \alpha\left(\sum X_i\right)^{\alpha-1}\left(\frac{-\sum X_i}{\alpha n}\right)^{\frac{\alpha}{\alpha+1}}$$

$$\left[\ln\left(\sum X_i\right) - \ln\left(\frac{-\sum X_i}{\alpha n}\right)^{\frac{\alpha}{\alpha+1}}\right]$$

There is no closed form solution for α and β. In this case, one can use numerical methods such as Newton-Raphson method to solve for α, and then with this value to solve for β.

5.3.9. $f(x) = \frac{\gamma(2\theta)}{\gamma(\theta)^2}[x(1-x)]^{(\theta-1)}, \theta \geq 0$

$\ln L(\theta, x) = n\left[\ln\gamma(2\theta) - \ln\gamma(\theta)^2\right] + (\theta-1)\sum_{i=1}^{n}\ln(X_i - 1)$

$\frac{\partial}{\partial\theta}\ln L(\theta, x) = n\left[\frac{2\gamma'(2\theta)}{\gamma(2\theta)} - \frac{2\gamma'(\theta)^2}{\gamma(0)^2}\right] + \sum_{i=1}^{n}\ln X_i(X_i - 1).$

5.3.13.

$$f(x) = \begin{cases} \frac{1}{3\theta+2} & \text{if, } 0 \leq x \leq 3\theta+2 \\ 0, & \text{otherwise.} \end{cases}$$

This implies $L(\theta, x) = \frac{1}{(3\theta+2)^2}$ for $0 \leq x \leq 3\theta + 2$

When $3\theta+2 \geq \max(X_i)$, the likelihood is $\frac{1}{(3\theta+2)^2}$. Which is positive and decreasing function of θ (for fixed n). However, for $\theta < \frac{\max(X_i)-2}{3}$, the likelihood drops to 0, creating discontinuity at point $\frac{\max(X_i)-2}{3}$.

Hence we will not be able to find the derivative. The MLE is the largest order statistic $\hat{\theta} = \frac{\max(X_i)-2}{3} = X_n$.

5.3.15. Here $X \sim N(\mu, \sigma^2)$

$$\ln L(\mu, \sigma, x) = -\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\sigma^2 - \sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}$$

$$\frac{\partial L}{\partial\mu} = \sum_{i=1}^{n}(X_i - \mu)$$

Similarly $\frac{\partial L}{\partial\sigma^2} = \frac{-n}{2\sigma^2} + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2$

For maximum likelihood estimates of μ and σ; $\sum(X_i - \mu) = 0$ implies $\hat{\mu} = \overline{X}$

Similarly, for σ^2,

$$\frac{-n}{2\sigma^2} + \frac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 = 0$$

$$\therefore \hat{\sigma} = \sqrt{\frac{\sum(X_i - \overline{X})}{n}}.$$

5.3.17. $f(x) = \frac{1}{\theta}e^{-\frac{x}{\theta}}$
It is given that the reliability $R(x) = 1 - F(x)$. This implies $F(x) = f'(x)$. Hence $F(x) = -\frac{1}{\theta^2}e^{-\frac{x}{\theta}}$.
Thus, $R(x) = 1 - F(x)$ and

$$L(x,\theta) = \prod_{i=1}^{n}\left(1 + \frac{1}{\theta^2}e^{-\frac{\sum X_i}{\theta}}\right).$$

EXERCISES 5.4

5.4.1. $E(\overline{X}) = E\left(\frac{\sum X_i}{n}\right)$ i.e $E(\overline{X}) = \frac{1}{n}\sum E(X_i)$. Where, $E(X) = \int_{\theta}^{\infty} xe^{-(x-\theta)}dx$. By integration by parts $E(X) = (1+\theta)$. Thus $E\left(\overline{X} = \frac{1}{n}\right)\sum(1+\theta)$. This implies $E(X) = 1+\theta$.

5.4.3. Sample standard deviation $s = \sqrt{\frac{1}{n-1}\sum(x_i - \overline{X})^2}$

$$E(s) = E\left\{\sqrt{\frac{1}{n-1}\sum(X_i - \mu + \mu - \overline{X})^2}\right\}$$

$$E(s) = E\left\{\sqrt{\frac{1}{n}\left[\sum(x_i - \mu)^2 - (\overline{X} - \mu)^2\right]}\right\}$$

$$E(s) = \left\{\sqrt{\frac{1}{n}\left[\sum E(x_i - \mu)^2 - E(\overline{X} - \mu)^2\right]}\right\}$$

$$E(s) = \left\{\sqrt{\frac{1}{n}\left[\sigma^2 - \frac{\sigma^2}{n}\right]}\right\}$$

5.4.5. Let $Y = C_1X_1 + C_2X_2 + \cdots\cdots\cdots + C_nX_n$.
For an unbiased estimate, we need to have $E(C_1E(X_1) + \cdots\cdots\cdots\cdots + C_nE(X_n)) = \mu$
That is, $C_1\mu + \cdots\cdots C_n\mu = \mu$
Which is possible if and only if $C_i's = \frac{1}{n}$ for all $i = 1, 2\ldots\ldots\ldots n$. This implies $\frac{1}{n}\mu + \frac{1}{n}\mu + \cdots\cdots\cdots + \frac{1}{n}\mu = \mu$
$\therefore \frac{n\mu}{n} = \mu$ i.e $\mu = \mu$. Verified.

5.4.7. $X_i \sim U(0.\theta)$.
$Y_n = \max X_1, \ldots .. X_n$
$\hat{\theta} = Y_n$ is the MLE of θ.
(a) By method of moment estimator $E(X) = \overline{X} = \frac{\theta+0}{2}$. This implies $\theta = 2\overline{X}$.
Hence the method of moment estimator $\hat{\theta} = 2\overline{X}$
(b) $E(\hat{\theta}) = E(Y_n)$
$E(\hat{\theta}) = E\{\max(X_1, \ldots X_n)\}$
$\therefore E(\hat{\theta}) = \frac{n\theta}{n+1}$
$E(\hat{\theta}) = E(2\overline{X})$.
$E(\hat{\theta}) = 2.E(\theta/2)$. That is, $E(\hat{\theta}) = \theta$. Hence, θ_2 ia an unbiased estimate of θ.
(c) $E(\hat{\theta_3}) = \frac{n+1}{n} E(\hat{\theta})$ This implies $E(\hat{\theta_3}) = \theta$.
$\therefore \hat{\theta}$ is an unbiased estimate of θ.

5.4.9. Here, $X_i \sim N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

We have $E(\hat{\mu}) = E(\overline{X}) = \mu$
μ is an unbiased estimate for μ. By definition, the unbiased estimate $\hat{\mu}$ that minimizes the mean square error is called the minimum variance unbiased estimate (MVUE) of μ.

$$MSE(\hat{\mu}) = E(\hat{\mu} - \mu)^2$$

That is $MSE(\hat{\mu}) = \text{var}(\hat{\mu})$. Minimizing the MSE implies that the minimization of $\text{Var}(\hat{\mu})$.
$\hat{\mu}$ is the MVUE for μ.

5.4.11. $E(M) = E(\overline{X}) = \mu$. Thus, sample median is an unbiased estimate of population mean μ.
Now, $\text{Var}(\overline{X}) = \frac{1}{n}\sum(\overline{X} - \mu)^2$
$\text{Var}(M) = \frac{1}{n}\sum(M - \mu)^2$
Where $\text{Var}(\overline{X}) \leq \text{Var} M$

5.4.13.

$$f_\sigma(X) = \frac{1}{2\sigma} \exp\left(-\frac{|X|}{\sigma}\right) \quad \text{for} \quad \infty < x < \infty$$

The likelihood function is given by:

$$f(X_1, X_2, \ldots \ldots .. X_n, \sigma) = \frac{1}{2^n} \frac{1}{\sigma^n} \exp\left\{-\frac{\sum |X_i|}{\sigma}\right\}.$$

Take $g\left(\sum |X_i|\right), \sigma = \frac{1}{\sigma^n} \exp\left\{-\frac{\sum |X_i|}{\sigma}\right\}$ and $h(X_1, X_2 \ldots \ldots X_n) = \frac{1}{2^n}$
$\therefore \sum |X_i|$ is sufficient for σ.

5.4.15. (a) The likelihood function is given by:

$$f(x_1, x_2 \ldots\ldots x_n, \theta) = \frac{1}{\theta^n} \exp\left\{-\frac{\sum X_i}{\theta}\right\}$$

Take $g\left(\sum X_i, \theta\right) = \frac{1}{\theta^n} \exp\left\{-\frac{\sum X_i}{\theta}\right\}$ and $h(x_1, \ldots\ldots x_n) = 1$ for if $x < 0$ and $h(x_1, \ldots\ldots x_n) = 0$ if $x \geq 0$.
$\therefore \sum X_i$ is sufficient for θ.
Similarly, $f(x_1, x_2 \ldots\ldots x_n, \theta) = \frac{1}{\theta^n} \exp\left\{-\frac{n\overline{X}}{\theta}\right\}$
Take $g\left(\overline{X}, \theta\right) = \frac{1}{\theta^n} \exp\left\{-\frac{n\overline{X}}{\theta}\right\}$ and $h(x_1, \ldots\ldots x_n) = 1$ for $x < 0$ and $h(x_1, \ldots\ldots x_n) = 0$ if $x \geq 0$.
$\overline{X}$ is sufficient for θ.
Sample mean is an unbiased estimate for population mean $\left(E\overline{X} = \theta\right)$.

5.4.17. The likelihood function is given by:

$$f(x_1, x_2 \ldots x_n) = \begin{cases} \frac{1}{\theta^n} & \text{if, } -\frac{\theta}{2} \leq x_i \leq \frac{\theta}{2}, i = 1, 2, \ldots. \\ 0, & \text{otherwise.} \end{cases}$$

Let us write

$$f_\theta\,(x_1, x_2 \ldots\ldots\ldots x_n) = h(x_1, x_2 \ldots\ldots\ldots x - n)\, g\left(x_{(1)}, x_{(n)}\right)$$

where

$$g\left(\min x_{(i)}, \max{}_{(i)}\right) = \begin{cases} \frac{1}{\theta^n} & \text{if, } -\frac{\theta}{2} \leq (x_{(1)}, x_{(n)}) \leq \frac{\theta}{2} i = 1, 2, \ldots. \\ 0, & \text{otherwise.} \end{cases}$$

Hence, $(\min X_i, \max X_i)$, $1 \leq i \leq n$, is sufficient for θ.

5.4.19. The likelihood function is given by:

$$f(x_1, x_2, \ldots\ldots\ldots x_n, \mu) = (2\pi)^{-\frac{n}{2}} \exp\left\{-\frac{\sum(x_i - \mu)^2}{2}\right\}$$

The above expression can be written as:

$$f(x_1, X_2, \ldots\ldots\ldots x_n, \mu) = \exp\left\{-x_i^2 - 2\mu x_1 + n\mu^2\right\} \exp\left\{-\sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=2}^{n} x_i\right\}$$

Let $g(X_1, \mu) = \exp\left\{-x_1^2 - 2\mu x_1 + n\mu^2\right\}$ This implies that we can not take the other function which is only the function of $X_i's$. Hence by factorization theorem, X_1 is not sufficient statistics for μ.

5.4.21. The likelihood function is given by

$$f(x_1, \ldots\ldots x_n) = \begin{cases} \theta^n \prod_{i=1}^{n} (x_i)^{\theta-1} & \text{for } 0 < x < 1, \theta > 0 \\ 0 \quad \text{otherwise} \end{cases}$$

Let $U = (X_1, X_2, \ldots\ldots\ldots X_n)$ then
$g(x_1, \ldots\ldots\ldots X_n, \theta) = \theta^n \prod_{i=1}^{n} x_i^{\theta-1}$ and $h(x_1, \ldots\ldots x_n) = 1$. Therefore U is sufficient for θ.

5.4.23. The likelihood function is given by

$$f(x_1, \ldots\ldots\ldots x_n) = \frac{2^n}{\alpha^n} \left\{ \prod_{i=1}^{n} X_i \right\} e^{-\frac{\sum x_i^2}{\alpha}}$$

Let $g\left(\sum x_i^2, \alpha\right) = \frac{2^n}{\alpha^n} e^{-\frac{\sum x_i^2}{\alpha}}$ and $h(x_1, \ldots\ldots\ldots x_n) = \prod_{i=1}^{n} x_i$
Hence, $\sum x_i^2$ is sufficient for the parameter α.

EXERCISES 5.5

5.5.1. $\ln L(p, x) = \ln \left\{ \prod_{i=1}^{n} \binom{n}{x_i} + \sum x_i \ln p \sum_{i=1}^{n} (n - x_i) \ln(1-p) \right\}$ For MLE $(1-p)\sum x_i - p\sum(n - x_i) = 0$. This implies $p = \frac{\sum x_i}{n^2}$. Suppose $Y_n = \frac{\sum x_i}{n}$. Thus $\hat{p} = \frac{Y_n}{n}$. Where $E\left(\frac{Y_n}{n}\right) = \frac{1}{n} E(Y_n) = \frac{1}{n^2} nnp = p$. $\hat{p}$ is an unbiased estimate of p. Similarly $\text{Var}\,\hat{p} = \text{Var}\,\frac{Y_n}{n} = \text{Var}\,\frac{\sum x_i}{n^2}$
This implies $\text{Var}\,\hat{p} = \frac{n}{n^4} np(1-p)$. Thus $\text{Var}\,\hat{p} \to 0 \quad as \quad n \to \infty$. $\frac{Y_n}{n}$ is consistent.

5.5.3. $E(X_i) = \mu_i$, $E(X_i^2) = \mu_2'$ and $E(X_i) = \mu_4'$.
$E(s^2) = \frac{1}{n} \sum E\left(X_i - \mu + \mu - \overline{X}\right)^2$
This implies $E(s^2) = \frac{n\sigma^2 - \sigma^2}{n}$. That is $E(s^2) = \frac{(n-1)\sigma^2}{n}$.
$\therefore$ S^2 is an biased estimator of σ^2
Here $S^2 = \frac{n-1}{n} S'^2$
$\text{Var}(S^2) = \frac{n-1}{n^2}\sigma^2 \to 0$ as $n \to \infty$.
Where $\frac{(n-1)s'^2}{\sigma^2} \sim \chi^2_{(n-1)}$.
Thus, $\text{Var}\left(\frac{(n-1)s''^2}{\sigma^2}\right) = 2(n-1)$. This implies $\frac{(n-1)^2}{\sigma^4} \text{var}(S'^2) = 2(n-1)$.
Finally, $\text{Var}(S'^2) = \frac{2}{n-1}\sigma^2$.
Moreover, $\text{Bias}(s^2) = E(s^2) - \sigma^2$. This implies $\frac{-\sigma^2}{n} \to 0$ as $n \to \infty$.
Thus s^2 is an unbiased estimate of σ^2.

5.5.5. Here $E(x) = \theta$ and $\text{var}(x) = \theta^2$ (For exponential distribution). Now $E(\overline{X} = E\theta)$ and $\text{Var}(\theta) = \frac{\theta^2}{n} \to 0$ as $n \to \infty$. $\therefore$ $\overline{X}$ is an unbiased estimate of θ.

5.5.7. Here, $\ln(\alpha, X) = -n \ln \alpha + \frac{1-\alpha}{\alpha} \sum_{i=1}^{n} \ln(X_i)$.
Differentiating above equation and equating to zero we get $\hat{\alpha} = \frac{-\sum_{i=1}^{n} \ln X_i}{n}$

5.5.9. Here, $\text{var}(\hat{\theta}_1) = \frac{1}{12n}$ and $\text{Var}(\hat{\theta}_2) = \frac{1}{12}$.
Thus the efficiency of $\hat{\theta}_2$ relative to $\hat{\theta}_1$ is $\epsilon(\hat{\theta}_2, \hat{\theta}_1) = \frac{1}{n} < 1$ for $n \geq 2$.
Here the efficiency depends on n. Thus $\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$ for $n \geq 2$.

5.5.11. By definition $E(x) = \theta$ and $\text{var}(X) = \theta^2$. Here, $\ln f(x) = -\ln(\theta) - \frac{x}{\theta}$, $E(X^2) = \theta^2$. Moreover $\text{Var}\left\{\frac{\partial}{\partial\theta} \ln f(x)\right\} = \text{var}\left(\frac{-1}{\theta} + \frac{X}{\theta^2}\right)$. Hence,

$$\frac{1}{n\text{Var}\left\{\frac{\partial}{\partial\theta} \ln f(x)\right\}} = \frac{\theta^2}{n} + \text{var}(\overline{X}).$$

$\therefore \overline{X}$ is efficient for θ.

5.5.13. Using given pdf $\ln f(X) = c - \frac{x-\mu}{2\sigma^2}$ where $c = \frac{-n\ln(2\pi\sigma^2)}{2}$.
Thus

$$E\left\{\frac{\partial^2}{\partial^2} \ln f(x)\right\} = \frac{-1}{\sigma^2}.$$

This implies

$$E\left[\frac{\partial}{\partial\mu} \ln f(x)\right]^2 = \frac{1}{\sigma^2}$$

Hence,

$$E\left[\frac{\partial}{\partial\mu} \ln f(x)\right]^2 = -E\left\{\frac{\partial^2}{\partial^2} \ln f(x)\right\}.$$

5.5.15. Because each X_i has a uniform distribution on the interval $(0.\theta)$, $\mu = E(X_i) = \theta/2$ and $\text{var}(Y_i) = \frac{\theta^2}{12}$.
$\therefore E(\hat{\theta}_2) = \theta$ and $\text{var}(\hat{\theta}_2) = \frac{\theta^2}{3n}$.
To find the variance of $\hat{\theta}_1$ and $\hat{\theta}_3$ we must have density of X_n.
$f_n(X) = n[F_X(x)]^n - 1 f_X(x)$. That is $f_n(X) = \frac{nx^{n-1}}{\theta^n}$ for $0 \leq x \leq \theta$.
This implies $E(\hat{\theta}) = \frac{n}{n+1}\theta$. Thus $\hat{\theta}_1$ is not an unbiased estimate of θ.
Similarly. $E(\hat{\theta}_3) = \theta$. $\hat{\theta}_3$ is an unbiased estimator of θ.
Now $\text{var}\,\hat{\theta}_1 = \frac{n}{(n+1)^2(n+2)}$ and $\text{var}\,\hat{\theta}_3 = \frac{\theta^2}{n(n+2)}$.
(b) The efficiency of $\hat{\theta}_1$relative to $\hat{\theta}_2$ is given by:

$$e\left(\hat{\theta}_1, \hat{\theta}_2\right) = \frac{(n+1)^3(n+1)}{3n^2}.$$

Hence $\hat{\theta}_1$ is more efficient than $\hat{\theta}_1$ if $\frac{(n+1)^3(n+1)}{3n^2} > 1$
Similarly, $e\left(\hat{\theta}_2, \hat{\theta}_3\right) = \frac{n+2}{3} > 1$ if $n > 1$.
$\hat{\theta}_2$ is efficient than $\hat{\theta}_3$ if $n > 1$.

5.5.17. It can be easily verified that $E(\hat{\theta_1}) = \theta$, $E(\hat{\theta_2}) = \theta$ and $E(\hat{\theta_3}) = \theta$
Similarly, $\mathrm{Var}(\hat{\theta_1}) = \frac{31}{81}\sigma^2$, $\mathrm{Var}(\hat{\theta_2}) = \frac{6n-17}{25(n-3)}\sigma^2$.
$\mathrm{Var}(\hat{\theta_3}) = \frac{\sigma^2}{n}$.
Now the corresponding efficiencies are given by $e(\hat{\theta_2}, \hat{\theta_1}) = \frac{31775(n-3)}{81(6n-17)}$, $e(\hat{\theta_3}, \hat{\theta_1}) = \frac{31n}{81}$.
$e(\hat{\theta_3}, \hat{\theta_2}) = \frac{(6n-17)n}{25(n-3)}$.

5.5.21. The ratio of the joint density function of two sample points is given by;

$$\frac{L(x_1, \ldots . x_n)}{L(y_1, \ldots y_n)} = \exp\left[\frac{-1}{2\sigma^2}\left\{\left(\sum_{i=1}^{n} X_i^2 - \sum_{i=1}^{n} Y_i^2\right) - 2\mu\left(\sum_{i=1}^{n} X_i - \sum_{i=1}^{n} Y_i\right)\right\}\right].$$

For this ratio to be free of μ and σ^2, we must have $\sum_{i=1}^{n} X_i = \sum_{i=1}^{n} Y_i$ and $\sum_{i=1}^{n} X_i^2 = \sum_{i=1}^{n} Y_i^2$.
Thus $\sum_{i=1}^{n} X_i$ and $\sum_{i=1}^{n} X_i^2$ are jointly minimal sufficient statistics for μ and σ^2. Since $\overline{X}$ is unbiased estimate for μ and s^2 is an unbiased estimate for σ^2. The estimators are functions of the minimal sufficient statistics. This implies the $\overline{X}$ and s^2 are MVUES for μ and σ^2.

5.5.23. The ratio of joint density function at two sample points, we have

$$\frac{L(x_1, \ldots . x_n)}{L(y_1, \ldots y_n)} = \frac{\prod_{i=1}^{n}(Y_i)}{\prod_{i=1}^{n}(X_i)!}\lambda^{\sum X_i - \sum Y_i}.$$

For the ratio to be free of λ we must have $\sum X_i - \sum Y_i = 0$. Thus $\sum X_i$, form the minimal sufficient statistics for μ.

5.5.25.

$$\frac{L(x_1, \ldots . x_n)}{L(y_1, \ldots y_n)} = e^{\sum X_i - \sum Y_i}$$

For the ratio to be independent of β, we need to have $\sum X_i = \sum Y_i$. Thus $\sum X_i$ is minimal sufficient for β. Now $E(\sum X_i) = n\overline{X}$ is UMVUE, by Rao–Blackwells theorem.

5.5.27.

$$\frac{L(x_1, \ldots . x_n)}{L(y_1, \ldots y_n)} = \frac{\prod_{i=1}^{n}(Y_i)}{\prod_{i=1}^{n}(Y_i)!}e^{\sum X_i^2 - \sum Y_i^2}.$$

The ratio to be free of β, we must have $\sum X_i^2 - \sum Y_i^2 = 0$. Therefore $\sum X_i^2$ is MVUE for β. Moreover s^2 is an unbiased estimator for σ^2.

Chapter 6

Interval Estimation

EXERCISES 6.1

6.1.1. **(a)** We are 99% confident that the estimate value of the parameter lies in the confidence interval.

(b) 99% confidence interval is wider

(c) When μ is known but σ^2 is unknown we use t-distribution for the sample size $n \le 30$. If the distribution is binomial and there are enough number of samples such that $np \ge 5$ and $np(1-p) \ge 5$, then we use normal approximation.

(d) More the information higher the confidence interval. So the sample size is inversely proportional to the width of the confidence interval.

6.1.3. **(a)** $p\left(-2.81 \le \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \le 2.75\right) = k$

$p\left(-2.81 \cdot \frac{\sigma}{\sqrt{n}} \le \bar{x} - \mu \le 2.75 \cdot \frac{\sigma}{\sqrt{n}}\right) = k$

$p\left(-2.81 \cdot \frac{\sigma}{\sqrt{n}} - \bar{x} \le -\mu \le 2.75 \cdot \frac{\sigma}{\sqrt{n}} + \bar{x}\right) = k$

$p\left(\bar{x} - 2.75 \cdot \frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + 2.81 \cdot \frac{\sigma}{\sqrt{n}}\right) = k$

(b) Confidence interval for μ is given by $\left(\bar{x} - 2.75 \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + 2.81 \cdot \frac{\sigma}{\sqrt{n}}\right)$

(c) Confidence level $= k$

6.1.5. **(a)** Here $x_i \sim N(\mu, \sigma^2)$

$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$

Pivot $= \frac{(n-1)s^2}{\sigma^2}$; where only σ^2 is unknown.

$p\left(a < \frac{(n-1)s^2}{\sigma^2} < b\right) = 1 - \alpha$

$p\left(\chi^2_{1-\alpha/2} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\alpha/2}\right) = 1 - \alpha$

$$p\left(\frac{1}{\chi^2_{1-\alpha/2}} < \frac{\sigma^2}{(n-1)s^2} < \frac{1}{\chi^2_{\alpha/2}}\right) = 1-\alpha$$

$$p\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right) = 1-\alpha$$

So $(1-\alpha)\cdot 100\%$ confidence interval for σ^2 is given by

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

(b) $n = 21, \overline{x} = 44.3, s = 3.96$

$\alpha = 0.1, 1-\alpha = 0.9, \alpha/2 = 0.05$

$\chi^2_{1-\alpha/2,20} = \chi^2_{0.95,20} = 10.851$

$\chi^2_{\alpha/2} = \chi^2_{0.05,20} = 31.410$

90% confidence interval is given by

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} = \frac{20(3.96)^2}{31.41} < \sigma^2 < \frac{20(3.96)^2}{10.851}$$

We are 90% confident that σ^2 lies in the interval (9.985, 28.903).

6.1.9. **(a)** $p\left(a \le \frac{\overline{x}-\mu}{\sigma/\sqrt{n}} \le b\right) = 1-\alpha$

$$p\left(-z_{\alpha/2} < \frac{\overline{x}-\mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1-\alpha$$

$$p\left(-z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} < \overline{x}-\mu < z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

$$p\left(-z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} - \overline{x} < -\mu < z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} - \overline{x}\right) = 1-\alpha$$

$$p\left(\overline{x} - z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

So the confidence interval is $\left(\overline{x} - z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}, \overline{x} + z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right)$.

(b) If σ^2 is unknown, we use sample variance for the estimation of confidence interval and t-distribution as a sampling distribution. Thus the confidence interval is $\left(\overline{x} - t_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}, \overline{x} + t_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right)$.

EXERCISES 6.2

6.2.1. **(a)** $n = 1200$

$\hat{p} = 0.35$

$z_{\alpha/2} = z_{0.25} = 1.96$

95% confidence interval is given by

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.35 \pm 1.96\sqrt{\frac{0.35(1-0.35)}{1200}} = (0.323, 0.377)$$

(b) $\hat{p} = 0.6$

$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.6 \pm 1.96\sqrt{\frac{0.6(1-0.6)}{1200}} = (0.572, 0.628)$

(c) $\hat{p} = 0.15$

$0.15 \pm 1.96\sqrt{\frac{0.15(1-0.15)}{1200}} = (0.13, 0.17)$

(d) We are 95% confident that the percentage of people who find political advertising to be true is (0.323, 0.377), the percentage of people who find political advertising to be untrue is (0.572, 0.628), the percentage of people who find falsehood in commercial is (0.13, 0.17).

6.2.3. **(a)** Here $f(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$

$L(x_1, x_2, \ldots, x_n, \mu, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}}\exp\left\{-\frac{\sum(x_i-\mu)^2}{2\sigma^2}\right\}$

$\ln L(x_1, x_2, \ldots, x_n, \mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{\sum(x_i-\mu)^2}{2\sigma^2}$

$\frac{\partial}{\partial\mu}\ln L(x_1, x_2, \ldots, x_n, \mu, \sigma^2) = \frac{\sum(x_i-\mu)^2}{2\sigma^2}$

For MLE estimator

$\frac{\partial}{\partial\mu}\ln L(x_1, x_2, \ldots, x_n, \mu, \sigma^2) = 0$

$\sum(x_i - \mu) = 0$

So $\hat{\mu} = \overline{x}$

(b) $(1-\alpha)100\%$ confidence interval for μ is given by $\left(\overline{x} - z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}, \overline{x} + z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right)$

$p\left(\overline{x} - z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$

$p\left(\overline{x} - 2\cdot\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + 2\cdot\frac{\sigma}{\sqrt{n}}\right) = 0.954$

$(1-\alpha) = 0.954$

$\alpha = 0.046$

$\alpha/2 = 0.023$

$z_{\alpha/2} = 2.0$ from z table

Thus verified.

(c) $p\left(\overline{x} - k\cdot\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + k\cdot\frac{\sigma}{\sqrt{n}}\right) = 0.90$

$(1-\alpha) = 0.90$

$\alpha = 0.10$

$\alpha/2 = 0.05$

$z_{\alpha/2} = 1.645$

$k = 1.645$

6.2.5. $n = 50$

$\tilde{p} = \frac{18}{50} = \frac{9}{25}$

$n\tilde{p} = 18 > 5$

$n(1 - \tilde{p}) = 32 > 5$

So the given data can be approximated as a normal distribution.

Here $1 - \alpha = 0.98$

$\alpha = 0.02$

$\alpha/2 = 0.01$

$z_{\alpha/2} = z_{0.01} = 2.325$

Thus the 98% confidence interval is given by

$$\left(\tilde{p} \pm z_{\alpha/2}\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n}}\right) = \left(\frac{9}{25} \pm 2.325\sqrt{\frac{\frac{9}{25}\cdot\frac{16}{25}}{50}}\right) = (0.202, 0.518)$$

6.2.7. $n = 50$

$\bar{x} = 11.4$

$\sigma = 4.5$

$1 - \alpha = 0.95$

$\alpha = 0.05$

$z_{\alpha/2} = 1.96$

95% confidence interval is

$$\left(\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = \left(11.4 \pm 1.96\left(\frac{4.5}{\sqrt{50}}\right)\right) = (10.153, 12.647)$$

6.2.9. $n = 400$

$\hat{p} = 0.3$

$n\hat{p} = 120 > 5$

$n(1 - \hat{p}) = 280 > 5$

95% confidence interval is given by

$$\left(\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = \left(0.3 \pm 1.96\sqrt{\frac{0.3\cdot 0.7}{400}}\right) = (0.255, 0.345)$$

6.2.11. Proportion of defection $\hat{p} = \frac{40}{500} = \frac{2}{25}$

$1 - \alpha = 0.9$

$\alpha/2 = 0.05$

$z_{\alpha/2} = 1.645$

90% confidence interval is given by

$$\left(\frac{2}{25} \pm 1.645\sqrt{\frac{\frac{2}{25}\cdot\frac{23}{25}}{500}}\right) = (0.06, 0.1)$$

6.2.13. $\overline{x} \sim N(\mu, 16)$

$p(\overline{x} - 2 < \mu < \overline{x} + 2) = 0.95$

$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 2$

$n = (z_{\alpha/2} \frac{\sigma}{2})^2 = \left(\frac{1.96 \cdot 4}{2}\right)^2 = 15.37 \approx 16$

6.2.15. $n = 425$

$\hat{p} = 0.45$

$n\hat{p} > 425 \cdot 0.45 > 5$

$n(1 - \hat{p}) = 425 \cdot 0.55 > 5$

95% confidence interval is given by

$$\left(\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = \left(0.45 \pm 1.96\sqrt{\frac{0.45 \cdot 0.55}{425}}\right) = (0.403, 0.497)$$

For 98% confidence interval

$1 - \alpha = 0.98$

$\alpha = 0.02$

$z_{\alpha/2} = 2.335$

$$\left(0.45 \pm 2.335\sqrt{\frac{0.45 \cdot 0.55}{425}}\right) = (0.394, 0.506)$$

6.2.19. $\hat{p} = \frac{52}{60}$

$1 - \alpha = 0.95$

$\alpha/2 = 0.025$

$z_{\alpha/2} = 1.96$

The 95% confidence interval is given by

$$\left(\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = \left(\frac{52}{60} \pm 1.96\sqrt{\frac{\frac{52}{60} \cdot \frac{8}{60}}{60}}\right) = (0.781, 0.953)$$

6.2.21. $\sigma = 35$

$E = 15$

$E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2 = \left(\frac{1.96 \cdot 35}{15}\right)^2 = 20.92 \approx 21$

6.2.23. $\overline{x} = 12.07$

$\sigma = 12.91$

$1 - \alpha = 0.98$

$\alpha/2 = 0.01$

$z_{\alpha/2} = 2.335$

98% confidence interval for mean is given by

$$\left(\overline{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right) = \left(12.07 \pm 2.335\frac{1.91}{\sqrt{35}}\right) = (11.32, 12.82)$$

EXERCISES 6.3

6.3.1. **(a)** When standard deviation is not given and there is not enough sample size, we use t-distribution.

(b) As differences decreases the sample size n increases which means we are closing in on the true parameter value of θ.

(c) The data are normally distributed, and the values of $\overline{x}$ and the sample standard deviation are known.

6.3.3. $\overline{x} = 20$

$s = 4$

$1 - \alpha = 0.95$

(a) $\left(\overline{x} \pm t_{\alpha/2,4} \cdot \frac{s}{\sqrt{n}}\right) = \left(20 \pm t_{0.025,4}\frac{4}{\sqrt{5}}\right)$

(b) $\left(20 \pm t_{0.025,9}\frac{4}{\sqrt{10}}\right)$

(c) $\left(20 \pm t_{0.025,19}\frac{4}{\sqrt{20}}\right)$

6.3.5. $\overline{x} = -2.22$

$s = 0.005$

$n = 26$

98% confidence interval for μ is

$$\left(\overline{x} \pm t_{\alpha/2,25} \cdot \frac{s}{\sqrt{n}}\right) = \left(-2.22 \pm 2.485\frac{1.67}{\sqrt{26}}\right) = (-3.03, -1.41)$$

6.3.7. $\overline{x} = 0.905$

$s = 1.67$

$1 - \alpha = 0.98$

$n = 10$

98% confidence interval for μ is

$$\left(0.905 \pm t_{0.025,9}\frac{0.005}{\sqrt{10}}\right)$$

6.3.9. Similar to 6.3.8

6.3.11. $\overline{x} = 410.93$

$s = 312.87$

95% confidence interval for μ is

$$\left(\bar{x} \pm t_{\alpha/2,14} \cdot \frac{s}{\sqrt{n}}\right) = \left(410.93 \pm 2.145 \frac{312.87}{\sqrt{15}}\right) = (237.65, 584.21)$$

6.3.13. $\bar{x} = 3.12$

$s = \sqrt{1.04}$

$n = 17$

99% confidence interval for μ is

$$\left(\bar{x} \pm t_{\alpha/2,4} \cdot \frac{s}{\sqrt{n}}\right) = \left(3.12 \pm 2.921 \frac{\sqrt{1.04}}{\sqrt{17}}\right) = (2.40, 3.84)$$

6.3.15. $\bar{x} = 3.85$

$s = \sqrt{4.55}$

$n = 20$

98% confidence interval for μ is

$$\left(\bar{x} \pm t_{\alpha/2,19} \cdot \frac{s}{\sqrt{n}}\right) = \left(3.85 \pm 2.539 \frac{\sqrt{4.55}}{\sqrt{20}}\right) = (2.64, 5.06)$$

6.3.17. $\bar{x} = 148.18$

$s = \sqrt{1.91}$

$n = 10$

95% confidence interval for μ is

$$\left(\bar{x} \pm t_{\alpha/2,9} \cdot \frac{s}{\sqrt{n}}\right) = \left(148.18 \pm 2.262 \frac{\sqrt{1.91}}{\sqrt{10}}\right) = (147.19, 149.17)$$

EXERCISES 6.4

6.4.1. $\bar{x} = -2.2$

$s = 1.42$

$1 - \alpha = 0.90$

$\alpha = 0.10$

$n = 20$

90% confidence interval for σ^2 is given by

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,19}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,19}}\right) = \left(\frac{19(1.42)^2}{32.85}, \frac{19(1.42)^2}{8.90655}\right) = (1.1663, 4.3015)$$

6.4.3. $\bar{x} = 60.908$

$s^2 = 12.66$

$1 - \alpha = 0.99$

$\alpha = 0.01$

$n = 10$

99% confidence interval for σ^2 is given by

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,9}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,9}}\right) = \left(\frac{9\cdot 12.66}{23.58}, \frac{9\cdot 12.66}{1.73}\right) = (4.8321, 65.8613)$$

6.4.5. $\bar{x} = 2.27$

$s^2 = 1.02$

$1 - \alpha = 0.99$

$\alpha = 0.01$

$n = 18$

99% confidence interval for σ^2 is given by

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,9}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,9}}\right) = \left(\frac{17\cdot 1.02}{27.58}, \frac{17\cdot 1.02}{5.69}\right) = (0.6287, 3.0475)$$

6.4.9. From excel or by calculation, sample variance $s^2 = 148.44$, sample mean $\bar{x} = 97.24$, $n = 25$

99% confidence interval for population variance is given by

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,24}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,24}}\right) = \left(\frac{24\cdot 148.44}{36.41}, \frac{24\cdot 148.44}{9.886}\right) = (97.8456, 360.3642)$$

6.4.11. $\bar{x} = 13.95$

$s^2 = 495.085$

$1 - \alpha = 0.98$

$\alpha = 0.02$

$n = 25$

98% confidence interval for σ^2 is given by

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,24}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,24}}\right) = \left(\frac{24\cdot 495.085}{42.97}, \frac{24\cdot 495.085}{10.85}\right) = (276.5194, 1095.1189)$$

EXERCISES 6.5

6.5.1. For procedure I, $\bar{x}_1 = 98.4$, $s_1^2 = 235.6$, $n_1 = 10$

For procedure II, $\bar{x}_2 = 95.4$, $s_2^2 = 87.15$, $n_2 = 10$

$\alpha = 0.02$

$\alpha/2 = 0.01$

$z_{\alpha/2} = 2.985$

98% confidence interval for difference of mean is

$$\left(\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right) = \left(98.4 - 95.4 \pm 2.985\sqrt{\frac{235.6}{10} + \frac{87.15}{10}}\right)$$
$$= (-13.9580, 19.9580)$$

6.5.3. $\bar{x}_1 = 16.0, \quad s_1 = 5.6, \quad n_1 = 42$

$\bar{x}_2 = 10.6, \quad s_2 = 7.9, \quad n_2 = 45$

$\alpha = 0.01$

$\alpha/2 = 0.005$

$z_{\alpha/2} = 2.575$

99% confidence interval for difference of mean is

$$\left(\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right) = \left(16.0 - 10.6 \pm 2.575\sqrt{\frac{(5.6)^2}{42} + \frac{(7.9)^2}{45}}\right) = (1.6388, 9.1612)$$

6.5.5. $\bar{x}_1 = 58,550, \; s_1 = 4,000, \; n_1 = 25$

$\bar{x}_2 = 53,700, \; s_2 = 3,200, \; n_2 = 23$

Since $\sigma_1^2 = \sigma_2^2$ but unknown, we can use pooled estimator

$S_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{(n_1+n_2-2)}$

$S_p^2 = \frac{24(4,000)^2+22(3,200)^2}{46}$

$S_p = 3639.398$

The 90% confidence interval is

$$\left(\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2,(n_1+n_2-2)} \cdot S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}\right)$$

$$= \left(58,550 - 53,700 \pm 2.326 \cdot 3639.398\sqrt{\frac{1}{25} + \frac{1}{23}}\right) = (2404, 7296)$$

6.5.7. $\bar{x}_1 = 28.4, \; s_1 = 4.1, \; n_1 = 40$

$\bar{x}_2 = 25.6, \; s_2 = 4.5, \; n_2 = 32$

(a) MLE of $\mu_1 - \mu_2$ is given by $(\bar{x}_1 - \bar{x}_2)$

(b) 99% confidence interval for $\mu_1 - \mu_2$ is

$$\left(\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right) = \left(28.4 - 25.6 \pm 2.565\sqrt{\frac{(4.1)^2}{40} + \frac{(4.5)^2}{32}}\right)$$

$$= (0.1678, 5.4322)$$

6.5.9. $\bar{x}_1 = 148,822, \; s_1 = 21,000, \; n_1 = 100$

$\bar{x}_2 = 155,908, \; s_2 = 23,000, \; n_2 = 150$

$1 - \alpha = 0.98$

$\alpha = 0.02$

$\alpha/2 = 0.01$

$z_{\alpha/2} = 2.575$

98% confidence interval for difference of mean is given by

$$\left(\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right) = \left(148,822 - 155,908 \pm 2.2\sqrt{\frac{(21{,}000)^2}{100} + \frac{(23{,}000)^2}{150}}\right)$$

$$= (-508, 13, 664)$$

6.5.11. $\bar{x}_1 = 35.18,\ s_1^2 = 19.76,\ n_1 = 11$

$\bar{x}_2 = 38.76,\ s_2^2 = 12.69,\ n_2 = 13$

$1 - \alpha = 0.9$

$\alpha = 0.1$

90% confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ is given by

$$\left(\frac{s_1^2}{s_2^2}\frac{1}{F_{n_1-1,n_2-1,1-\alpha/2}}, \frac{s_1^2}{s_2^2}\frac{1}{F_{n_1-1,n_2-1,\alpha/2}}\right) = \left(\frac{19.76}{12.69}\frac{1}{F_{10,12,0.95}}, \frac{19.76}{12.69}\frac{1}{F_{10,12,0.05}}\right)$$

$$= \left(\frac{19.76}{12.69}\frac{1}{2.75}, \frac{19.76}{12.69}\cdot 2.91\right) = (0.5662, 4.5313)$$

6.5.13. $\bar{x}_1 = 68.91,\ s_1^2 = 287.17,\ n_1 = 12$

$\bar{x}_2 = 80.66,\ s_2^2 = 117.87,\ n_2 = 12$

$1 - \alpha = 0.95$

$\alpha = 0.05$

$\alpha/2 = 0.025$

(a) 95% confidence interval of difference of mean is

$$\left(\bar{x}_1 - \bar{x}_2 \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}\right) = \left(68.91 - 80.66 \pm 1.96\sqrt{\frac{281.17}{12} + \frac{117.87}{12}}\right)$$

$$= (-23.0525, -0.4475)$$

(b) 95% confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ is given by

$$\left(\frac{s_1^2}{s_2^2}\frac{1}{F_{n_1-1,n_2-1,1-\alpha/2}}, \frac{s_1^2}{s_2^2}\frac{1}{F_{n_1-1,n_2-1,\alpha/2}}\right) = \left(\frac{281.17}{117.87}\frac{1}{F_{11,11,0.975}}, \frac{281.17}{117.87}\frac{1}{F_{11,12,0.025}}\right)$$

$$= \left(\frac{281.17}{118.87}\cdot\frac{1}{3.48}, \frac{281.17}{118.87}\cdot 3.48\right) = (0.6855, 8.3055)$$

Chapter 7

Hypothesis Testing

EXERCISES 7.1

7.1.1. **(a)** $H_0 : \mu = \mu_0$

$H_1 : \mu > \mu_0$

(b) $H_0 : \mu = \mu_0$

$H_1 : \mu > 1.2\mu_0$

7.1.3. $H_0 : p = 0.5$

$H_1 : p > 0.5$

$n = 15$

(a) $\alpha =$ probability of type I error $= p(\text{reject } H_0 | H_0 \text{ is true})$

$$= p(y \geq 10 | p = 0.5) = 1 - p(y \leq 10 | p = 0.5)$$

$$= 1 - \sum_{y=10}^{15} c(15, y)(0.5)^y (0.5)^{15-y}$$

$$= 1 - 0.941$$

$$= 0.059$$

(b) $\beta = p(\text{accept } H_0 | H_0 \text{ is false})$

$$= p(y \leq 9 | p = 0.7)$$

$$= \sum_{y=0}^{9} c(15, y)(0.7)^y (0.3)^{15-y}$$

$$= 0.278$$

(c) $\beta = p(y \leq 9 | p = 0.6)$

$$= \sum_{y=0}^{9} c(15, y)(0.6)^y (0.4)^{15-y}$$

$$= 0.597$$

(d) For $\alpha = 0.01$

$$0.01 = p(y \geq k|p = 0.5)$$

From binomial table $\alpha = 0.01$ falls between $k = 2$ and $k = 3$. However, for $k = 3$, $\alpha = 0.018$ which exceeds 0.01. If we want to limit α to be no more than 0.01, we will take $k = 2$. That is we reject H_0 if $k \geq 2$.
For $\alpha = 0.01$

$$0.03 = p(y \geq k|p = 0.5)$$

From binomial table $\alpha = 0.03$ falls between $k = 3$ and $k = 4$. However, for $k = 4$, $\alpha = 0.059$ which exceeds 0.05. If we want to limit α to be no more than 0.05, we will take $k = 3$. That is we reject H_0 if $k \geq 3$.

(e) When $\alpha = 0.01$. From part d, rejection region is of the form $(y \geq 2)$.
For $p = 0.7$

$$\begin{aligned}\beta &= p(y \geq 2|p = 0.7)\\ &= 1 - p(y < 1|p = 0.7)\\ &= 1 - 0.000\\ &= 1\end{aligned}$$

7.1.5. $n = 25$

$\sigma = 4$

$H_0 : \mu = 10$

$H_1 : \mu > 10$

(a) $\alpha =$ probability of type I error $= p(\text{reject } H_0|H_0 \text{ is true})$

$$\begin{aligned}&= p(\overline{x} > 11.2|\mu = 10)\\ &= P\left(\frac{\overline{x}-\mu}{\sigma/\sqrt{n}} > \frac{11.2-10}{4/\sqrt{25}}|\mu = 10\right)\\ &= P(z > 1)\\ &= 0.1587\end{aligned}$$

(b) $\beta = p(\text{accept } H_0|H_0 \text{ is false})$

$$\begin{aligned}&= p(\overline{x} \leq 11.2|\mu = 11)\\ &= P\left(\frac{\overline{x}-\mu}{\sigma/\sqrt{n}} \leq \frac{11.2-11}{4/\sqrt{25}}|\mu = 11\right)\\ &= P(z \leq 0.25)\\ &= 0.5787\end{aligned}$$

(c) $\mu_0 = 10$

$\mu_a = 11$

$z_\alpha = z_{0.01} = 2.33$

$z_\beta = z_{0.8} = -0.525$

$$\begin{aligned}n &= \frac{(z_\alpha+z_\beta)^2\sigma^2}{(\mu_a-\mu_0)^2}\\ &= \frac{(2.33-0.525)^2 \cdot 4^2}{(11-10)^2}\\ &= 52.13 \text{ rounded up to } 53\end{aligned}$$

7.1.9. $\sigma^2 = 16$

$H_0 : \mu = 25$

$H_1 : \mu \neq 25$

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{(\mu_a - \mu_0)^2}$$
$$= \frac{(1.645 + 1.645)^2 \cdot 16}{(1)^2}$$
$= 173.19$ rounded up to 174

EXERCISES 7.2

7.2.1. $H_0 : \mu = \mu_0$

$H_1 : \mu = \mu_1$

$$L(\mu) = \frac{1}{\left(\sqrt{2\pi}\right)^n \sigma^n} \exp\left\{-\frac{\sum (x_i - \mu)^2}{2\sigma^2}\right\}$$

$$L(\mu_0) = \frac{1}{\left(\sqrt{2\pi}\right)^n \sigma^n} \exp\left\{-\frac{\sum (x_i - \mu_0)^2}{2\sigma^2}\right\}$$

$$L(\mu_1) = \frac{1}{\left(\sqrt{2\pi}\right)^n \sigma^n} \exp\left\{-\frac{\sum (x_i - \mu_1)^2}{2\sigma^2}\right\}$$

$$\frac{L(\mu_0)}{L(\mu_1)} = \exp\left\{-\frac{\sum (x_i - \mu_0)^2}{2\sigma^2} + \frac{\sum (x_i - \mu_1)^2}{2\sigma^2}\right\}$$
$$= \exp\left\{\frac{\sum (x_i - \mu_1)^2 - \sum (x_i - \mu_0)^2}{2\sigma^2}\right\}$$

$$\ln\left\{\frac{L(\mu_0)}{L(\mu_1)}\right\} = \frac{\sum (x_i - \mu_1)^2 - \sum (x_i - \mu_0)^2}{2\sigma^2}$$
$$= \frac{\sum x_i^2 - 2n\bar{x}\mu_1 + \mu_1^2 - \sum x_i^2 - 2n\bar{x}\mu_0 - \mu_0^2}{2\sigma^2}$$
$$= \frac{2n\bar{x}(\mu_0 - \mu_1) - (\mu_0 - \mu_1)(\mu_0 + \mu_1)}{2\sigma^2}$$
$$= \frac{(\mu_0 - \mu_1)\sum x_i}{\sigma^2} - \frac{(\mu_0^2 - \mu_1^2)}{2\sigma^2}$$

Therefore, the most powerful test is to reject H_0 if

$$\frac{(\mu_0 - \mu_1)\sum x_i}{\sigma^2} - \frac{\left(\mu_0^2 - \mu_1^2\right)}{2\sigma^2} \leq \ln k$$
$$\frac{(\mu_0 - \mu_1)\sum x_i}{\sigma^2} \leq \ln k + \frac{\left(\mu_0^2 - \mu_1^2\right)}{2\sigma^2}$$
$$\sum x_i \leq \left\{\frac{\sigma^2 \ln k + \frac{\left(\mu_0^2 - \mu_1^2\right)}{2}}{(\mu_0 - \mu_1)}\right\} = C$$

Assume $\mu_0 < \mu_1$, rejection region for $\mu = \mu_1$ is given by

$$\sum x_i \leq C$$

Where $C = \left\{\frac{\sigma^2 \ln k + \frac{\left(\mu_0^2 - \mu_1^2\right)}{2}}{(\mu_0 - \mu_1)}\right\}$

Rejection region for $\mu = \mu_0$ is given by

$$\sum x_i \geq C$$

7.2.5. $H_0 : \eta = \eta_0$

$H_1 : \eta < \eta_0$

$f(y) = \frac{2y}{\eta^2} e^{-y^2/\eta^2} x > 0$

$L(\eta) = \prod_{i=1}^{n} \left(\frac{2y_i}{\eta^2}\right)^n e^{-\sum y_i^2/\eta^2}$

$L(\eta_0) = \prod_{i=1}^{n} \left(\frac{2y_i}{\eta_0^2}\right)^n e^{-\sum y_i^2/\eta_0^2}$

$L(\eta_1) = \prod_{i=1}^{n} \left(\frac{2y_i}{\eta_1^2}\right)^n e^{-\sum y_i^2/\eta_1^2}$

$$\frac{L(\eta_0)}{L(\eta_1)} = \frac{\prod_{i=1}^{n} \left(\frac{2y_i}{\eta_0^2}\right)}{\prod_{i=1}^{n} \left(\frac{2y_i}{\eta_1^2}\right)^n} e^{\sum y_i^2/\eta_1^2 - \sum y_i^2/\eta_0^2}$$

$$= \left(\frac{\eta_1}{\eta_0}\right)^{2n} e^{\sum y_i^2/\eta_1^2 - \sum y_i^2/\eta_0^2}$$

$\ln\left\{\frac{L(\eta_0)}{L(\eta_1)}\right\} = 2n \ln\left(\frac{\eta_1}{\eta_0}\right) + \sum y_i^2/\eta_1^2 - \sum y_i^2/\eta_0^2 \leq \ln k$

$\sum y_i^2 \leq C$

Where $C = \left\{\ln k - 2n \ln\left(\frac{\eta_1}{\eta_0}\right)\right\} \frac{\eta_0^2 \eta_1^2}{\eta_0^2 - \eta_1^2}$

7.2.7. $H_0 : p = p_0$

$H_1 : p = p_1$ where $p_1 > p_0$

$f(p) = p^x (1 - p)^{1-x}$

$L(p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}$

$L(p_0) = p_0^{\sum x_i} (1 - p_0)^{n - \sum x_i}$

$L(p_1) = p_1^{\sum x_i} (1 - p_1)^{n - \sum x_i}$

$\frac{L(p_0)}{L(p_1)} = \left(\frac{p_0}{p_1}\right)^{\sum x_i} \left(\frac{1-p_0}{1-p_1}\right)^{n - \sum x_i} \leq k$

Taking natural logarithm, we have

$$\sum x_i \ln\left(\frac{p_0}{p_1}\right) + \left(n - \sum x_i\right) \ln\left(\frac{1 - p_0}{1 - p_1}\right) \leq \ln k$$

$$\left\{\ln\left(\frac{p_0}{p_1}\right) - \ln\left(\frac{1 - p_0}{1 - p_1}\right)\right\} \sum x_i + n \ln\left(\frac{1 - p_0}{1 - p_1}\right) \leq \ln k$$

$$\sum x_i \leq \left\{\ln k - n \ln\left(\frac{1 - p_0}{1 - p_1}\right)\right\} \Big/ \left\{\ln\left(\frac{p_0}{p_1}\right) - \ln\left(\frac{1 - p_0}{1 - p_1}\right)\right\}$$

To find the rejection region for a fixed value of α, write the region as $\sum x_i \leq C$, where $C = \left\{\ln k - n \ln\left(\frac{1-p_0}{1-p_1}\right)\right\} \Big/ \left\{\ln\left(\frac{p_0}{p_1}\right) - \ln\left(\frac{1-p_0}{1-p_1}\right)\right\}$

EXERCISES 7.3

7.3.1. $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$

$L(\sigma_1) = \frac{1}{\left(\sqrt{2\pi}\right)^n \sigma_1^n} \exp\left\{-\frac{\sum(x_i-\mu)^2}{2\sigma_1^2}\right\}$

Here $\Theta_0 = \left\{\sigma_0^2\right\}$ and $\Theta_a = R - \left\{\sigma_0^2\right\}$

Hence

$$L\left(\sigma_0^2\right) = \max\left(\frac{1}{\sqrt{2\pi}\sigma_0}\right)^n \exp\left\{-\frac{\sum(x_i-\mu)^2}{2\sigma_0^2}\right\}$$

Since the only unknown parameter in the parameter space Θ is σ^2, $-\infty < \sigma^2 < \infty$; maximum likelihood function is achieved when σ^2 equals to its maximum likelihood estimator

$$\hat{\sigma}^2_{\text{mle}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

$$\lambda = \left(\frac{\sigma_1^2}{\sigma_0^2}\right)^{n/2} \exp\left\{\frac{\sum(x_i-\mu)^2}{2\sigma_1^2} - \frac{\sum(x_i-\mu)^2}{2\sigma_0^2}\right\}$$

$$= \left(\frac{\sum(x_i-\overline{x})^2}{n\sigma_0^2}\right)^{n/2} \exp\left\{\frac{n\sum(x_i-\mu)^2}{2\sum(x_i-\overline{x})^2} - \frac{\sum(x_i-\mu)^2}{2\sigma_0^2}\right\}$$

The likelihood ratio test has the rejection region:

Reject if $\lambda \le k$, which is equivalent to

$\frac{n}{2}\ln\left\{\sum(x_i-\overline{x})^2\right\} - \frac{n}{2}\ln\left(n\sigma_0^2\right) + \frac{n\sum(x_i-\mu)^2}{2\sum(x_i-\overline{x})^2} - \frac{\sum(x_i-\mu)^2}{2\sigma_0^2} \le \ln k$

7.3.3. $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$

$L(\sigma_1) = \frac{1}{\left(\sqrt{2\pi}\right)^n \sigma_1^n} \exp\left\{-\frac{\sum(x_i-\mu)^2}{2\sigma_1^2}\right\}$

$L(\sigma_2) = \frac{1}{\left(\sqrt{2\pi}\right)^n \sigma_2^n} \exp\left\{-\frac{\sum(y_i-\mu)^2}{2\sigma_2^2}\right\}$

Let $\lambda = \frac{L(\sigma_2^2)}{L\left(\sigma_1^2\right)} = \left(\frac{\sigma_2}{\sigma_1}\right)^n \exp\left\{\frac{\sum(y_i-\mu)^2}{2\sigma_2^2} - \frac{\sum(x_i-\mu)^2}{2\sigma_1^2}\right\}$

Thus the likelihood ratio test has the rejection region

Reject H_0 if $\lambda \le k$

$$n\ln\left(\frac{\sigma_2}{\sigma_1}\right) + \frac{\sum(y_i-\mu)^2}{2\sigma_2^2} - \frac{\sum(x_i-\mu)^2}{2\sigma_1^2} \le \ln k$$

$$\frac{\sum(y_i-\mu)^2}{\sigma_2^2} - \frac{\sum(x_i-\mu)^2}{\sigma_1^2} \le 2\ln k - 2n\ln\left(\frac{\sigma_2}{\sigma_1}\right)$$

$$\hat{\sigma}_1^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

$$\hat{\sigma}_2^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

$$\frac{n\sum(y_i - \mu_2)^2}{\sum(y_i - \overline{y})^2} - \frac{n\sum(x_i - \mu_1)^2}{\sum(x_i - \overline{x})^2} \le 2\ln k - 2n\ln\left(\frac{\sigma_2}{\sigma_1}\right)$$

The rejection region is $\dfrac{n\sum(y_i - \mu_2)^2}{\sum(y_i - \overline{y})^2} - \dfrac{n\sum(x_i - \mu_1)^2}{\sum(x_i - \overline{x})^2} \le C$

7.3.5. $f(x) = \frac{1}{\theta}e^{-x/\theta}$ for $x > 0$

$L(\theta) = \frac{1}{\theta^n}\exp\left\{-\frac{\sum x_i}{\theta}\right\}$

$L(\theta_0) = \frac{1}{\theta_0^n}\exp\left\{-\frac{\sum x_i}{\theta_0}\right\}$

$L(\theta_1) = \frac{1}{\theta_1^n}\exp\left\{-\frac{\sum x_i}{\theta_1}\right\}$

$$\begin{aligned}\frac{L(\theta_0)}{L(\theta_1)} &= \frac{\theta_1^n}{\theta_0^n}\exp\left\{\frac{\sum x_i}{\theta_1} - \frac{\sum x_i}{\theta_0}\right\}\\ &= \left(\frac{\theta_1}{\theta_0}\right)^n \exp\left\{\frac{\sum x_i}{\theta_1} - \frac{\sum x_i}{\theta_0}\right\}\end{aligned}$$

We reject the null hypothesis if

$$n\ln\left(\frac{\theta_1}{\theta_0}\right) + \left(\frac{1}{\theta_1} - \frac{1}{\theta_0}\right)\sum x_i \le \ln k$$

$$\sum x_i \le \left\{\ln k - n\ln\left(\frac{\theta_1}{\theta_0}\right)\right\}\left(\frac{\theta_1\theta_0}{\theta_0 - \theta_1}\right)$$

Where we reject null hypothesis if $\sum x_i \le m_1$ or $\sum x_i \ge m_2$

Provided $m_1 = \left\{\ln k - n\ln\left(\frac{\theta_1}{\theta_0}\right)\right\}\left(\frac{\theta_1\theta_0}{\theta_0-\theta_1}\right)$ and $m_2 = \frac{1}{\ln k - n\ln\left(\frac{\theta_1}{\theta_0}\right)}\left(\frac{\theta_0-\theta_1}{\theta_1\theta_0}\right)$

EXERCISES 7.4

7.4.1. $n = 50, \alpha = 0.02, \bar{x} = 62, s = 8$

$H_0 : \mu \ge 64$

$H_1 : \mu < 64$

(a) The observed test statistic ($n \ge 30$) is

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{62 - 64}{8/\sqrt{50}} = \frac{-2}{1.13} = -1.769$$

(b) p-value $= p(z < -1.769) = p(z > 1.769) = 0.0384$

(c) Smallest $\alpha = 0.0384$
p-value > 0.02
We fail to reject the null hypothesis.

7.4.3. $H_0 : \mu = 0.45$
$H_1 : \mu < 0.45$
Here $\bar{x} = \frac{22.8}{78} = 0.2923$
$s^2 = \frac{20.5}{77} = 0.2666$
$s = 0.16309$

(a) Test statistic $z = \dfrac{\bar{x}}{s/\sqrt{n}} = \dfrac{0.2923 - 0.45}{0.16309/\sqrt{78}} = -0.85$

Rejection region is $\{z < -z_{0.01}\} = \{z < -2.33\}$

Since $z = -2.33 < -0.85$, the null hypothesis is rejected at $\alpha = 0.01$

p-value $= p\{z < -0.85\} = p\{z > 0.85\} = 0.1977$

(b) Here $z = -0.85$

$z_{\alpha/2} = z_{0.005} = 2.58$

Rejection region is $\{z < -z_{0.005}, z > z_{0.005}\}$, i.e. $\{z < -2.58, z > 2.58\}$

p-value $= \min\{z < -0.85, z > 0.85\} = 0.3954$

(c) Assumptions: even though population standard deviation is unknown, because of the large sample size, normal distribution is assumed.

7.4.5. $\bar{x} = \$58,800, n = 15, s = \$8,300$

(a) $P(\text{reject } H_0 | H_0 \text{ is true}) = 0$
Since the probability of rejecting null hypothesis equals to zero. Therefore, the null hypothesis is accepted.

(b) $\alpha = 0.01$

$H_0 : \mu = 55,648$

$H_1 : \mu > 55.648$

$t = \frac{\bar{x}-\mu}{s/\sqrt{n}} = \frac{58,800-55,648}{8,300/\sqrt{15}} = \frac{3152}{2143.05} = 1.47$

Rejection region is $\{t > t_{0.01,14}\}$

$t_{0.01,14} = 2.624$
Since $t = 2.642$ is greater than 1.47, we fail to reject the null hypothesis

7.4.7. $H_0 : p_0 = 0.3$
$H_1 : p_0 > 0.3$
$\hat{p} = \frac{550}{1500} = 0.366$
$z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.366-0.3}{\sqrt{\frac{0.3 \cdot 0.7}{1500}}} = \frac{0.066}{0.0118} = 5.593$
Rejection region is $\{z > z_{0.01}\}$
$z_{0.01} = 2.33$

i.e. $\{z > 2.33\}$

Yes, customer has preference over ivory color

7.4.9. **(a)** $\bar{x} = 42.9, s = 6.3674, \alpha = 0.1$

$H_0 : \mu = 44$

$H_1 : \mu \neq 44$

The data is normally distributed.

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{42.9 - 44}{6.3674/\sqrt{10}} = \frac{-1.1}{2.013} = -0.5644$$

Rejection region for z is $\{|z| < z_{0.05}\}$ where $z_{0.05} = 1.645$

$-z_{0.05} = -1.645 < -0.56444$

We fail to reject the null hypothesis

(b) 90% confidence interval for μ is $\left(\bar{x} - z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}, \bar{x} - z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}\right) = (42.9 - 1.645 \cdot 2.013, 42.9 + 1.645 \cdot 2.013) = (39.588, 46.21138)$

(c) From a, we can see that it is reasonable to take $\mu = 44$. The argument is supported by the confidence interval in b.

7.4.11. $\bar{x} = 13.7, s = 1.655, n = 20$

$H_0 : \mu = 14.6$

$H_1 : \mu < 14.6$

$t = \frac{13.7-14.6}{1.655/\sqrt{10}} = \frac{-0.9}{0.523} = -1.7198$

Rejection region is $\{z > z_{0.01}\}$

$-z_{0.01} = -2.33$

We reject the null hypothesis. Thus there is a statistical evidence to support this claim.

7.4.13. $\bar{x} = 32{,}277, s = 1{,}200, n = 100, \alpha = 0.05$

$H_0 : \mu = 30{,}692$

$H_1 : \mu > 30{,}692$

$z = \frac{32{,}277-30{,}692}{1{,}200/\sqrt{100}} = 13.20$

Rejection region is $\{z > z_{0.05}\}$

$z_{0.05} = 1.645$

Hence we reject the null hypothesis that the expenditure per consumer is increased from 1994 to 1995

7.4.15. $H_0 : \mu = 1.129$

$H_1 : \mu \neq 1.129$

$t = \frac{1.24-1.129}{0.01/\sqrt{24}} = \frac{0.111}{0.00204} = 54.41$

$t_{0.05,23} = 2.807$

Where $t = 54.41 > t_{0.05,23}$

Thus we reject the null hypothesis. That is price of gas is changed recently.

EXERCISES 7.5

7.5.1. $H_0 : \mu_1 - \mu_2 = 0$

$H_1 : \mu_1 - \mu_2 \neq 0$

$z = \frac{(\bar{y}_1 - \bar{y}_2) - (\bar{\mu}_1 - \bar{\mu}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{74-71}{\sqrt{\frac{81}{50} + \frac{100}{50}}} = \frac{3}{1.9026} = 1.5767$

Rejection region for z is $\left\{|z| > z_{0.025}\right\}$

$z_{0.025} = 1.96$

Since $z_{0.025} > 1.57$, we fail to reject the null hypothesis. To see the significant difference we need to have $\alpha = 0.0582$ level of significance.

7.5.3. $\bar{x}_1 = 58,550, \bar{x}_2 = 53,700, s_1 = 4000, s_2 = 3200, n_1 = 25, n_2 = 23$

$H_0 : \mu_1 - \mu_2 = 0$

$H_1 : \mu_1 - \mu_2 > 0$

$S_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1+n_2-2)} = \frac{24(4000)^2 + 22(3200)^2}{25+23-2} = 13245217$

$S_p = 3639.3979$

$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(58,550 - 53,700) - 0}{3639\sqrt{\frac{1}{25} + \frac{1}{23}}} = 4.61288$

Rejection region is $\{t > t_{0.05,46}\}$ i.e. $\{t > 1.679\}$

Since $1.679 < 4.61288$, we reject the null hypothesis. Thus implies that there exists significant evidence to show that the male's salary is higher than that of female.

7.5.5. $\bar{x}_1 = 105.9, \bar{x}_2 = 100.5, s_1^2 = 0.21, s_2^2 = 0.19, n_1 = 80, n_2 = 100$

$H_0 : \mu_1 - \mu_2 = 0$

$H_1 : \mu_1 - \mu_2 \neq 0$

Use $t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ and use two sided t-test.

7.5.7. $\bar{x}_1 = 7.65, \bar{x}_2 = 9.75, s_1 = 0.9312, s_2 = 0.852, n_1 = 10, n_2 = 10$

(a) $H_0 : \mu_1 - \mu_2 = 0$

$H_1 : \mu_1 - \mu_2 > 0$

$S_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1+n_2-2)} = \frac{9(0.9312)^2 + 9(0.852)^2}{18}$

$S_p = 0.892479$

$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(7.65 - 9.75) - 0}{0.89\sqrt{\frac{1}{10} + \frac{1}{10}}} = -5.276$

Rejection region is $\{t > t_{0.05,18}\}$ i.e. $\{t > 1.734\}$

Thus fail to reject the null hypothesis.

(b) $H_0 : \sigma_1^2 = \sigma_2^2$

$H_1 : \sigma_1^2 \neq \sigma_2^2$

Test statistic $F = \frac{s_1^2}{s_2^2} = \frac{(0.9312)^2}{(0.852)^2} = 1.1945$

From the F-table $F_{0.025(9,9)} = 4.03$

$F_{0.95(9,9)} = \frac{1}{4.03} = 0.248$

Rejection region is $F > 4.03$ and $F < 0.248$

Since the observed value of the statistic $1.1945 < 4.03$, we fail to reject the null hypothesis

(c) $H_0 : \mu_d = 0$

$H_1 : \mu_d > 0$

$\bar{d} = -2.1$

$S_d = 1.1670$

$t = \frac{\bar{d}-d_0}{S_d/\sqrt{n}} = \frac{-2.1}{1.670/\sqrt{10}} = -3.16$

From t-table $t_{0.05,9} = 1.833$

We reject the null hypothesis, this implies hat the down stream is less than the upstream.

7.5.9. (a) $\bar{x}_1 = 2.04, \bar{x}_2 = 3.55, s_1 = 0.551, s_2 = 0.6958, n_1 = 14, n_2 = 14$

$H_0 : \mu_1 - \mu_2 = 0$

$H_1 : \mu_1 - \mu_2 < 0$

Since the variances are equal and unknown

$S_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{(n_1+n_2-2)} = \frac{13(2.04)^2+13(3.55)^2}{26} = 8.38$

$S_p = \sqrt{8.38} = 2.89$

$t = \frac{(\bar{x}_1-\bar{x}_2)-0}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{(2.04-3.55)}{2.89\sqrt{\frac{1}{13}+\frac{1}{13}}} = -1.13321$

Rejection region is $\{t < -t_{0.05,13}\}$ i.e. $\{t < -1.77\}$

Here $-1.13321 > -1.77$, we fail to reject the null hypothesis.

(b) Test statistic $F = \frac{s_1^2}{s_2^2} = 0.6281$

From the F-table $F_{0.025(13,13)} = 3.11$

$F_{0.95(13,13)} = \frac{1}{3.11} = 0.321$

Rejection region is $F > 3.11$ and $F < 0.321$, we fail to reject the null hypothesis

(c) $H_0 : \mu_d = 0$

$H_1 : \mu_d < 0$

$\bar{d} = -1.5071$

$S_d = 0.7467$

$t = \frac{\bar{d}-d_0}{S_d/\sqrt{n}} = \frac{-1.5071}{0.7467/\sqrt{14}} = -7.55196$

From t-table $-t_{0.05,9} = -1.833$

$t = -7.5598 < t_{0.05,9} = -1.83$

We reject the null hypothesis.

7.5.11. $\bar{x}_1 = 106, \bar{x}_2 = 109, s_1 = 10, s_2 = 7, n_1 = 17, n_2 = 14$

Test statistic $F = \frac{s_1^2}{s_2^2} = \frac{100}{49} = 2.0408$

From the F-table $F_{0.01(10,7)} = 2.585$

$F_{0.90(10,7)} = \frac{1}{2.585} = 0.3868$

Rejection region is $F > 2.585$ and $F < 0.3868$

Since observed value of test statistic $= 2.0408 < 2.585$, we fail to reject the null hypothesis.

EXERCISES 7.6

7.6.1. $c = 3, r = 3$

$(c-1)(r-1) = 4$

$\chi^2_{0.05,4} = 9.48$

Hence the rejection region is $\varphi^2 > 9.48$

By using contingency table

$\varphi^2 = 43.86$

φ^2 falls in the rejection region at $\alpha = 0.05$, we reject the null hypothesis. That is collective bargaining is dependent on employee classification.

7.6.3. $O_1 = 12, O_2 = 14, O_3 = 78, O_4 = 40, O_5 = 6$

We now compute $\pi_i (i = 1, 2, 3, 4, 5)$ using continuity correction

$\pi_1 = p(x \leq 55)$

$\pi_2 = p\left(z \leq \frac{65.5-70}{4}\right)$

$\pi_3 = p\left(z \leq \frac{75.5-70}{4}\right)$

$\pi_4 = p\left(z \leq \frac{85.5-70}{4}\right)$

$\pi_5 = p\left(z \leq \frac{95.5-70}{4}\right)$

Taking above probability we need to find E_i and follow 6.6.5.

7.6.5. $E_1 = 950 \cdot 0.35, E_2 = 950 \cdot 0.15, E_3 = 950 \cdot 0.20, E_4 = 950 \cdot 0.30$

$O_1 = 950 \cdot 0.45, O_2 = 950 \cdot 0.25, O_3 = 950 \cdot 0.02, O_4 = 950 \cdot 0.28$

$\varphi^2 = 834.7183$

From Chi-Square table

$\chi^2_{0.05,3} = 7.81$

Thus $\varphi^2 = 834.71 > 7.81$, we reject the null hypothesis, at least one probabilities is different from hypothesized value.

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Chapter 8

Linear Regression Models

EXERCISES 8.2

8.2.1. **(a)** Proof see example 8.2.1

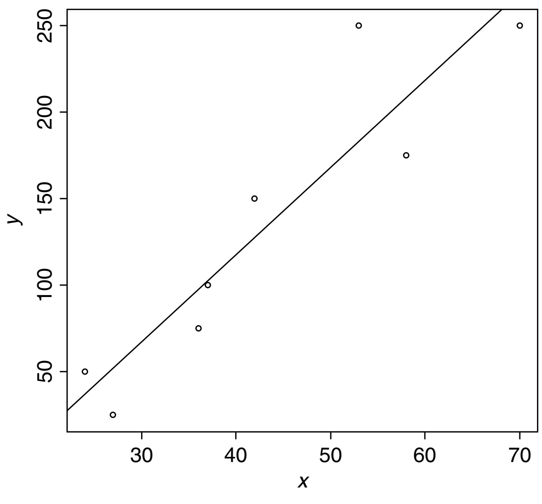
(b) $\frac{\text{SSE}}{\sigma^2}$ follows central Kai square with degree of freedom $n-2$

$E\left(\frac{\text{SSE}}{\sigma^2}\right) = n-2$ and σ^2 is a constant

Therefore $E(\text{SSE}) = (n-2)\sigma^2$

8.2.3. **(a)** Least-squares regression line is $\hat{y} = -84.1674 + 5.0384x$

(b)



8.2.5. **(a)** Check the proof in Derivation of $\hat{\beta}_0$ and $\hat{\beta}_1$.

(b) We know the line of best fit is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ and plug in the point $(\bar{x}, \bar{y})$ we get $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Complete the proof.

8.2.7.
$$y = \beta_1 x + \varepsilon$$
$$\text{SSE} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} [y_i - (\hat{\beta}_1 x_i)]^2$$

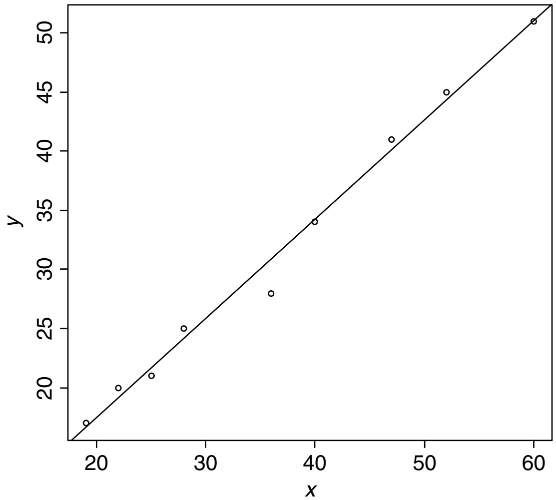
$$\frac{\partial(\text{SSE})}{\partial\beta_1} = \frac{\partial \sum_{i=1}^{n} [y_i - (\hat{\beta}_1 x_i)]^2}{\partial\beta_1} = -2\sum_{i=1}^{n}[y_i - (\hat{\beta}_1 x_i)]x_i$$

$$= -2\sum_{i=1}^{n}[x_i y_i - (\hat{\beta}_1 x_i^2)] = 0$$

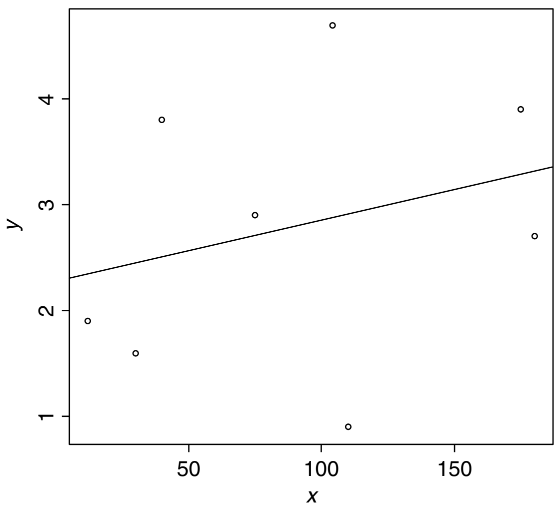
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i y_i)}{\sum_{i=1}^{n}(x_i^2)}$$

$$\hat{y} = -40.175 + .9984x$$

8.2.9. Least-squares regression line is $\hat{y} = .62875 + .83994x$



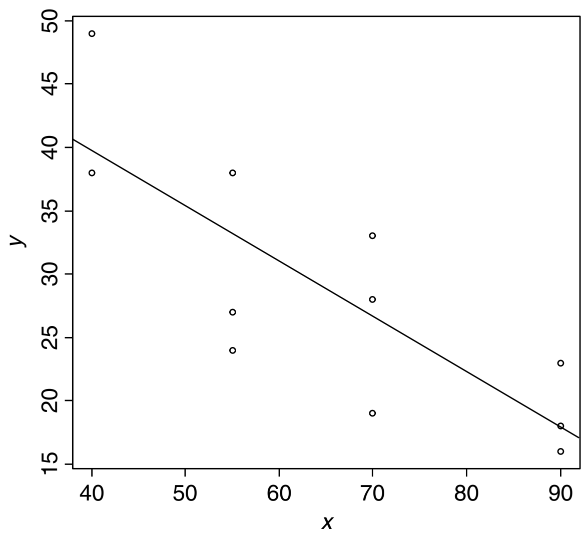
8.2.11. Least-squares regression line is $\hat{y} = 2.2752 + .00578x$



EXERCISES 8.3

8.3.1. **(a)** Least-squares regression line is $\hat{y} = 57.2383 - .4367x$

(b)



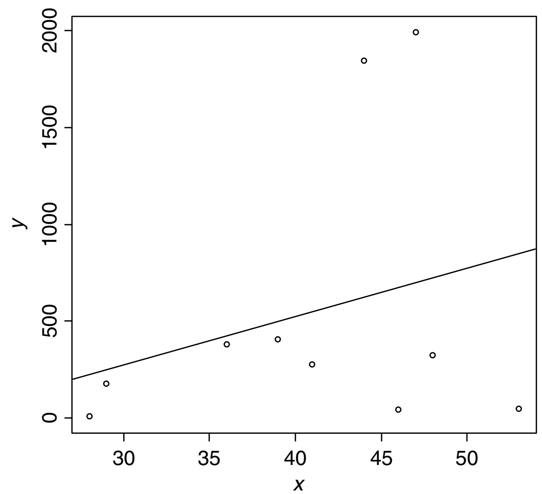
(c) The 95% confidence intervals for β_0 is $(40.5929, 73.8837)$
The 95% confidence intervals for β_1 is $(-.6806, -.1928)$

8.3.3. $\hat{\beta}_1$ and $\bar{y}$ are both normally distributed. In order to show these two are independent we just need to show the covariance of them is 0. (Property of normal distribution)

$$\begin{aligned}
\operatorname{Cov}\left(\hat{\beta}_1, \bar{y}\right) &= E\left(\hat{\beta}_1 \times \bar{y}\right) - E\left(\hat{\beta}_1\right) \times E\left(\bar{y}\right) \\
&= E\left(\frac{S_{xy}}{S_{xx}} \times \bar{y}\right) - \beta_1 \times E\left(\frac{1}{n}\sum_{i=1}^{n} y_i\right) \\
&= E\left(\frac{\sum_{i=1}^{n}(y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \times \bar{y}\right) - \beta_1 \times \frac{1}{n}\left(\sum_{i=1}^{n}(\beta_0 + \beta_1 \bar{x})\right) \\
&= \beta_1 \times (\beta_0 + \beta_1 \bar{x}) - \beta_1 \times \beta_0 - \beta_1^2 \bar{x} \\
&= 0
\end{aligned}$$

8.3.5. **(a)** Least-squares regression line is $\hat{y} = -474.76 + 24.95x$

(b)



(c) The 95% confidence intervals for β_0 is $(-3418.974, 2469.454)$

The 95% confidence intervals for β_1 is $(-45.429, 95.329)$

8.3.7. By assumption and from normal equation we know $\sum_{i=1}^{n} \varepsilon_i = 0$ and $\sum_{i=1}^{n} \varepsilon_i x_i = 0$

$$\begin{aligned}
\sum_{i=1}^{n} (y_i - \bar{y})^2 &= \sum_{i=1}^{n} (y_i - \hat{y} + \hat{y} - \bar{y})^2 \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2 + 2\sum_{i=1}^{n} (y_i - \hat{y})(\hat{y} - \bar{y}) \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2 + 2\sum_{i=1}^{n} (\varepsilon_i)(\hat{y} - \bar{y}) \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2 + 2\sum_{i=1}^{n} (\varepsilon_i)(\hat{y}) - 2\sum_{i=1}^{n} (\varepsilon_i)(\bar{y}) \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2 + 2\sum_{i=1}^{n} (\varepsilon_i)(\hat{\beta}_0 + \hat{\beta}_1 x_i) - 2\sum_{i=1}^{n} (\varepsilon_i)(\bar{y}) \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2 + 2\hat{\beta}_0 \sum_{i=1}^{n} (\varepsilon_i) + 2\hat{\beta}_1 \sum_{i=1}^{n} (\varepsilon_i)(x_i) - 2(\bar{y})\sum_{i=1}^{n} (\varepsilon_i) \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2 + 2\hat{\beta}_0 \times 0 + 2\hat{\beta}_1 \times 0 - 2(\bar{y}) \times 0 \\
&= \sum_{i=1}^{n} (y_i - \hat{y})^2 + \sum_{i=1}^{n} (\hat{y} - \bar{y})^2
\end{aligned}$$

EXERCISES 8.4

8.4.1. The 95% prediction interval for $x = 92$ is from 83.6195 to 111.7445
We can conclude with 95% confidence that the true value of Y at the point $x = 92$ will be somewhere between 83.6195 and 111.7445

8.4.3. The 95% prediction interval for $x = 12$ is from 84.20125 to 113.2464

8.4.5. The 95% prediction interval for $x = 85$ is from 128.7849 to 191.9912
We can conclude with 95% confidence that the true value of Y at the point $x = 85$ will be somewhere between 128.7849 and 191.9912
The assumption is the linear regression still valid for beyond our domain and therefore our y do make sense.

EXERCISES 8.5

8.5.1. **(a)** At 95% confidence level, we got the z is 1.3853 which is less than critical value. Therefore we do not reject Ho and it means X and Y are independent.
(b) P-value is 0.08297713.
(c) The assumption is (x, y) follows the bivariate normal distribution and this test procedure is approximate.

8.5.3. $E(\gamma) = E\left(\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}\right) \neq \frac{\sigma_{xy}}{\sqrt{\sigma_{xx}\sigma_{yy}}} = \rho$
Therefore γ is not an unbiased estimator of the population coefficient ρ.

8.5.5. **(a)** At 95% confidence level, we got the z is 1.237125 which is less than critical value. Therefore we do not reject Ho and it means X and Y are independent.
(b) P-value is 0.108.
(c) $\hat{y} = 77.87 + .624x$
(d) The usefulness of this model is we can use this model to make prediction.
(e) The assumption is (x, y) follows the bivariate normal distribution and this test procedure is approximate.

EXERCISES 8.6

8.6.1. **(a)** $\hat{y} = X\hat{\beta}$ where $\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \hat{y}_4 \end{bmatrix}_{4\times 1}$, $X = \begin{bmatrix} 1 & X_1 & X_2 \end{bmatrix}_{4\times 3}$ and $\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \hat{\beta}_3 \end{bmatrix}_{3\times 1}$

(b) $X^T X = \begin{bmatrix} 4 & 9 & 11 \\ 9 & 23 & 24 \\ 11 & 24 & 39 \end{bmatrix}$

$$(X^T X)^{-1} = \begin{bmatrix} 3.41489 & -.92553 & -.3936 \\ -.92553 & .37234 & .0319 \\ -.3936 & .0319 & .117 \end{bmatrix}$$

$$(X^T Y) = \begin{bmatrix} 18 \\ 41 \\ 47 \end{bmatrix}$$

(c) $\hat{\beta} = \begin{bmatrix} 5.02128 \\ .10638 \\ -.2766 \end{bmatrix}_{3\times 1}$

(d) The estimation of error variance is 2.14286.

8.6.3. $$\hat{y} = -84.1674 + 5.0384x$$

$$X^T X = \begin{bmatrix} 8 & 347 \\ 347 & 16807 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 1.19648 & -.0247 \\ -.0247 & .0005695 \end{bmatrix}$$

$$(X^T Y) = \begin{bmatrix} 1075 \\ 55475 \end{bmatrix}$$

Chapter 9

Design of Experiments

EXERCISES 9.2

9.2.1. Response is amount of fat was absorbed by has-brown potatoes
Factors are frying durations and different type of fats
Factor types are frying durations which quantitative and continuous and different type of fats which is qualitative and discrete.

Treatments

There are 16 treatments

2 min with animal fat I, 2 min with animal fat II, 2 min with vegetable fat I and 2 min with vegetable fat II

3 min with animal fat I, 3 min with animal fat II, 3 min with vegetable fat I and 3 min with vegetable fat II

4 min with animal fat I, 4 min with animal fat II, 4 min with vegetable fat I and 4 min with vegetable fat II

5 min with animal fat I, 5 min with animal fat II, 5 min with vegetable fat I and 5 min with vegetable fat II

9.2.3. Procedure for random assignment

1. Number the experimental units from 1 to 30.
2. Use a random number table or a statistical software to get a list of numbers that is a random permutation of the numbers 1 to 30.
3. Give treatment 1 to the experimental units having the first 10 numbers in the list. Treatment 2 will be given to the next 10 numbers in the list, and so on, give treatment 3 to the last 10 units in the list.

Here response is rose bushes and factor is different fertilizers. 24, 12, 30, 21, 8, 3, 20, 1, 11,18, 13, 15, 28, 5, 25, 29, 4, 10, 14, 19, 26, 9, 2, 6, 22, 16, 23, 7, 27,17

Brand	Subject									
A	24	12	30	21	8	3	20	1	11	18
B	13	15	28	5	25	29	4	10	14	19
C	26	9	2	6	22	16	23	7	27	17

9.2.5. Procedure for a randomized complete block design with 3 replications

1. Group the experimental units into 3 groups (called blocks), each containing 3*3 homogeneous experimental units.
2. In group 1, number the experimental units from 1 to 9 and generate a list of numbers which are random permutation of the numbers 1 to 9.
3. In group 1, assign treatment 1 to the experimental units having numbers given by the first 3 numbers in the list. Assign treatment 2 to the experiments having next 3 numbers in the list, and so on until treatment 3 receives 3 experimental units.
4. Repeat steps 2 and 3 for the remaining blocks of experimental units.

G	R	J
3(A)	5(A)	1(A)
1(A)	4(A)	2(A)
2(A)	3(A)	4(A)
6(B)	9(B)	7(B)
8(B)	1(B)	8(B)
7(B)	6(B)	5(B)
9(C)	8(C)	6(C)
5(C)	7(C)	3(C)
4(C)	2(C)	9(C)

9.2.7. 19, 37, 52, 42, 13, 34, 56, 48, 44, 43, 24, 12, 5, 32, 40, 23, 41, 11, 10, 6, 30, 26, 18, 8, 2, 29, 21, 36, 1, 54, 20, 39, 33, 27, 49, 16, 51, 15, 28, 47, 53, 35, 31, 3, 38, 25, 17, 55, 4, 50, 14, 22, 9, 46, 45, 7

Brand	Subject													
A	19	37	52	42	13	34	56	48	44	43	24	12	5	32
B	40	23	41	11	10	6	30	26	18	8	2	29	21	36
C	1	54	20	39	33	27	49	16	51	15	28	47	53	35
D	31	3	38	25	17	55	4	50	14	22	9	46	45	7

9.2.9. Start

	New material				
Days	**1**	**3**	**2**	**4**	**5**
2	A	B	C	D	E
3	B	C	D	E	A
5	C	D	E	A	B
1	D	E	A	B	C
4	E	A	B	C	D

Then

	New material				
Days	**1**	**3**	**2**	**4**	**5**
1	D	E	A	B	C
2	A	B	C	D	E
3	B	C	D	E	A
4	E	A	B	C	D
5	C	D	E	A	B

The final is

	New material				
Days	**1**	**2**	**3**	**4**	**5**
1	D	A	E	B	C
2	A	C	B	D	E
3	B	D	C	E	A
4	E	B	A	C	D
5	C	E	D	A	B

9.2.11.

	Grid			
Grid	**1**	**2**	**3**	**4**
1	D	A	B	C
2	C	B	D	A
3	A	D	C	B
4	B	C	A	D

EXERCISES 9.3

9.3.1. One factor at a time experiment to predict average amount of profit
Assume we fix proportion at 40% then if we increase the quality from ordinary to fine we get average profit decrease from 25,000 to 10,000. If we fix proportion at 60% then if we increase the quality from ordinary to fine we get average profit decrease from 9500 to 3000.

Then if we change the setting from 40% and ordinary to 60% and fine then we get from 25,000 to 3000.

9.3.3. In fractional factorial experiment, only a fraction of the possible treatments are actually used in the experiment. A full factorial design is most ideal design through which we could obtain information on all main effects and interactions. Due to prohibitive size of the experiments, such designs are not practical to run. The total number of distinct treatments will be $2 \times 2 = 4$.

Fractional factorial experiments are used in which trials are conducted at only a well-balanced subset of the possible combinations of levels of factors. This allows the experimenter to obtain information about all main effects and interactions while maintaining the size of the experiment manageable.

The experiment is carried out in a single systematic effort.

EXERCISES 9.4

9.4.1.

$$\sigma_1 = \sqrt{4} = 2 \text{ and } \sigma_2 = \sqrt{9} = 3$$

$$n_1 = \frac{2}{2+3}100 = 40 \text{ and } n_2 = \frac{3}{2+3}100 = 60$$

EXERCISES 9.5

9.5.1.

$$\bar{L} = k[(\bar{X} - T)^2 + S^2] = k[(14.15 - 14.5)^2 + .42^2] = .2946k.$$

Chapter 10

Analysis of Variance

EXERCISES 10.2

10.2.1. (a) We need to test $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$

From the random sample, we obtain the following needed estimates $n_1 = n_2 = y$, $\overline{y}_1 = 1.888889, \overline{y}_2 = 2.777778, \sum_i \sum_j y_{ij}^2 = 120, \sum_i \sum_j y_{ij} = 42$, Total SS $=\sum_{i=1}^{2} \sum_{j=1}^{n_i} \left(y_{ij} - \overline{y}\right)^2 = 22$, SSE $=\sum_{i=1}^{2} \sum_{j=1}^{n_i} \left(y_{ij} - \overline{y}\right)^2 = 18.4444$, SST $= 3.55556$

Where Total SS = SSE+SST, then MST $= \frac{\text{SST}}{1} = 3.55556$, MSE $= \frac{\text{SSE}}{n_1+n_2-2} = 1.152778$, and $F = \frac{\text{MST}}{\text{MSE}} = 3.084337$

With $\alpha = 0.05$, $F_{\alpha,(1,n_1+n_2-2)} = 4.49$

Since 3.0843 is not greater than 4.49, H_0 is not rejected.

There is not enough evidence to indicate that the means differs for the tho populations.

(b) $S^2 = S_p^2 = \text{MSE} = 1.152778$, $\overline{y}_1 = 1.888889$, $\overline{y}_2 = 2.777778$
Then, the t-statistic is

$$T = \frac{\overline{y}_1 - \overline{y}_2}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = -1.7562$$

Now, $t_{0.025,14} = 2.12$, and the rejections region is $\{|t| > 2.12\}$

Since -1.7562 is not less than -2.12, H_0 is not rejected, which implies that there is no significant difference between the mean for the two populations. $t^2 = F$, implying that in the two sample case, t-test and F-test lead to the same result.

10.2.3. (a) At $\alpha = 0.01$, we need to test $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$
We need the estimates $\overline{y}_1 = 48.84615$, $\overline{y}_2 = 45.91667$, $\sum_i \sum_j y_{ij}^2 = 56582$, $\sum_i \sum_j y_{ij} = 1186$, Total SS $=\sum_{i=1}^{2} \sum_{j=1}^{n_i} \left(y_{ij} - \overline{y}\right)^2 = 318.16$, SSE $=\sum_{i=1}^{2} \sum_{j=1}^{n_i} \left(y_{ij} - \overline{y}\right)^2 = 264.609$, SST $= 53.551$, where Total SS $=$ SSE $+$ SST, then MST $= \frac{\text{SST}}{1} = 53551$, MSE $= \frac{\text{SSE}}{n_1+n_2-2} = 11.50474$, and $F = 4.654693$. At $\alpha = 0.01$, $F_{\alpha,(1,n_1+n_2-2)} = 7.881134$, then $F_{0.01}\,(1, 23) = 7.88$

(b) Assumptions: The samples are assumed to be independent from the Normal population with respective means μ_1, μ_2 and equal but unknown variances.

(c) $S^2 = S_p^2 = \text{MSE} = 11.50474$, $\overline{y}_1 = 48.84615$, $\overline{y}_2 = 4591667$. Then, the t-statistic is:

$$t = \frac{\overline{y}_1 - \overline{y}_2}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = 2.1574$$

Now, $t_{0.005,23} = 2.807$, and the rejection region is $\{|t| > 2.807\}$. Since 2.1574 is not greater than 2.807, H_0 is not rejected, which implies that there is no significant difference between the mean time to relief for the two populations, and $t^2 = F$ implies that in the two sample case, t-test and F =test lead to the same result

10.2.5. Let X_i, with $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, 3, \ldots, n_1$, and let Y_j, with $Y_j \sim N(\mu, \sigma^2)$ for $j = 1, 2, 3, \ldots, n_2$, be two set of independent random variables. To test $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 \neq \mu_2$ we reject H_0 when $\left|\frac{\overline{X}-\overline{Y}}{\sqrt{\left(\frac{1}{n_1}+\frac{1}{n_2}\right)S_p^2}}\right| > t_{n_1+n_2-1;\alpha/2}$. Now, for ANOVA, with $\overline{\mu} = \frac{\sum_{i=1}^{n_1} x_i + \sum_{i=1}^{n_2} y_i}{n_1+n_2}$, we have

$$F = \frac{\text{MST}}{\text{MSE}} = \frac{\frac{\sum_{i=1}^{n_1}(\overline{x}-\overline{\mu})^2+\sum_{i=1}^{n_2}(\overline{y}-\overline{\mu})^2}{2-1}}{\frac{\sum_{i=1}^{n_1}(x_i-\overline{x})^2+\sum_{i=1}^{n_2}(y_i-\overline{y})^2}{n_1+n_2-2}} = \frac{n_1(\overline{x}^2 - 2\overline{x}\cdot\overline{\mu} + \overline{\mu}^2) + n_2(\overline{y}^2 - 2\overline{y}\cdot\overline{\mu} + \overline{\mu}^2)}{S_p^2}$$

$$= \frac{n_1\overline{x}^2 + n_2\overline{y}^2 - (n_1+n_2)\overline{\mu}^2}{S_p^2}$$

$$= \frac{\frac{n_1 n_2}{n_1+n_2}[\overline{x}^2 - 2\overline{x}\cdot\overline{y} + \overline{y}^2]}{S_p^2}, \quad \text{since } \overline{\mu} = \frac{n_1\overline{x} + n_2\overline{y}}{n_1+n_2}$$

$$= \frac{(\overline{x} - \overline{y})^2}{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)S_p^2}$$

Therefore $F = \frac{(\overline{X}-\overline{Y})^2}{\left(\frac{1}{n_1}+\frac{1}{n_2}\right)S_p^2}$. Then, we reject H_0 if $\frac{(\overline{X}-\overline{Y})^2}{\left(\frac{1}{n_1}+\frac{1}{n_2}\right)S_p^2} > F_{1,n_1+n_2-2,\alpha}$.

Since $(t_{n_1+n_2-2,\alpha/2})^2 \equiv F_{1,n_1+n_2-2,\alpha}$ and $\frac{(\overline{x}-\overline{y})^2}{\left(\frac{1}{n_1}+\frac{1}{n_2}\right)S_p^2} > k \Leftrightarrow \left|\frac{\overline{X}-\overline{Y}}{\sqrt{\left(\frac{1}{n_1}+\frac{1}{n_2}\right)S_p^2}}\right| > k'$ for appropriate values k and k', the probability for this events are the same. Hence, the two sample t-test and the analysis of variance are equivalent for testing $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 \neq \mu_2$.

Note: In the text, x_i is y_{1i}, $\overline{x}$ is $\overline{y}_1$, y_i is y_{2i}, and $\overline{y}$ is $\overline{y}_2$.

EXERCISES 10.3

10.3.1. **(a)** Assuming that the samples are from populations which are normally distributed with equals variances and means μ_1, μ_2, μ_3. In our case, $n_1 = n_2 = n_3 = 4$, $k = 3$, $N = \sum_{i=1}^{k} n_i = 12, T_i = \sum_{j=1}^{n_i} y_{ij}, \overline{T}_i = \frac{t_i}{n_i}, T_1 = 1488, T_2 = 1704, T_3 = 1434$, $\overline{T}_1 = 372, \overline{T}_2 = 426, \overline{T}_3 = 358.3, \overline{y} = \frac{\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}}{N} = 385.5, \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 = 1835388$, $\text{CM} = N\overline{y}^2 = 1783323$, Total SS $= \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \overline{y}\right)^2 = 52065$, or Total SS $= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \text{CM} = 52065$, $\text{SSE} = \sum_{i-1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \overline{T}_i\right)^2 = 41859$, $\text{SST} = \sum_{i=1}^{k} n_i\left(\overline{T}_i - \overline{y}\right)^2 = \sum_{i=1}^{k}\frac{T_i^2}{n_i} - \text{CM} = 10206, S^2 = \text{MSE} = \frac{\text{SSE}}{n_1+n_2+\cdots+n_k-k} = \frac{\text{SSE}}{N-k} = 4651, \text{MST} = \frac{\text{SST}}{k-1} = 5103, F = \frac{\text{MST}}{\text{MSE}} = 1.0972$. At $\alpha = 0.05 F_{\alpha,(k-1,N-k)} = 4.2565$

Therefore, the ANOVA table is

ANOVA Table					
Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F-Statistic	p-Value
Treatments	2	10206	5103	1.0972	0.37462
Error	9	41859	4651		
Total	11	52065			

From the table, since the p-value is more than 0.05, we reject at $\alpha = 0.05$ the null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3$

(b) Letting H_0: The mean auto insurance premium paid per six months by all drivers insured for each of these companies is the same. Based on the data, there is evidence to suggest that the mean auto insurance premium pay per six months by all drivers insured for each of these companies is the same.

10.3.3. $n_1 = n_2 = \cdots = n_k = n'$

$$\sum_{j=1}^{n'} (y_{ij} - \overline{y})^2 = \sum_{j=1}^{n'} \left[(y_{ij} - \overline{T}_i) + (\overline{T}_i - \overline{y})\right]^2 = \sum_{j=1}^{n'} (y_{ij} - \overline{T}_i)^2 + n'(\overline{T}_i - \overline{y})^2,$$

because $\sum_{j=1}^{n'} (y_{ij} - \overline{T}_i) = \sum_{j=1}^{n'} y_{ij} - n'\overline{T}_i = \sum_{j=1}^{n_i} y_{ij} - n_i\overline{T}_i = \sum_{j=1}^{n_i} y_{ij} - \sum_{j=1}^{n_i} y_{ij} = 0$.
Therefore, $\sum_{i=1}^{k}\sum_{j=1}^{n'} (y_{ij} - \overline{y})^2 = \sum_{i=1}^{k}\sum_{j=1}^{n'} (y_{ij} - \overline{T}_i)^2 + n'\sum_{i=1}^{k}\left(\overline{T}_i - \overline{y}\right)^2$

10.3.5. **(a)** From exercise 10.3.4 we know that $\text{SST} = \sum_{i=1}^{k} n_i\left(\overline{T}_i - \overline{y}\right)^2$ (where T stand for "Treatment"), and SS Total $= \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \overline{y}\right)^2$. Then,

$$\text{SSE} = \text{SS Total} - \text{SST} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \overline{y}\right)^2 - \sum_{i=1}^{k} n_i\left(\overline{T}_i - \overline{y}\right)^2$$

$$= \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij} - \overline{y}\right)^2 - \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(\overline{T}_i - \overline{y}\right)^2$$

and

$$\sum_{j=1}^{n_i}(y_{ij}-\overline{y})^2=\sum_{j=1}^{n_i}[(y_{ij}-\overline{T}_i)+(\overline{T}_i-\overline{y})]^2$$

$$=\sum_{j=1}^{n_i}(y_{ij}-\overline{T}_i)^2+\sum_{j=1}^{n_i}(\overline{T}_i-\overline{y})^2$$

Since $\sum_{j=1}^{n_i}(y_{ij}-\overline{T}_i)=\sum_{j=1}^{n_i}y_{ij}-n_i\overline{T}_i=\sum_{j=1}^{n_i}y_{ij}-\sum_{j=1}^{n_i}y_{ij}=0$
Then $\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\overline{y})^2=\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\overline{T}_i)^2+\sum_{i=1}^{k}\sum_{j=1}^{n_i}(\overline{T}_i-\overline{y})^2$
Therefore, SSE $=\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\overline{T}_i)^2$

(b) Since $y_{ij}\sim N(\mu,\sigma^2)$, $\frac{\sum_{j=1}^{n_i}(y_{ij}-\overline{T}_i)^2}{\sigma^2}\sim\chi^2_{n_i-1}$, and since they are independent, $\frac{\text{SSE}}{\sigma^2}=\sum_{i=1}^{k}\sum_{j=1}^{n_i}\frac{(y_{ij}-\overline{T}_i)^2}{\sigma^2}$ follows a chi-square distribution with $\sum_{i=1}^{k}(n_i-1)$, or $N-k$, degrees of freedom.

10.3.7. $n_1=n_4=5$, $n_2=n_3=6$, $k=4$, $N=\sum_{i=1}^{k}n_i=22$, $T_i=\sum_{j=1}^{n_i}y_{ij}$, $\overline{T}_i=\frac{t_i}{n_i}$, $T_1=92$, $T_2=69$, $T_3=75$, $T_4=94$, $\overline{T}_1=18.4$, $\overline{T}_2=11.5$, $\overline{T}_3=12.5$, $\overline{T}_4=18.8$, $\overline{y}=\frac{\sum_{i=1}^{k}\sum_{j=1}^{n_i}y_{ij}}{N}=15$, $\sum_{i=1}^{k}\sum_{j=1}^{n_i}y_{ij}^2=5338$, $\text{CM}=N\overline{y}^2=4950$, Total SS $=\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\overline{y})^2=388$, or Total SS $=\sum_{i=1}^{k}\sum_{j=1}^{n_i}y_{ij}^2-\text{CM}=388$, SSE $=\sum_{i-1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\overline{T}_i)^2=147$, SST $=\sum_{i=1}^{k}n_i(\overline{T}_i-\overline{y})^2=\sum_{i=1}^{k}\frac{T_i^2}{n_i}-\text{CM}=241$, $S^2=\text{MSE}=\frac{\text{SSE}}{n_1+n_2+\cdots+n_k-k}=\frac{\text{SSE}}{N-k}=8.16667$, $\text{MST}=\frac{\text{SST}}{k-1}=80.3333$, $F=\frac{\text{MST}}{\text{MSE}}=9.8367$

Therefore, the ANOVA table is

(a)

ANOVA Table					
Source of Variation	Degrees of Freedom	Sum of Squares	Mean Square	F-Statistic	p-Value
Treatments	3	241	80.3333	9.8367	0.00046
Error	18	147	8.166667		
Total	21	388			

Assumptions: The samples are randomly selected from the 4 populations in an independent manner. The populations are normally distributed with equal variances σ^2 and means μ_1,μ_2,μ_3,μ_4

(b) Since F is greater than critical value at $\alpha=0.05$, there is sufficient evidence to indicate a difference between the mean number of customers served by the 4 employees.

10.3.9. Assumptions: The samples are normally selected from the population in an independent manner. The populations are assumed to be normally distributed with common variances. $n_1=n_2=n_3=n_4=5$, $k=4$, $N=\sum_{i=1}^{k}n_i=20$, $T_i=\sum_{j=1}^{n_i}y_{ij}$, $\overline{T}_i=\frac{t_i}{n_i}$, $T_1=34.3$, $T_2=39.6$, $T_3=45.9$, $T_4=34.2$, $\overline{T}_1=6.86$, $\overline{T}_2=7.92$, $\overline{T}_3=9.18$, $\overline{T}_4=6.84$, CM $= N\overline{y}^2 = 1185.8$, Total SS $= \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(y_{ij}-\overline{y}\right)^2 = 23.78$, or Total SS $= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \text{CM} = 23.78$, SSE $= \sum_{i-1}^{k}\sum_{j=1}^{n_i}\left(y_{ij}-\overline{T}_i\right)^2 = 5.36$, SST $= \sum_{i=1}^{k} n_i\left(\overline{T}_i-\overline{y}\right)^2 = \sum_{i=1}^{k}\frac{T_i^2}{n_i} - \text{CM} = 18.42$, $S^2 = \text{MSE} = \frac{\text{SSE}}{n_i+n_2+\cdots+n_k-k} = \frac{\text{SSE}}{N-k} = 0.335$, MST $= \frac{\text{SST}}{k-1} = 6.14$, $F = \frac{\text{MST}}{\text{MSE}} = 18.3284$

At $\alpha = 0.01\, F_{\alpha,(k-1,N-k)} = 5.2922$

Since $F > 5.2922$, the sample evidence supports the alternative hypothesis that the true rental and homeowner vacancy rates by area indeed different for all five years at 0.01 l of significance level.

10.3.11. $n_1 = n_2 = n_3 = 6, k = 3, N = \sum_{i=1}^{k} n_i = 18, T_i = \sum_{j=1}^{n_i} y_{ij}, \overline{T}_i = \frac{t_i}{n_i}, T_1 = 1273, T_2 = 1275, T_3 = 1257, \overline{T}_1 = 212.16667, \overline{T}_2 = 212.5, \overline{T}_3 = 209.5$ CM $= N\overline{y}^2 = 804334.7222$, Total SS $= \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij}-\overline{y})^2 = 5994.27778$, or Total SS $= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \text{CM} = 5994.27778$, SSE $= \sum_{i-1}^{k}\sum_{j=1}^{n_i}\left(y_{ij}-\overline{T}_i\right)^2 = 5961.8333$, SST $= \sum_{i=1}^{k} n_i\left(\overline{T}_i-\overline{y}\right)^2 = \sum_{i=1}^{k}\frac{T_i^2}{n_i} - \text{CM} = 32.4444$, $S^2 = \text{MSE} = \frac{\text{SSE}}{n_i+n_2+\cdots+n_k-k} = \frac{\text{SSE}}{N-k} = 397.456$, MST $= \frac{\text{SST}}{k-1} = 16.2222$, $F = \frac{\text{MST}}{\text{MSE}} = 0.0408$

At $\alpha = 0.01\ F_{\alpha,(k-1,N-k)} = 6.3589$

Since $F < 6.3589$, based on the data there is not enough evidence to support the alternative hypothesis that the true mean cholesterol levels for all races in the United States during 1978–1980 are different at 0.01 of significance level.

EXERCISES 10.4

10.4.1.

$$y_{ij} - \overline{y} = \left(y_{ij} - \overline{T}_i - \overline{B}_j + \overline{y}\right) + \left(\overline{T}_i - \overline{y}\right) + \left(\overline{B}_j - \overline{y}\right)$$

Then

$$\begin{aligned}\left(y_{ij}-\overline{y}\right)^2 &= \left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)^2 + \left(\overline{T}_i-\overline{y}\right)^2 + \left(\overline{B}_j-\overline{y}\right)^2 + 2\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\left(\overline{T}_i-\overline{y}\right)\\ &\quad + 2\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\left(\overline{B}_j-\overline{y}\right) + 2\left(\overline{T}_i-\overline{y}\right)\left(\overline{B}_j-\overline{y}\right)\end{aligned}$$

Then

$$\begin{aligned}\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{i_j}-\overline{y}\right)^2 &= \sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)^2 + \sum_{j=1}^{b}\sum_{i=1}^{k}\left(\overline{T}_i-\overline{y}\right)^2 + \sum_{j=1}^{b}\sum_{i=1}^{k}\left(\overline{B}_j-\overline{y}\right)^2\\ &\quad + 2\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\left(\overline{T}_i-\overline{y}\right)\\ &\quad + 2\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\left(\overline{B}_j-\overline{y}\right)\\ &\quad + 2\sum_{j=1}^{b}\sum_{i=1}^{k}\left(\overline{T}_i-\overline{y}\right)\left(\overline{B}_j-\overline{y}\right)\end{aligned}$$

Now, $\sum_{j=1}^{b}\left(y_{ij}-\overline{T}_i\right)=\sum_{j=1}^{b}y_{ij}-b\overline{T}_i=0$, since $b\overline{T}_i = \sum_{j=1}^{b}y_{ij}$, and $b\overline{y} = b\cdot\frac{1}{n}$ $\sum_{i=1}^{k}\sum_{j=1}^{b}y_{ij}=\frac{b}{bk}\sum_{i=1}^{k}\sum_{j=1}^{b}y_{ij}=\frac{1}{k}\sum_{j=1}^{b}\sum_{i=1}^{k}y_{ij}=\sum_{j=1}^{b}\left[\frac{1}{k}\sum_{i=1}^{k}y_{ij}\right]=\sum_{j=1}^{b}\overline{B}_j$

Then, $\sum_{j=1}^{b}\left(\overline{B}_j-\overline{y}\right)=\sum_{j=1}^{b}\overline{B}_j-b\overline{y}=0$, and $\sum_{j=1}^{b}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)=\sum_{j=1}^{b}\left(y_{ij}-\overline{T}_i\right)$ $-\sum_{j=1}^{b}\left(\overline{B}_j-\overline{y}\right)=0$

Then, $\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\left(\overline{T}_i-\overline{y}\right)=\sum_{i=1}^{k}\left[\left(\overline{T}_i-\overline{y}\right)\sum_{j=1}^{b}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\right]$ $=0$

and $\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\left(\overline{B}_j-\overline{y}\right)=\sum_{j=1}^{k}\left[\left(\overline{B}_j-\overline{y}\right)\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)\right]$ $=0$

and $\sum_{j=1}^{b}\sum_{i=1}^{k}\left(\overline{T}_i-\overline{y}\right)\left(\overline{B}_j-\overline{y}\right)=\sum_{i=1}^{k}\left[\left(\overline{T}_i-\overline{y}\right)\sum_{j=1}^{b}\left(\overline{B}_j-\overline{y}\right)\right]=0.$

Therefore, $\sum_{j=1}^{b}\sum_{j=1}^{k}\left(y_{ij}-\overline{y}\right)^2=\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)^2+\sum_{j=1}^{b}\sum_{i=1}^{k}\left(\overline{T}_i-\overline{y}\right)^2$ $+\sum_{i=1}^{k}\sum_{j=1}^{b}\left(\overline{B}_j-\overline{y}\right)^2=\sum_{j=1}^{b}\sum_{i=1}^{k}\left(y_{ij}-\overline{T}_i-\overline{B}_j+\overline{y}\right)^2+b\sum_{i=1}^{k}\left(\overline{T}_i-\overline{y}\right)^2+k\sum_{j=1}^{b}$ $\left(\overline{B}_j-\overline{y}\right)^2$

10.4.3. $W=\sum_{i=1}^{k}\sum_{j=1}^{b}(y_{ij}-\mu-\tau_i-\beta_j)^2$, then $\frac{\partial W}{\partial\mu}=-2\sum_{i=1}^{k}\sum_{j=1}^{b}(y_{ij}-\mu-\tau_i-\beta_j)$

If $\frac{\partial W}{\partial\mu}=0$, then $\sum_{i=1}^{k}\sum_{j=1}^{b}(y_{ij}-\tau_i-\beta_j)-\sum_{i=1}^{k}\sum_{j=1}^{b}\mu=0$, and since by the restriction $\sum_{j=1}^{b}\beta_j=\sum_{i=1}^{k}\tau_i=0$, the solution is given by $\hat{\mu}=\overline{y}$

Now, for any fixed $i\,\frac{\partial W}{\partial\tau_i}=-\sum_{j=1}^{b}(y_{ij}-\mu-\tau_i-\beta_j)$

if $\frac{\partial W}{\partial\tau_i}=0$, then $\frac{\partial W}{\partial\mu}=\sum_{j=1}^{b}(y_{ij}-\mu-\tau_i-\beta_j)=0$. Then, since $\sum_{j=1}^{b}\beta_j=0$ then $\hat{\tau}_i=\overline{T}_i-\hat{\mu}$, for any $i=1,2,\ldots,k$; i.e., $\hat{\tau}_i=\overline{T}_i-\overline{y}$

For any fixed j, $\frac{\partial W}{\partial\beta_i}=-2\sum_{i=1}^{k}(y_{ij}-\mu-\tau_i-\beta_j)$. If $\frac{\partial W}{\partial\beta_i}=0$, since $\sum_{i=1}^{k}\tau_i=0$ the solution is $\hat{\beta}_j=\overline{B}_j-\overline{y}$, $j=1,2,\ldots,b$.

10.4.5. $T_1=602, T_2=619, T_3=427, T_4=439 B_1=386, B_2=390, B_3=414, B_4=437,$ $B_5=460\ b=5, k=4, n=bk=20$

$\text{CM}=\frac{1}{n}\left(\sum_{j=1}^{b}B_j\right)^2=217778.5$, $\text{SSB}=\frac{1}{k}\sum_{j=1}^{b}B_j^2-\text{CM}=986.8$, $\text{MSB}=\frac{\text{SSB}}{b-1}=246.7$, $\text{SST}=\frac{1}{b}\sum_{i=1}^{k}T_i^2-\text{CM}=6344.55$, $\text{MST}=\frac{\text{SST}}{k-1}=2114.85$, Total SS $=\sum_{i=1}^{k}\sum_{j=1}^{n_i}y_{ij}^2-\text{CM}$ $=7390.55$, SSE $=$ Total SS $-$ SSB $-$ SST $=59.2$, $\text{MSE}=\frac{\text{SSE}}{n-b-k+1}=4.9333$

To test if the true income lower limits of top 5 percent of U.S. households for each races are the same, $F=\frac{\text{MST}}{\text{MSE}}=5.35283$, $F_{0.05,3,12}=3.49$. Since the observed value $F>3.49$, we reject the null hypothesis and conclude that there is difference in the true income lower limits of top 5 percent of U.S. households for each races.

To test if the true income lower limits of top 5 percent of U.S. households for each year between 1994–1998 are the same, $F=\frac{\text{MSB}}{\text{MSE}}=50.00676$, and $F_{0.05,4,12}=3.2592$. Since the observed value $F>3.2592$, we conclude that there is difference in the true income lower limits of top 5 percent of U.S. households for each year among 1994–1998

10.4.7. $T_1 = 112, T_2 = 110, T_3 = 133, T_4 = 157,\ B_1 = 201, B_2 = 165, B_3 = 146\ b = 3, k = 4,$ $n = bk = 12$
$\text{CM} = \frac{1}{n}\left(\sum_{j=1}^{b} B_j\right)^2 = 21845.33, \text{SSB} = \frac{1}{k}\sum_{j=1}^{b} B_j^2 - \text{CM} = 390.1667, \text{MSB} = \frac{\text{SSB}}{b-1} = 195.0833, \text{SST} = \frac{1}{b}\sum_{i=1}^{k} T_i^2 - \text{CM} = 482, \text{MST} = \frac{\text{SST}}{k-1} = 160.667, \text{Total SS} = \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \text{CM} = 1234.667, \text{SSE} = \text{Total SS} - \text{SSB} - \text{SST} = 362.5, \text{MSE} = \frac{\text{SSE}}{n-b-k+1} = 60.41667$

To test if the true mean performance for different hours of sleep are the same, $F = \frac{\text{MST}}{\text{MSE}} = 2.6593$, $F_{0.05,3,6} = 4.7571$. Since the observed value $F < 4.7571$, there is evidence to conclude that there is no difference in the true mean performance for different hours of sleep

To test if the true mean performance for each category of the test are the same, $F = \frac{\text{MSB}}{\text{MSE}} = 3.2290$, and $F_{0.05,2,6} = 5.1433$. Since the observed value $F < 5.1433$, there is evidence to conclude that there is no difference in the true mean performance for each category of the test.

EXERCISES 10.5

10.5.1. **(a)** For simplicity of computation, we will use SPSS. The following is the output.
Oneway

ANOVA Average Time

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.900	3	.300	.919	.461
Within Groups	3.919	12	.327		
Total	4.818	15			

(b) Since there is no significant difference, Tukey's method is not necessary.

(c) Since F is smaller than the critical value, $F_{0.05,3,12} = 3.4903$, there is evidence to conclude there is no difference in the average time to process claim forms among the four processing facilities.
Assumptions: The samples are randomly selected from the 4 populations in an independent manner. The population are normally distributed with equal variances σ^2 and mean $\mu_1, \mu_2, \mu_3, \mu_4$.

10.5.3. **(a)** Oneway

ANOVA Income

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	6344.550	3	2114.850	32.350	.000
Within Groups	1046.000	16	65.375		
Total	7390.550	19			

Since F is greater than critical value F, $F_{0.05,3,16} = 3.24$, based on data provided, there is evidence to conclude that there is difference in the income lower limits of top 5 percents of U.S. households for each races for all five years at 0.05 level of significance.

(b) Post Hoc Tests

Multiple Comparisons Income Tukey HSD						
					95% Confidence Interval	
(I) race_num	(J) race_num	Mean Difference (I–J)	Std. Error	Sig.	Lower Bound	Upper Bound
1	2	−3.4	5.11371	0.91	−18.0304	11.2304
	3	35.00000*	5.11371	0	20.3696	49.6304
	4	32.60000*	5.11371	0	17.9696	47.2304
2	1	3.4	5.11371	0.91	−11.2304	18.0304
	3	38.40000*	5.11371	0	23.7696	53.0304
	4	36.00000*	5.11371	0	21.3696	50.6304
3	1	−35.00000*	5.11371	0	−49.6304	−20.3696
	2	−38.40000*	5.11371	0	−53.0304	−23.7696
	4	−2.4	5.11371	0.97	−17.0304	12.2304
4	1	−32.60000*	5.11371	0	−47.2304	−17.9696
	2	−36.00000*	5.11371	0	−50.6304	−21.3696
	3	2.4	5.11371	0.97	−12.2304	17.0304

(c)

$\mu_i - \mu_j$	$\overline{T}_i - \overline{T}_j$	Tukey Interval	Reject or N.R.	Conclusion
$\mu_1 - \mu_2$	120.4 – 123.8	(−18.03,11.23)	N.R.	$\mu_1 = \mu_2$
$\mu_1 - \mu_3$	120.4 – 85.4	(20.37, 49.63)	R	$\mu_1 \neq \mu_3$
$\mu_1 - \mu_4$	120.4 – 87.8	(17.97, 47.23)	R	$\mu_1 \neq \mu_4$
$\mu_2 - \mu_3$	123.8 – 85.4	(23.77, 53.03)	R	$\mu_2 \neq \mu_3$
$\mu_2 - \mu_4$	123.8 – 87.8	(21.37, 50.63)	R	$\mu_2 \neq \mu_4$
$\mu_3 - \mu_4$	85.4 – 87.8	(−17.03, 12.23)	N.R.	$\mu_3 = \mu_4$

Assuming the samples are randomly selected in an independent manner, the populations are normally distributed with equal variances σ^2, and based on 95% Tukey intervals, All races is similar to White, and Black is similar to Hispanic. All other true income lower limits for each races are different.

EXERCISE 10.7

10.7.1. Oneway

ANOVA Cholesterol					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	32.444	2	16.222	.041	.960
Within Groups	5961.833	15	397.456		
Total	5994.278	17			

Since F is smaller than critical value F, $F_{0.01,2,15} = 6.36$, based on data provided, there is not enough evidence to conclude that there is difference in the true cholesterol levels for all races in United States during 1987–1980.

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Chapter 11

Bayesian Estimation and Inference

EXERCISES 11.2

11.2.1. By Bayes' rule,

$$
\begin{aligned}
&P(\text{The die is loaded}\,|\,\text{3 consecutive fives})\\
&= \frac{P(\text{3 consecutive fives}\,|\,\text{Loaded})\,P(\text{Loaded})}{P(\text{3 consecutive fives}\,|\,\text{Loaded})\,P(\text{Loaded}) + P(\text{3 consecutive fives}\,|\,\text{Fair})\,P(\text{Fair})}\\
&= \frac{(0.6)^3 0.02}{(0.6)^3 0.02 + (1/6)^3 0.98} = 0.488
\end{aligned}
$$

11.2.3. **(a)** $f(p|x) \propto \pi(p) f(x|p)$
$\propto p^2\{p^x(1-p)^{n-x}\}$
$\propto p^{x+2}(1-p)^{n-x}$, which is the kernel density of $\beta(x+3, n-x+1)$.
Hence, the posterior distribution of p is $\beta(x+3, n-x+1)$.

(b) $f(p|x) \propto \pi(p) f(x|p)$
$\propto \{p^{a-1}(1-p)^{b-1}\}\{p^x(1-p)^{n-x}\}$
$\propto p^{x+a-1}(1-p)^{n-x+b-1}$, which is the kernel density of $\beta(x+a, n-x+b)$.
Hence, the posterior distribution of p is $\beta(x+a, n-x+b)$.

11.2.5. **(a)** Note that

$$
\begin{aligned}
f(\lambda|x) &\propto \pi(\lambda) f(x|\lambda)\\
&\propto e^{-\mu\lambda} \prod_i (\lambda e^{-\lambda x_i})\\
&\propto \lambda^n e^{-(\mu+\sum x_i)\lambda}, \text{ which is the kernel density of } \Gamma(n+1, \mu + \textstyle\sum_{i=1}^n x_i).
\end{aligned}
$$

Hence, the posterior distribution of λ is $\Gamma(n+1, \mu + \sum_{i=1}^n x_i)$.

(b) $E(\lambda|x) = (n+1)/(\mu + \sum_{i=1}^n x_i)$.

11.2.7. **(a)** $f(\lambda|x) \propto \pi(\lambda) f(x|\lambda)$

$$\propto e^{-\lambda} \prod_i (e^{-\lambda} \lambda^{x_i})$$

$$\propto \lambda^{\sum x_i} e^{-(n+1)\lambda}, \text{ which is the kernel density of } \Gamma\left(\sum_{i=1}^n x_i + 1, n+1\right).$$

Hence, the posterior distribution of λ is $\Gamma\left(\sum_{i=1}^n x_i + 1, n+1\right)$.

(b) $E(\lambda|x) = \left(\sum_{i=1}^n x_i + 1\right) / (n+1)$.

11.2.9.

$$f(\mu|x) \propto \pi(\mu) f(x|\mu) \propto \exp\left\{-\frac{\mu^2}{2\sigma^2}\right\} \prod \exp\left\{-\frac{(x_i-\mu)^2}{2\cdot 2}\right\}$$

$$\propto \exp\left\{-\frac{\mu^2}{2\sigma^2} - \frac{\sum (x_i-\mu)^2}{2\cdot 2}\right\}$$

$$\propto \exp\left\{-\frac{1}{2}\left[\left(\frac{1}{\sigma^2} + \frac{n}{2}\right)\mu^2 - n\bar{x}\mu\right]\right\}$$

$$\propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma^2} + \frac{n}{2}\right)\left(\mu - \frac{\frac{n}{2}\bar{x}}{\frac{1}{\sigma^2} + \frac{n}{2}}\right)^2\right\},$$

which is the kernel density of $N\left(\frac{\frac{n}{2}\bar{x}}{\frac{1}{\sigma^2}+\frac{n}{2}}, \frac{1}{\frac{1}{\sigma^2}+\frac{n}{2}}\right)$.

Hence the posterior distribution of μ is $N\left(\frac{\frac{n}{2}\bar{x}}{\frac{1}{\sigma^2}+\frac{n}{2}}, \frac{1}{\frac{1}{\sigma^2}+\frac{n}{2}}\right)$.

EXERCISES 11.3

11.3.1. **(a)** We have seen from **Example 11.2.7** that the posterior distribution of μ given $x_1, x_2, \ldots, x_n$ is normally distributed with

$$\text{Mean} = \frac{\frac{n}{\sigma^2}\bar{x}}{\frac{1}{1} + \frac{n}{\sigma^2}} = \frac{\frac{n}{9}\bar{x}}{1 + \frac{n}{9}} = \frac{n\bar{x}}{9+n}, \text{ and Variance} = \frac{1}{\frac{1}{1} + \frac{n}{\sigma^2}} = \frac{1}{1+\frac{n}{9}} = \frac{9}{9+n}.$$

Hence, a 95% credible interval for μ is

$$\frac{n\bar{x}}{9+n} \pm z_{0.025}\sqrt{\frac{9}{9+n}} = \frac{n\bar{x}}{9+n} \pm 1.96\sqrt{\frac{9}{9+n}}.$$

(b) First note that $n = 20$ and $\bar{x} = \dfrac{0.92 + 1.05 + \cdots + (-4.78)}{20} = 0.8725$.

Using the results of part (a), a 95% credibel interval for μ is

$$\frac{20(0.8725)}{9+20} \pm 1.96\sqrt{\frac{9}{9+20}} = (-0.490, 1.694)$$

11.3.3. We have seen from **Exercise 11.2.8** that the posterior distribution of μ is $\Gamma\left(\sum_{i=1}^{50} x_i + \alpha, 50 + \beta\right) = \Gamma(12.1, 52)$ since $\alpha = 0.1$, $\beta = 2$, and $x = \sum_{i=1}^{50} x_i = 12$. Since $(\mu|x = 12) \sim \Gamma(12.1, 52)$. Using the procedures summarized in **Section 11.3** we can show that $\Pr(0.121 < \mu < 0.381) = 0.95$, thus $(0.121, 0.381)$ is a 95% credible interval for μ.

11.3.5. Let x = sodium intake in this ethnic group, and μ = the mean sodium intake for this group. Then $\bar{x} \sim N(\mu, 300^2)$ and $\mu \sim N(2700, 250^2)$. From **Example 11.2.7** the posterior distribution of μ given $\bar{x} = 3000$ is normally distributed with

$$\text{Mean} = \frac{\frac{1}{250^2}2700 + \frac{1}{300^2}\bar{x}}{\frac{1}{250^2} + \frac{1}{300^2}} = \frac{\frac{1}{250^2}2700 + \frac{1}{300^2}3000}{\frac{1}{250^2} + \frac{1}{300^2}} = 2822.95, \text{ and}$$

$$\text{Variance} = \frac{1}{\frac{1}{250^2} + \frac{1}{300^2}} = 36885.25.$$

Hence, a 95% credible interval for μ is $2822.95 \pm z_{0.025}\sqrt{36885.25} = 2822.95 \pm 1.96\sqrt{36885.25} = (2446.52, 3199.38)$.

11.3.7. Let x = the number of calls received in five minutes. Then $x \sim \text{Poi}(5\lambda)$

$$\begin{aligned}\pi(\lambda|x = 25) &\propto \pi(\lambda) f(x = 25|\lambda)\\ &\propto e^{-2\lambda}\{e^{-5\lambda}(5\lambda)^{25}\}\\ &\propto \lambda^{25}e^{-7\lambda}, \text{ which is the kernel density of } \Gamma(25+1, 7).\end{aligned}$$

Hence, the posterior distribution of λ is $\Gamma(26, 7)$.
Knowing the posterior distribution of λ we can show that $\Pr(2.43 < \lambda < 5.27) = 0.95$. Thus $(2.43, 5.27)$ is a 95% credible interval of λ.

EXERCISES 11.4

11.4.1. **(a)** Referring to **Exercise 11.3.1.** we know that $\bar{x} = 0.8725$ and $n = 20$ and the posterior distribution of μ is normal with

$$\text{Mean} = \frac{\frac{n}{\sigma^2}\bar{x}}{\frac{1}{4} + \frac{n}{\sigma^2}} = \frac{\frac{20}{9}0.8725}{\frac{1}{4} + \frac{20}{9}} = 0.784, \text{ and Variance } = \frac{1}{\frac{1}{4} + \frac{n}{\sigma^2}} = \frac{1}{\frac{1}{4} + \frac{20}{9}} = 0.4045.$$

We can now compute

$$\begin{aligned}\alpha_0 = P(\mu \le 0|\bar{x} = 0.784) &= P\left(\frac{\mu - 0.784}{\sqrt{0.4045}} \le \frac{0 - 0.784}{\sqrt{0.4045}}\right)\\ &= P\left(z \le \frac{-0.784}{\sqrt{0.4045}}\right) = P(z \le -1.2327) = 0.109\end{aligned}$$

and

$$\alpha_1 = P(\mu > 0|\bar{x} = 0.784) = 1 - \alpha_0$$
$$= 1 - 0.109 = 0.891$$

Thus, $\alpha_0/\alpha_1 = 0.109/0.891 = 0.122 < 1$, and we reject H_0.

(b) First compute

$$z = \frac{\bar{x} - 0}{\sigma/\sqrt{n}} = \frac{0.8725}{\sqrt{9/20}} = 1.3.$$

Thus, $z = 1.3 < z_{0.95} = 1.645$, and we do not reject H_0.
In this case, we see that we obtain different decisions from using the Bayesian approach than from using the classical approach.

11.4.3. We have seen from **Exercise 11.3.3.** the posterior distribution of μ is $\Gamma(12.1, 52)$. Using any statistical software, we can compute

$$\alpha_0 = P(\mu \le 0.1|x = 12) = P(\mu \le 0.1 \text{ and } \mu \sim \Gamma(12.1, 52)) = 0.0067,$$

and $\alpha_1 = P(\mu > 0.1|x = 12) = 1 - \alpha_0 = 1 - 0.0067 = 0.9933$.
Thus, $\alpha_0 < \alpha_1$, and we reject H_0.

11.4.5. We have seen from **Exercise 11.3.5.** the posterior distribution of μ is $N(2822.95, 36885.25)$. We can now compute

$$\alpha_0 = P(\mu \le 2400|\bar{x} = 3000) = P\left(\frac{\mu - 2822.95}{\sqrt{36885.25}} \le \frac{2400 - 2822.95}{\sqrt{36885.25}}\right)$$
$$= P(z \le -2.202) = 0.014$$
$$\text{and } \alpha_1 = P(\mu > 2400|\bar{x} = 3000) = 1 - \alpha_0 = 1 - 0.014 = 0.986.$$

Thus, $\alpha_0 < \alpha_1$, and we reject H_0.

EXERCISES 11.5

11.5.1.
$$\begin{aligned} \textit{Expected Return} &= 25P(H)P(H) + 15P(H)P(T) + 15P(T)P(H) + (-15)P(T)P(T) \\ &= 25(1/2)(1/2) + 15(1/2)(1/2) + 15(1/2)(1/2) - 15(1/2)(1/2) \\ &= 10 \end{aligned}$$

Since the expected return is positive, in a long run we should win. Thus, we should play this game.

11.5.3. According to the government forecast, we write $P(G) = 0.7$ to denote that the economy will expand with a 70% chance, and $P(B) = 0.3$ to denote that the economy will decline.

$$\begin{aligned} \textit{Expected Earning} &= 300000P(G) + (-200000)P(B) \\ &= 300000(0.7) - 200000(0.3) \\ &= 150000 \end{aligned}$$

Since the expected earning is greater than 50000, the optimal decision is to open a new office.

Here we made an assumption that the government forecast will be correct with 100% certainty.

11.5.5. **(a)** Let G and B denote the true state of nature, and let G' and B' denote the weather person's prediction. Based on the record we have $P(G'|G) = 6/8 = 3/4$ and $P(G'|B) = 3/7$.

Initially assume the prior as $P(G) = P(B) = 1/2$.

Using the Bayes' theorem, we obtain the likelihood as

$$\begin{aligned} P(G|G') &= \frac{P(G'|G)P(G)}{P(G'|G)P(G) + P(G'|B)P(B)} \\ &= \frac{(3/4)(1/2)}{(3/4)(1/2) + (3/7)(1/2)} \\ &= 7/11 \end{aligned}$$

and

$$\begin{aligned} P(G|B') &= \frac{P(B'|G)P(G)}{P(B'|G)P(G) + P(B'|B)P(B)} \\ &= \frac{(1/4)(1/2)}{(1/4)(1/2) + (4/7)(1/2)} \\ &= 7/23 \end{aligned}$$

Updated prior when the weather person predicts good weather:

$$\pi(G) = P(G|G') = 7/11;\ \pi(B) = 1 - \pi(G) = 4/11.$$

Updated prior when the weather person predicts bad weather:

$$\pi(G) = P(G|B') = 7/23;\ \pi(B) = 1 - \pi(G) = 16/23.$$

(b) When the weather person predicts good weather:

$$\begin{aligned} \text{Expected gain if we insure } &= 125\pi(G) + 135\pi(B) \\ &= 125(7/11) + 135(4/11) \\ &= 128.64, \end{aligned}$$

and

$$\begin{aligned} \text{Expected gain if we do not insure } &= 200\pi(G) \\ &= 200(7/11) \\ &= 127.27. \end{aligned}$$

Therefore our decision, given that the weather person predicts good weather, is to insure.

When the weather person predicts bad weather:

$$\begin{aligned}\text{Expected gain if we insure } &= 125\pi(G) + 135\pi(B)\\ &= 125(7/23) + 135(16/23)\\ &= 131.96,\end{aligned}$$

and

$$\begin{aligned}\text{Expected gain if we do not insure } &= 200\pi(G)\\ &= 200(7/23)\\ &= 60.87.\end{aligned}$$

Therefore our decision, given that the weather person predicts bad weather, is also to insure.

11.5.7. By assuming a uniform prior we have $\pi(\theta_1) = \pi(\theta_2) = \pi(\theta_3) = 1/3$.

$$\begin{aligned}\text{The expected utility for decision } d_1 &= 0 \cdot \pi(\theta_1) + 10 \cdot \pi(\theta_2) + 4 \cdot \pi(\theta_3)\\ &= 0(1/3) + 10(1/3) + 4(1/3)\\ &= 4.67.\end{aligned}$$

$$\begin{aligned}\text{The expected utility for decision } d_2 &= (-2) \cdot \pi(\theta_1) + 5 \cdot \pi(\theta_2) + 1 \cdot \pi(\theta_3)\\ &= (-2)(1/3) + 5(1/3) + (1/3)\\ &= 1.33\end{aligned}$$

Therefore our decision is d_1.

11.5.9. **(a)**

States of Nature → Decision Space ↓	p_1	p_2	……	p_k
Predicting p_1 (d_1)	g	$-l$	……	$-l$
Predicting p_2 (d_2)	$-l$	g	……	$-l$
……	……	……	……	……
Predicting p_k (d_k)	$-l$	$-l$	……	g

(b) The expected utility for decision d_i is given by

$$E(U|d_i) = \sum_{j=1}^{k} U(d_i, p_j)\pi(p_j) = g\frac{1}{k} + (k-1)(-l)\frac{1}{k} = \frac{g-(k-1)l}{k}.$$

Therefore the expected utility is the same for all decision d_i.

(c) When $X = x_1$, compute the updated prior as

$$\pi(p_i|X = x_1) = \frac{P(X = x_1|p_i)\pi(p_i)}{\sum\limits_j P(X = x_1|p_j)\pi(p_j)} = \frac{a_i(1/k)}{\sum\limits_j a_j(1/k)} = \frac{a_i}{\sum\limits_j a_j}, \quad i = 1, 2, \ldots, k.$$

The expected utility for decision $d_i = \sum\limits_{j=1}^{k} U(d_i, p_j)\pi(p_j|X = x_1)$

$$= g\frac{a_i}{\sum\limits_j a_j} - l\frac{\sum\limits_{j\neq i} a_j}{\sum\limits_j a_j} = (g + l)\frac{a_i}{\sum\limits_j a_j} - l, \quad i = 1, 2, \ldots, k.$$

Since the above expression only depends on the term a_i, therefore our decision is d_i where i is such that a_i is the largest among $a_1, \ldots, a_k$.

When $X = x_2$, compute the updated prior as

$$\pi(p_i|X = x_2) = \frac{P(X = x_2|p_i)\pi(p_i)}{\sum\limits_j P(X = x_2|p_j)\pi(p_j)} = \frac{(1 - a_i)(1/k)}{\sum\limits_j (1 - a_j)(1/k)} = \frac{(1 - a_i)}{\sum\limits_j (1 - a_j)}, \quad i = 1, 2, \ldots, k.$$

The expected utility for decision $d_i = \sum\limits_{j=1}^{k} U(d_i, p_j)\pi(p_j|X = x_2)$

$$= g\frac{(1 - a_i)}{\sum\limits_j (1 - a_j)} - l\frac{\sum\limits_{j\neq i} (1 - a_j)}{\sum\limits_j (1 - a_j)} = (g + l)\frac{(1 - a_i)}{\sum\limits_j (1 - a_j)} - l, \quad i = 1, 2, \ldots, k.$$

Since the above expression only depends on the term $(1 - a_i)$, therefore our decision is d_i where i is such that a_i is the smallest among $a_1, \ldots, a_k$.

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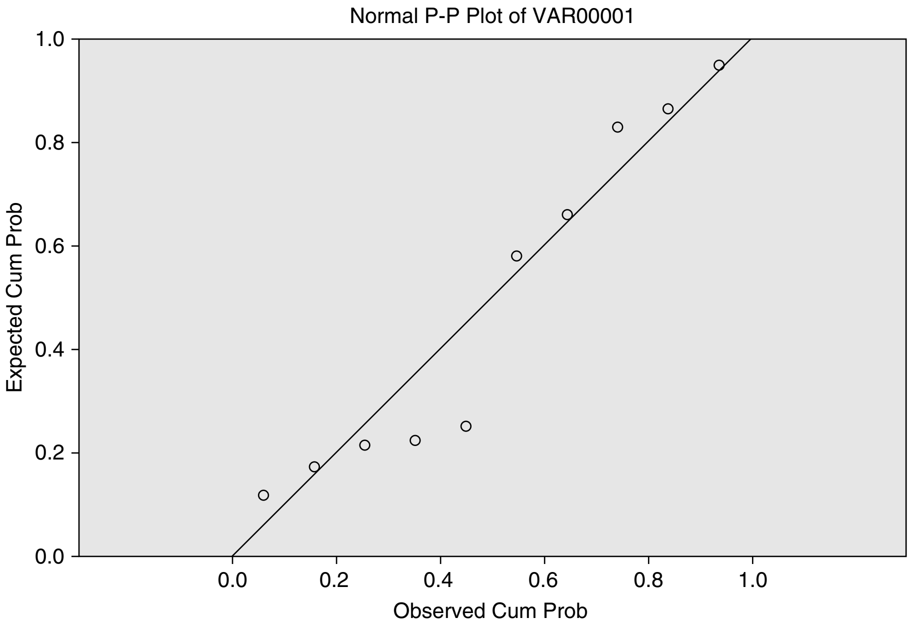
Chapter 12

Nonparametric Tests

EXERCISES 12.2

12.2.1. In this case $n = 9$, $p = 0.5$, $X \sim \text{Bin}(n, p)$. If $P(X \leq a) = 0.025$, using the Binomial table, $a = 1$, then $b = n+1-a = 9$. Then the first and ninth value in the order list, an approximate 95% confidence interval is $2.7 < M < 8.5$

12.2.3. **(a)** The next normal probability plot generated by a SPSS statistical software shows that the normality assumption may not be satisfied.



(b) Looking in the table for $n = 10$, $p = 0.5$, $X \sim \text{Bin}(n, p)$. If $P(X \leq a) = 0.025$, using the Binomial table, $a = 1$, then $b = n + 1 - a = 10$. Then, an approximate 95% confidence interval for the central median is greater than 57.3 and less than 66.7. That is, we have at least 95% chance that the true median air pollution index for the city will fall in the interval (57.3, 66.7).

12.2.5. Looking in the table for $n = 6$, $p = 0.5$, $X \sim \text{Bin}(n, p)$. If $P(X \leq a) = 0.005$, using the Binomial table, $a = 0$.

12.2.7. Looking in the table for $n = 15$, $p = 0.5$, $X \sim \text{Bin}(n, p)$. If $P(X \leq a) = 0.005$, using the Binomial table, $a = 2$, then $b = n + 1 - a = 14$. That is, we have 99% chance that the true median time required to prune an acre of grapes will fall in the interval (4.2, 5.8).

12.2.9. Looking in the table for $n = 15$, $p = 0.5$, $X \sim \text{Bin}(n, p)$. If $P(X \leq a) = 0.025$, using the Binomial table, $a = 3$, then $b = n + 1 - a = 13$. Then, an approximate 95% confidence interval for the median in-state tuition costs is $3683 < M < 5212$. That is, we have 95% chance that the median is-state tuition costs fall in (3683, 5212).

EXERCISES 12.3

12.3.1. Let $m_0 = 7.75$
We test $H_0: M = m_0$ vs. $H_a: M \neq m_0$ at $\alpha = 0.01$

(i) Since there is a tie, $n = 8$. In this case, $n^+ = 3$ and $N^+ \sim \text{Bin}(n, p = \frac{1}{2})$. Then $P(N^+ \geq 3) = 0.85547$, which is not less than $\frac{\alpha}{2}$. Then H_0 is not rejected. Based on the sample, the median interest rate in the city is not significantly different from 7.75

(ii) Eliminating the tie, it can be obtained the next table

x_i	$z_i = \lvert x_i - 65\rvert$	Sign	Rank
7.625	0.125	−	2
7.875	0.125	+	2
7.625	0.125	−	2
8	0.25	+	5
7.5	0.25	−	5
8	0.25	+	5
7.375	0.375	−	7
7.25	0.5	−	8

thus $W^+ = 12$, $n = 8$. Then H_0 is rejected since the rejection region is $W^+ \leq 3$ or $W^+ \geq 33$. Then the same conclusion as the Sign test.

12.3.3. Let $m_0 = 1000$
We test $H_0: M = m_0$ vs. $H_a: M > m_0$ at $\alpha = 0.05$

(i) Since there is no tie, $n = 10$. In this case, $n^+ = 6$ and $N^+ \sim \text{Bin}\left(n, p = \frac{1}{2}\right)$. Then $P\left(N^+ \geq 6\right) = 0.37695$, which is not less than $\frac{\alpha}{2}$. Then H_0 is not rejected. Based on the sample, the median SAT scores is not significantly greater than 1000

(ii) It can be obtained the next table

x_i	$z_i = \lvert x_i - 65\rvert$	Sign	Rank
986	14	−	1
1065	65	+	2
1089	89	+	3
890	110	−	4
1128	128	+	5
1157	157	+	6
1224	224	+	7
765	235	−	8
1355	355	+	9
567	433	−	10

thus $W^+ = 32, n = 10$. Then $H_0$is rejected since the rejection region is $W^+ \geq 10$. Based on the sample we conclude that the median SAT scores is significantly greater than 1000.

12.3.5. Let $m_0 = 250$. We test $H_0: M = m_0$ vs. $H_a: M > m_0$ at $\alpha = 0.05$

(i) Since there is no tie, $n = 20$. Using large sample approximation, $N^+ \sim N(\mu = np, \sigma^2 = np(1-p))$, where $p = 1/2$ then $Z = \frac{2N^+ - n}{\sqrt{n}}$ follows standard normal distribution. In this case $n^+ = 11$, then $z = 0.447214$. Since $z_\alpha = 1.6448$, the rejection region is $|z| \geq z_\alpha$. Based on the sample we conclude that the median weight of NFL players is not significantly greater than 250 pounds.

(ii) We have

x_i	$z_i = \lvert x_i - 65\rvert$	Sign	Rank
254	4	+	1.5
246	4	−	1.5
259	9	+	3
234	16	−	4
232	18	−	5
269	19	+	6
229	21	−	7
274	24	+	8
276	26	+	9
285	35	+	10.5
285	35	+	10.5
288	38	+	12
211	39	−	13
296	46	+	14
298	48	+	15
193	57	−	16
192	58	−	17
311	61	+	18.5
189	61	−	18.5
178	72	−	20

thus $W^+ = 108$, $n = 20$. Using normal approximation, $Z = \frac{W^+ - \frac{1}{4}n(n+1)}{\sqrt{n(n+1)(2n+1)/24}}$ then $z = 0.111998$. Since $z_\alpha = 1.6448$, the rejection region es $|z| \geq z_\alpha$. We reach the same conclusion as the sign test.

12.3.7. Using the difference as "after"-"before" we test $H_0: M = 0$ vs. $H_a: M < 0$ at $\alpha = 0.05$

(i) We have

Before	185	222	235	198	224	197	228	234
After	188	217	229	190	226	185	225	231
Difference	3	-5	-6	-8	2	-12	-3	-3
Sign	+	−	−	−	+	−	−	−

Then $n^+ = 2$. Using large sample approximation, $N^+ \sim N(\mu = np, \sigma^2 = np(1-p))$, where $p = 1/2$ then $Z = \frac{2N^+ - n}{\sqrt{n}}$ follows standard normal distribution. In this case, $n = 8$ and $n^+ = 2$, then $z = -1.4142$. Since $z_\alpha = -1.645$, the rejection region is $z \leq z_\alpha$. Based on the sample, there is not enough evidence to conclude that the new diet reduces the systollic blood pressure on individuals of over 40 years old.

(ii) We have

x_i	$z_i = \lvert x_i - 65 \rvert$	Sign	Rank
2	2	+	1
3	3	+	3
-3	3	−	3
-3	3	−	3
-5	5	−	5
-6	6	−	6
-8	8	−	7
-12	12	−	8

thus $W^+ = 4$, $n = 8$. Using normal approximation, $Z = \frac{W^+ - \frac{1}{4}n(n+1)}{\sqrt{n(n+1)(2n+1)/24}}$ then $z = -1.96$. Since $z_\alpha = -1.645$, the rejection region es $z \leq z_\alpha$. We reach the same conclusion as the sign test.

EXERCISES 12.4

Assumptions: Observations are randomly selected and $n_1 \leq n_2$

12.4.1. We need to test $H_0: m_1 = m_2$ vs. $H_a: m_1 \neq m_2$, where m_1 is the median for American conference, and m_2 is the median for National conference. In this case $n_1 = n_2 = 6$.

Combining and keeping track the next table is obtained

Value	Population	Rank
0.455	N	1
0.545	A	2.5
0.545	N	2.5
0.636	N	5
0.636	N	5
0.636	N	5
0.727	A	7.5
0.727	A	7.5
0.818	A	10
0.818	A	10
0.818	N	10
0.909	A	12

Then $R = 28.5$, $W = R - \frac{1}{2}n_2(n_2+1)$, so $w = 28.5 - \frac{1}{2}(6)(6+1) = 7.5$. For $\alpha = 0.05$ the rejection region is $W \le 28$ or $W \ge 50$. There is enough evidence to conclude that the two samples comes from population with different medians.

12.4.3. If we select n_2 numbers from $\{1, 2, \ldots, n_1 + n_2\}$ at random without replacement and if X_i is the ith number selected then

$$E\left(\sum_{i=1}^{n_2} X_i\right) = \frac{n_2(n_1+n_2+1)}{2}$$

and

$$\mathrm{Var}\left(\sum_{i=1}^{n_2} X_i\right) = \frac{n_2(n_1+n_2+1)n_1}{12}.$$

R is the sum of the ranks $r(X_i)$ of observations from population 2. Under the null hypothesis they have the same distribution.

Since the rank $r(X_i)$ takes one of $1, 2, \ldots, n_1 + n_2$, if (without loss in generality) all observations are distinct then $R = \sum_{i=1}^{n_2} r(X_i)$ has

$$E(R) = \frac{n_2(n_1+n_2+1)}{2}$$

and

$$\mathrm{Var}(R) = \frac{n_2(n_1+n_2+1)n_1}{12}.$$

We have $W = R - \frac{1}{2}n_2(n_2+1)$, then

$$\begin{aligned} E(W) &= E(R) - \frac{1}{2}n_2(n_2+1) \\ &= \frac{n_2(n_1+n_2+1)}{2} - \frac{1}{2}n_2(n_2+1) \\ &= \frac{n_2 n_1}{2} \end{aligned}$$

and

$$\begin{aligned} \mathrm{Var}(W) &= \mathrm{Var}(R) \\ &= \frac{n_2(n_1+n_2+1)n_1}{12} \end{aligned}$$

12.4.5. Using Wilcoxon-Rank-Sum test:
We need to test $H_0: m_1 = m_2$ vs. $H_a: m_1 < m_2$, where m_1 is the median net conversion in female rat, and m_2 is the median net conversion in male rat. In this case $n_1 = 12, n_2 = 14$. Combining and keeping track the next tables are obtained

Value	5.1	5.5	6.5	7.2	7.5	9.5	9.8	9.8	10.4	11.2	11.6	12.8	13.1
Population	F	F	F	F	F	F	M	F	F	F	M	M	M
Rank	1	2	3	4	5	6	7.5	7.5	9	10	11	12	13

Value	13.5	13.8	13.8	14.2	14.5	15.1	15.8	15.9	16.0	16.0	16.7	16.9	17.3
Population	M	M	F	M	F	M	F	M	M	M	M	M	M
Rank	14	15.5	15.5	17	18	19	20	21	22.5	22.5	24	25	26

Then $R = 250$, $W = R - \frac{1}{2}n_2(n_2+1)$, so $w = 28.5 - \frac{1}{2}(14)(14+1) = 145$. We will use Wilcoxon-Rank-Sum test for large sample, then we use the statistic $Z = \frac{W - n_1 n_2/2}{\sqrt{n_1 n_2(n_1+n_2+1)/12}}$, for which the realization is $z = 3.1375$. For $\alpha = 0.05$, $z_\alpha = -1.64485$, and the rejection region is $z < -1.64485$. There is not enough evidence to conclude that the median net conversion of progesterone in male rats is larger than in female rats.

Using Median test:
Testing the same hypothesis, the grand median is 13.3. The next tables can be obtained

Sample	5.1	5.5	6.5	7.2	7.5	9.5	9.8	9.8	10.4
Population	F	F	F	F	F	F	M	F	F
Above/Below Median	B	B	B	B	B	B	B	B	B

Sample	11.2	11.6	12.8	13.1	13.5	13.8	13.8	14.2	14.5
Population	F	M	M	M	M	M	F	M	F
Above/Below Median	B	B	B	B	A	A	A	A	A

Sample	15.1	15.8	15.9	16	16	16.7	16.9	17.3
Population	M	F	M	M	M	M	M	M
Above/Below Median	A	A	A	A	A	A	A	A

Then, it can be obtained the next table

	Below	Above	Total
sample 1	9	3	12
sample 2	4	10	14
total	13	13	26

The total above is $N_a = 13$, the total below is $N_b = 13$ the sample sizes are $n_1 = 12$, the statistic for large sample is given by $Z = \frac{N_{1a} - E(N_{1a})}{\sqrt{\text{Var}(N_{1a})}}$, which realization is $z = -2.31455$, since $N_{1a} = 3$ is the number of observation in sample 1 above the median, $E(N_{1a}) = \frac{N_a n_1}{n} = \frac{(13)(12)}{26} = 6$, and $\text{Var}(N_{1a}) = \frac{N_a n_1 n_2 N_b}{n^2(n-1)} = 1.68$.
At $\alpha = 0.05$, $z_\alpha = -1.64485$. Then, the rejection region is $z_\alpha < -1.64485$. Therefore, the same conclusion is reached.

12.4.7. We need to test $H_0 : m_1 = m_2$ vs. $H_a : m_1 > m_2$, where m_1 is the median for sample II, and m_2 is the median for sample I. In this case, the sample sizes are $n_1 = 8$ and $n_2 = 12$. Combining and keeping track the next table can be obtained

Value	4	6	7	8	8	10	11	12	12	13	13	13
Population	S1	S1	S1	S1	S1	S1	S1	S1	S1	S1	S2	S2
Rank	1	2	3	4.5	4.5	6	7	8.5	8.5	16.5	16.5	16.5

Value	14	15	15	16	17	17	18	19
Population	S1	S1	S2	S2	S2	S2	S2	S2
Rank	13	14.5	14.5	16	17.5	17.5	19	20

Then $R = 89$, $W = R - \frac{1}{2}n_2(n_2+1)$, so $w = 89 - \frac{1}{2}(12)(12+1) = 11$. For $\alpha = 0.01$, the rejection region is $W \geq 115$. There is no enough evidence to suggest that the median for sample I is less than the median for sample II.

EXERCISES 12.5

12.5.1. We need to test $H_0 : M_1 = M_2 = M_3 = M_4 = 0$ vs. H_a: Not all M_i's equal 0, where M_i is the true median for group i, $i = 1, 2, 3, 4$. In this case $n_1 = n_2 = n_3 = n_4 = 8$, $N = \sum_{i=1}^{4} n_i = 32$, $r_1 = 80$, $r_2 = 116$, $r_3 = 115$, and $r_4 = 177$. Then $H = \frac{12}{N(N+1)} \sum_{i=1}^{2} \frac{r_i^2}{n_i} - 3(N+1) = 7.832386$

(i) At $\alpha = 0.05$, $\chi^2_{\alpha,k-1} = \chi^2_{0.05,3} = 7.8147$. There is enough evidence to suggest that not all M_i's are equal to 0

(ii) At $\alpha = 0.01$, $\chi^2_{\alpha,k-1} = \chi^2_{0.01,3} = 11.3449$. There is enough evidence to suggest that not all M_i's are equal to 0

12.5.3. For $n=2$, $H=\frac{12}{N(N+1)}\sum_{i=1}^{2}\frac{r_i^2}{n_i}-3(N+1)=\frac{12}{N(N+1)}\sum_{i=1}^{2}\frac{1}{n_i}\left(r_i-\frac{n_i(N+1)}{2}\right)^2$. Then, since $r_1+r_2=\frac{N(N+1)}{2}$, $N=n_1+n_2$, and $(-1)^2=1$, it can be obtained $H=\frac{12}{N(N+1)}\left[\frac{1}{n_1}\left(r_1-\frac{n_1(N+1)}{2}\right)^2+\frac{1}{n_2}\left(\frac{N(N+1)}{2}-r_1-\frac{n_2(N+1)}{2}\right)^2\right]=\frac{12}{N(N+1)}\left[\frac{1}{n_1}\left(r_1-\frac{n_1(N+1)}{2}\right)^2+\frac{1}{n_2}\left(r_1-\frac{n_1(N+1)}{2}\right)^2\right]$. Therefore, $H=\frac{12}{N(N+1)}\cdot\frac{N}{n_1n_2}\left(r_1-\frac{n_1(N+1)}{2}\right)^2=\frac{12}{n_1n_2(N+1)}\left(r_1-\frac{n_1(N+1)}{2}\right)^2$.
Now, we reject H_0 if, and only if $H>k$, for certain value according the rejection rule using the Kruskall-Wallis test. And, $H>k\Leftrightarrow\left|r_1-\frac{n_1(N+1)}{2}\right|^2>k'$, for appropriate $k'\Leftrightarrow r_1>c_1$ or $r_1>c_1$, for some c_1 and c_2, which correspond to the rejection rule using Willcoxon-Rank-Sum test.
Thus, they are equivalent.

12.5.5. We need to test H_0: Yields of corn for fertilizers are equal vs. H_a: Not all yields of corn for fertilizer are equal. The corresponding table is

	None:	Fertilizer I:	Fertilizer II:	Fertilizer III:	Fertilizer IV:
	11	15.5	26	31.5	44
	1	14	23	36	41
	2.5	10	23	23	38
	5	12	21	33	43
	6	7	29.5	31.5	40
	2.5	13	20	25	34
	8.5	18	19	29.5	42
	8.5	15.5	28	38	38
	4		17	27	35
r_i	49	105	206.5	274.5	355

In this case, $n_1=n_3=n_4=n_5=9$, $n_2=8$, $k=5$, $N=\sum_{i=1}^{k}n_i=44$, $r_1=49$, $r_2=105$, $r_3=206.5$, $r_4=274.5$, and $r_5=355$. Then $H=\frac{12}{N(N+1)}\sum_{i=1}^{k}\frac{r_i^2}{n_i}-3(N+1)=39.29066$
At $\alpha=0.01$, $\chi^2_{\alpha,k-1}=13.2767$. There is enough evidence to suggest a difference in yields of corn from the different fertilizers.

Chapter 13

Empirical Methods

CHAPTER 13

Statistical software **R** is used for this chapter. All outputs and codes given are in **R**. **R** is a free statistical software, and it can be downloaded from the website: *www.r-project.org*

EXERCISES 13.2

13.2.1. **(a)** Let $\theta = \mu$ and $\hat{\theta} =$ sample mean. Use the command "**jackknife**" in **R**, we obtain the results below:

```
$jack.se
[1] 12.66607

$jack.bias
[1] 0

$jack.values
 [1]  287.1818 289.2727 276.3636 279.4545 287.4545 284.0909
284.8182 288.0909
 [9] 286.7273 289.0000 287.0000 288.5455
```

Using the same notations defined in **Section 13.2**, we have the relation between our notations and the results in **R** as follows:

$$\text{jack.values} = \left(\hat{\theta}_{-1}, \hat{\theta}_{-2}, \hat{\theta}_{-3}, \ldots, \hat{\theta}_{-n}\right)$$

$$\text{jack.bias} = (n-1)\left(\frac{1}{n}\sum_{k=1}^{n}\hat{\theta}_{-k} - \hat{\theta}\right)$$

$$\text{jack.se} = \text{standard error of the jackknife esitmate} = \frac{s^*}{\sqrt{n}}$$

Thus, the jackknife estimate can be obtained by

$$\begin{aligned}\hat{\theta}^* &= \frac{1}{n}\sum_{k=1}^{n}\hat{\theta}_k^* = \frac{1}{n}\sum_{k=1}^{n}\left(n\hat{\theta} - (n-1)\hat{\theta}_{-k}\right)\\ &= \frac{1}{n}\sum_{k=1}^{n}\left(\hat{\theta} + (n-1)\hat{\theta} - (n-1)\hat{\theta}_{-k}\right)\\ &= \hat{\theta} - (n-1)\left(\frac{1}{n}\sum_{k=1}^{n}\hat{\theta}_{-k} - \hat{\theta}\right)\\ &= \hat{\theta} - \text{jack.bias}\end{aligned}$$

Since $\hat{\theta}$ = the sample mean of the complete data = 285.67, we have the jackknife estimate of μ as $\hat{\theta}^* = \hat{\theta} - \text{jack.bias} = 285.67 - 0 = 285.67$.

(b) A 95% jackknife confidence interval for μ is

$$\hat{\theta}^* \pm t_{\alpha/2,n-1} \cdot \text{jack.se} = 2.85.67 \pm 2.201 \cdot 12.666 = (257.789, 313.545).$$

(c) Compare the 95% jackknife confidence interval with **Example 6.3.3**, where the confidence interval is (257.81,313.59). Thus, in this exercise through the two methods, we get a very close confidence interval for μ.

13.2.3. Let $\theta = \mu$ and $\hat{\theta}$ = sample mean. From **R** we can obtain the following results:

```
$jack.se
[1] 1.050376

$jack.bias
[1] 0
```

Since $\hat{\theta}$ = the sample mean of the complete data = 61.22, we have the jackknife estimate of μ as $\hat{\theta}^* = \hat{\theta} - \text{jack.bias} = 61.22 - 0 = 61.22$.
A 95% jackknife confidence interval for μ is

$$\hat{\theta}^* \pm t_{\alpha/2,n-1} \cdot \text{jack.se} = 61.22 \pm 2.262 \cdot 1.05 = (58.844, 63.596).$$

There is a 95% chance that the true mean falls in (58.844, 63.596).

13.2.5. Let $\theta = \sigma^2$ and $\hat{\theta}$ = sample variance. From **R** we can obtain the following results:

```
$jack.se
[1] 2247.042

$jack.bias
[1] 0
```

Since $\hat{\theta}$ = the sample variance of the complete data = 9386, we have the jackknife estimate of σ^2 as $\hat{\theta}^* = \hat{\theta} - \text{jack.bias} = 9386 - 0 = 9386$.
A 95% jackknife confidence interval for σ^2 is

$$\hat{\theta}^* \pm t_{\alpha/2,n-1} \cdot \text{jack.se} = 9386 \pm 2.262 \cdot 2247.042 = (4302.838, 14469.16).$$

There is a 95% chance that the true variance falls in (4302.838, 14469.16). Compare the 95% jackknife confidence interval with **Example 6.4.2**, where the confidence interval is (4442.3, 31299). Thus, in this case the jackknife confidence interval for σ^2 is much shorter than the classical one in **Example 6.4.2.**

13.2.7. **(a)** Let $\theta = \mu$ and $\hat{\theta} =$ sample mean. From **R** we can obtain the following results:

```
$jack.se
[1] 0.1837461

$jack.bias
[1] 0
```

Since $\hat{\theta} =$ the sample mean of the complete data $= 2.317$, we have the jackknife estimate of μ as $\hat{\theta}^* = \hat{\theta} -$ jack.bias $= 2.317 - 0 = 2.317$.
A 95% jackknife confidence interval for μ is

$$\hat{\theta}^* \pm t_{\alpha/2,n-1} \cdot \text{jack.se} = 2.317 \pm 2.201 \cdot 0.184 = (1.912, 2.721).$$

(b) Let $\theta = \sigma^2$ and $\hat{\theta} =$ sample variance. From **R** we can obtain the following results:

```
$jack.se
[1] 0.1682317

$jack.bias
[1] 0
```

Since $\hat{\theta} =$ the sample variance of the complete data $= 0.405$, we have the jackknife estimate of σ^2 as $\hat{\theta}^* = \hat{\theta} -$ jack.bias $= 0.405 - 0 = 0.405$.
A 95% jackknife confidence interval for σ^2 is

$$\hat{\theta}^* \pm t_{\alpha/2,n-1} \cdot \text{jack.se} = 0.405 \pm 2.201 \cdot 0.168 = (0.035, 0.775).$$

(c) There is a 95% chance that the true mean falls in (1.912, 2.721); and there is a 95% chance that the true variance falls in (0.035, 0.775).

EXERCISES 13.3

Please note that resampling procedure may produce different bootstrap samples every time. Hence the results might be different. You can perform the bootstrapping using any statistical software.

13.3.1. Using statistical software **R** we have created $N = 8$ bootstrap samples of size 20. Next we calculated the mean of each bootstrap samples, denoted by $\overline{X}_1^*, \ldots, \overline{X}_N^*$. Then we have the following results:

$$\text{The bootstrap mean} = \overline{X}^* = \frac{1}{N}\sum_{i=1}^{N} \overline{X}_i^* = 5.875;$$

$$\text{The standard error} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N} (\overline{X}_i^* - \overline{X}^*)^2} = 1.137.$$

13.3.3. Using statistical software **R** we have created $N = 199$ bootstrap samples of size 10. Then $0.025 \times (199 + 1) = 5$ and $0.975 \times (199 + 1) = 195$. Thus, respectively, the 0.025 and 0.975 quantiles of the sample means are the 5^{th} and 195^{th} values of ascending-ordered sample means from the bootstrap samples. Then we get the 95% bootstrap confidence interval for μ as $(59.07, 63.16)$.

13.3.5. **(a)** Using statistical software **R** we have created $N = 199$ bootstrap samples of size 6. Then $0.025 \times (199 + 1) = 5$ and $0.975 \times (199 + 1) = 195$. Thus, respectively, the 0.025 and 0.975 quantiles of the sample means are the 5^{th} and 195^{th} values of ascending-ordered sample means from the bootstrap samples. Then we get the 95% bootstrap confidence interval for μ as $(150.667, 1149.5)$.

(b) And, respectively, the 0.025 and 0.975 quantiles of the sample medians are the 5^{th} and 195^{th} values of ascending-ordered sample medians from the bootstrap samples. Then we get the 95% bootstrap confidence interval for the population median as $(110, 1366.5)$.

EXERCISES 13.4

13.4.1. **(a)** Since $S \sim N(0, \theta^2)$ and $N \sim N(0, \sigma^2)$, then $Y = S + N \sim N(0, \sigma^2 + \theta^2)$. Then the likelihood function of Y is

$$L(\theta; y) = f_Y(y|\theta) = \frac{1}{\sqrt{2\pi(\sigma^2 + \theta^2)}} \exp\left(-\frac{1}{2(\sigma^2 + \theta^2)} y^2\right).$$

And

$$\frac{\partial \ln L(\theta; y)}{\partial \theta} = -\frac{\theta}{\sigma^2 + \theta^2} + \frac{y^2 \theta}{(\sigma^2 + \theta^2)^2}.$$

Solving $\dfrac{\partial \ln L(\theta; y)}{\partial \theta} = 0$, we obtain the MLE as

$$\hat{\theta}_{MLE} = \left(\max\left\{0, y^2 - \sigma^2\right\}\right)^{1/2}.$$

(b) The complete likelihood for Y and S is

$$L_C(\theta; y, s) = f_{Y,S}(y, s|\theta) = f_{S,N}(s, y - s|\theta)|J| \text{ where } J = \begin{vmatrix} \dfrac{\partial s}{\partial s} & \dfrac{\partial s}{\partial y} \\ \dfrac{\partial n}{\partial s} & \dfrac{\partial n}{\partial y} \end{vmatrix}$$

$$= f_S(s|\theta) f_N(y - s|\theta) = \frac{1}{2\pi\theta\sigma} \exp\left(-\frac{s^2}{2\theta^2} - \frac{(y - s)^2}{2\sigma^2}\right)$$

The conditional probability density of S given $Y = y$ is

$$h(s|\theta, y) = \frac{f_{Y,S}(y, s|\theta)}{f_Y(y|\theta)} \propto \exp\left(-\frac{s^2}{2\theta^2} - \frac{(y-s)^2}{2\sigma^2}\right)$$

$$\propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\theta^2} + \frac{1}{\sigma^2}\right)\left(s - \frac{\frac{y}{\sigma^2}}{\frac{1}{\theta^2} + \frac{1}{\sigma^2}}\right)^2\right\}$$

Thus $S|\theta, y \sim N\left(\frac{\frac{y}{\sigma^2}}{\frac{1}{\theta^2} + \frac{1}{\sigma^2}}, \frac{1}{\frac{1}{\theta^2}+\frac{1}{\sigma^2}}\right)$.

Fix θ_0, consider

$$Q(\theta|\theta_0, y) = E_{\theta_0}[\ln L_C(\theta; y, S)|\theta_0, y]$$

$$= E_{\theta_0}\left[\text{constant } - 2\ln\theta - \frac{s^2}{2\theta^2} - \frac{(y-s)^2}{2\sigma^2}\right]$$

$$= (\text{constant of } \theta) - 2\ln\theta - \frac{1}{2\theta^2}\left[\frac{1}{\frac{1}{\theta^2} + \frac{1}{\sigma^2}} + \left(\frac{\frac{y}{\sigma^2}}{\frac{1}{\theta^2} + \frac{1}{\sigma^2}}\right)^2\right]$$

Solving $\dfrac{\partial Q(\theta|\theta_0, y)}{\partial\theta} = 0$, we have

$$\hat{\theta} = \left\{\frac{1}{2}\left[\frac{1}{\frac{1}{\theta_0^2} + \frac{1}{\sigma^2}} + \left(\frac{\frac{y}{\sigma^2}}{\frac{1}{\theta_0^2} + \frac{1}{\sigma^2}}\right)^2\right]\right\}^{1/2}$$

Then we obtain the EM algorithm as

$$\hat{\theta}_{(k+1)} = \left\{\frac{1}{2}\left[\frac{1}{\frac{1}{\theta_{(k)}^2} + \frac{1}{\sigma^2}} + \left(\frac{\frac{y}{\sigma^2}}{\frac{1}{\theta_{(k)}^2} + \frac{1}{\sigma^2}}\right)^2\right]\right\}^{1/2}.$$

13.4.3. Let $n = n_1 + n_2 + n_3$ and $\theta = (p, q)$. Let $x = (n_1, n_2, n_3)$ be the observed data and $z = (n_{11}, n_{12}, n_{21}, n_{22}, n_3)$ be the compete data where n_{11} is the number of male identical pairs, n_{21} is the number of female identical pairs, and n_{21} and n_{21} are the non-identical pairs respectively for males and females. Here, the complete data set z has a multinomial distribution with the likelihood given by

$$L(\theta; z) = \binom{n}{n_{11}, n_{12}, n_{21}, n_{22}, n_3}[p(1-q)]^{n_{11}}[(1-p)(1-q)^2]^{n_{12}}[pq]^{n_{21}}$$
$$\times [(1-p)q^2]^{n_{22}}[2(1-p)q(1-q)]^{n_3}$$

Then

$$\ln L(\theta; z) = (\text{constant of } \theta) + (n_{11} + n_{21}) \ln p + (n_{12} + n_{22} + n_3) \ln(1-p) \\ + (n_{21} + 2n_{22} + n_3) \ln q + (n_{11} + 2n_{12} + n_3) \ln(1-q)$$

For multinomial distribution the expected value of each class is n multiplied by the probability of that class.
Then, for $\hat{\theta}_{(k)} = (\hat{p}_{(k)}, \hat{q}_{(k)})$ as the k^{th} step estimate, using the Bayes' rule we have

$$n_{11}^{(k)} = E\left(n_{11} | \hat{\theta}_{(k)}, x\right) = n_1 \frac{\hat{p}_{(k)}(1-\hat{q}_{(k)})}{\hat{p}_{(k)}(1-\hat{q}_{(k)}) + (1-\hat{p}_{(k)})(1-\hat{q}_{(k)})^2},$$

$$n_{12}^{(k)} = E\left(n_{12} | \hat{\theta}_{(k)}, x\right) = n_1 - n_{11}^{(k)},$$

$$n_{21}^{(k)} = E\left(n_{21} | \hat{\theta}_{(k)}, x\right) = n_2 \frac{\hat{p}_{(k)}\hat{q}_{(k)}}{\hat{p}_{(k)}\hat{q}_{(k)} + (1-\hat{p}_{(k)})(\hat{q}_{(k)})^2},$$

$$n_{22}^{(k)} = E\left(n_{22} | \hat{\theta}_{(k)}, x\right) = n_2 - n_{21}^{(k)}.$$

Then,

$$Q\left(\theta | \hat{\theta}_{(k)}, x\right) = E_{\hat{\theta}_{(k)}}[\ln L(\theta; Z) | \hat{\theta}_{(k)}, x]$$

$$= (\text{constant of } \theta) + \left(n_{11}^{(k)} + n_{21}^{(k)}\right) \ln p + \left(n_{12}^{(k)} + n_{22}^{(k)} + n_3\right) \ln(1-p) \\ + \left(n_{21}^{(k)} + 2n_{22}^{(k)} + n_3\right) \ln q + \left(n_{11}^{(k)} + 2n_{12}^{(k)} + n_3\right) \ln(1-q)$$

Solving $\frac{\partial Q}{\partial p} = 0$ and $\frac{\partial Q}{\partial q} = 0$, we then obtain the EM algorithm as

$$\hat{p}_{(k+1)} = \frac{n_{11}^{(k)} + n_{21}^{(k)}}{n},$$

$$\hat{q}_{(k+1)} = \frac{n_{21}^{(k)} + 2n_{22}^{(k)} + n_3}{n + n_{12}^{(k)} + n_{22}^{(k)} + n_3}.$$

13.4.5. **(a)** The survival function is $S(y) = \Pr(Y > y) = 1 - \Phi(y - \theta)$, where $Y \sim N(\theta, 1)$ and Φ is the cdf of $N(0, 1)$. Then the likelihood of X and Y is

$$L(\theta; x, y) = \prod_{i=1}^{n_1} \left[\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}(x_i - \theta)^2\right\}\right] \prod_{i=1}^{n_2} [1 - \Phi(y_i - \theta)\}], \text{ and}$$

$$\ln L(\theta; x, y) = \text{constant} - \frac{1}{2} \sum_{i=1}^{n_1} (x_i - \theta)^2 + \sum_{i=1}^{n_2} \ln[1 - \Phi(y_i - \theta)\}].$$

Solving $\frac{\partial \ln L(\theta;x,y)}{\partial\theta} = 0$, then the MLE, $\hat{\theta}_{MLE}$, is the solution such that

$$n_1\left(\bar{x} - \hat{\theta}_{MLE}\right) + \sum_{i=1}^{n_2} \frac{\phi\left(y_i - \hat{\theta}_{MLE}\right)}{1 - \Phi\left(y_i - \hat{\theta}_{MLE}\right)} = 0, \text{ where } \phi \text{ is the pdf of } N(0,1).$$

(b) Let $z = (x_1, \ldots, x_{n_1}, z_1, \ldots, z_{n_2})$ be the complete data set. Then the likelihood is

$$L_C(\theta; z) = \left(\frac{1}{\sqrt{2\pi}}\right)^{n_1+n_2} \exp\left[-\frac{1}{2}\sum_{i=1}^{n_1}(x_i-\theta)^2 - \frac{1}{2}\sum_{i=1}^{n_2}(z_i-\theta)^2\right], \text{ and}$$

$$\ln L_C(\theta; z) = \text{constant} - \frac{1}{2}\sum_{i=1}^{n_1}(x_i-\theta)^2 - \frac{1}{2}\sum_{i=1}^{n_2}(z_i-\theta)^2.$$

For $i = 1, \ldots, n_2$, the conditional pdf of Z_i given $X = x, Y = y$ and $\theta = \theta_0$ is

$$h(z_i|\theta_0, x, y) = \frac{f(z_i, z_i \geq y_i|\theta_0)}{f(z_i \geq y_i|\theta_0)} = \frac{\phi(z_i - \theta_0)}{1 - \Phi(y_i - \theta_0)}, \quad z_i \geq y_i.$$

Then,

$$\begin{aligned} Q(\theta|\theta_0, x, y) &= E_{\theta_0}[\ln L_C(\theta; Z)|\theta_0, x, y] \\ &= \text{constant} - \frac{1}{2}\sum_{i=1}^{n_1}(x_i-\theta)^2 - \frac{1}{2}\sum_{i=1}^{n_2}E_{\theta_0}(z_i-\theta)^2 \\ &= \text{constant} - \frac{1}{2}\sum_{i=1}^{n_1}(x_i-\theta)^2 - \frac{1}{2}\sum_{i=1}^{n_2}\int_{y_i}^{\infty}(z_i-\theta)^2 h(z_i|\theta_0, x, y)dz_i. \end{aligned}$$

Solving $\frac{\partial Q}{\partial\theta} = 0$, after a lengthy computation we then obtain

$$\theta = \frac{n_1}{n_1+n_2}\bar{x} + \frac{n_2}{n_1+n_2}\theta_0 + \frac{1}{n_1+n_2}\sum_{i=1}^{n_2}\frac{\phi(y_i-\theta_0)}{1-\Phi(y_i-\theta_0)}.$$

Therefore, we obtain the EM algorithm as

$$\hat{\theta}_{(k+1)} = \frac{n_1}{n_1+n_2}\bar{x} + \frac{n_2}{n_1+n_2}\hat{\theta}_{(k)} + \frac{1}{n_1+n_2}\sum_{i=1}^{n_2}\frac{\phi(y_i-\hat{\theta}_{(k)})}{1-\Phi(y_i-\hat{\theta}_{(k)})}.$$

EXERCISES 13.5

All the algorithms and generations of simple distributions in this section can be done by any statistical software. The statistical software, **R**, is used here.

13.5.1. The 1st iteration:
We have $x_0 = 6$.

Step 1: Generate j from $A = \{a_{ij}\}$. Suppose the software generated $j = 7$.
Step 2: $r = \frac{\pi(7)}{\pi(6)} = \frac{e^{-3}3^7/7!}{e^{-3}3^6/6!} = 0.4286$
Step 3: Generate u from $U(0, 1)$. Suppose the software generated $u = 0.5494$.
Since $r < u$, we reject the new state 7 and stay at state 6. Set $x_1 = x_0 = 6$.

The 2nd iteration:
Start with $x_1 = 6$.

Step 1: Generate j from $A = \{a_{ij}\}$. Suppose the software generated $j = 5$.
Step 2: $r = \frac{\pi(5)}{\pi(6)} = \frac{e^{-3}3^5/5!}{e^{-3}3^6/6!} = 2$
Step 3: Since $r > 1$, set $x_2 = j = 5$.

The 3rd iteration:
Start with $x_2 = 5$.

Step 1: Generate j from $A = \{a_{ij}\}$. Suppose the software generated $j = 6$.
Step 2: $r = \frac{\pi(6)}{\pi(5)} = \frac{e^{-3}3^6/6!}{e^{-3}3^5/5!} = 0.5$
Step 3: Generate u from $U(0, 1)$. Suppose the software generated $u = 0.7594$.
Since $r < u$, set $x_3 = x_2 = 5$.

The first 3 iterations are given above. The readers can follow the same algorithm to obtain more sample points. Note that different results may appear due to the different generated values each time.

13.5.3. The Metropolis–Hastings algorithm is given below:
For $t = 0$, start with an arbitrary point, $x_{(0)}$.

Step 1: Generate y from the proposal density $\Gamma([\alpha], [\alpha]/\alpha)$.
Step 2: $r = \dfrac{\pi(y)}{\pi(x_{(t)})} = \dfrac{\frac{1}{\Gamma(\alpha)\beta^\alpha} y^{\alpha-1} \exp\left(-\frac{y}{\beta}\right)}{\frac{1}{\Gamma(\alpha)\beta^\alpha} x_{(t)}^{\alpha-1} \exp\left(-\frac{x_{(t)}}{\beta}\right)} = \left(\frac{y}{x_{(t)}}\right)^{\alpha-1} \exp\left(\frac{x_{(t)}-y}{\beta}\right).$
Step 3: Acceptance/Rejection.
Generate u from $U(0, 1)$.
If $r \geq u$, set $x_{(t+1)} = y$ (i.e. accept the proposed new state);
else set $x_{(t+1)} = x_{(t)}$ (i.e. reject the proposed new state).
Step 4: Set $t = t + 1$, go to step 1.

13.5.5. Use the nominating matrix below

$$A = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

The Metropolis–Hastings algorithm is given below:
For $k = 0$, start with an arbitrary point, $x_k = i$.

Step 1: Generate j from the proposal matrix as follows:
Generate u_1 from $U(0, 1)$.
For $i = 0$, if $u_1 \geq 0.5$, set $j = 1$; else set $j = 0$.
For $i = 1$ or 2, if $u_1 \geq 0.5$, set $j = i + 1$; else set $j = i - 1$.
For $i = 3$, if $u_1 \geq 0.5$, set $j = 3$; else set $j = 2$.
Step 2: Calculate $r = \frac{\pi(j)}{\pi(i)}$ according to the target distribution $\pi(x)$.
Step 3: Acceptance/Rejection.
Generate u_2 from $U(0, 1)$.
If $r \geq u_2$, set $x_{k+1} = j$;
else set $x_{k+1} = x_k$.
Step 4: Set $k = k + 1$, go to step 1.

13.5.7. π is an exponential random variable with parameter θ, i.e. $\pi(x) = \frac{1}{\theta}\exp\left(-\frac{x}{\theta}\right), x > 0$.
Let the proposal density be $q_x(y) \propto \exp\left[-\frac{(y-x)^2}{2(0.5)^2}\right]$.
The Metropolis–Hastings algorithm is given below:
For $t = 0$, start with an arbitrary point, $x_{(0)} > 0$.

Step 1: Generate y from the proposal density $q_{x(t)}(y) \propto \exp\left[-\frac{(y-x_{(t)})^2}{2(0.5)^2}\right]$.
That is to generate y from $N(x_{(t)}, (0.5)^2)$.

Step 2: $$r = \frac{\pi(y)q_y(x_{(t)})}{\pi(x_{(t)})q_{x(t)}(y)} = \frac{\frac{1}{\theta}\exp\left(-\frac{y}{\theta}\right)\exp\left[-\frac{(x_{(t)}-y)^2}{2(0.5)^2}\right]}{\frac{1}{\theta}\exp\left(-\frac{x_{(t)}}{\theta}\right)\exp\left[-\frac{(y-x_{(t)})^2}{2(0.5)^2}\right]} = \exp\left(\frac{x_{(t)}-y}{\theta}\right).$$
Let $\alpha = \min\{1, r\}$.
Step 3: Acceptance/Rejection.
Generate u from $U(0, 1)$.
If $\alpha \geq u$, set $x_{(t+1)} = y$;
else set $x_{(t+1)} = x_{(t)}$.
Step 4: Set $t = t + 1$, go to step 1.

13.5.9. From **Example 13.5.5** with $n = 15$, $\alpha = 1$, and $\beta = 2$ recall that
$X|Y = y \sim \text{Binomial}(n, y) = \text{Binomial}(15, y)$, and
$Y|X = x \sim \beta(x + \alpha, n - x + \beta) = \beta(x + 1, 17 - x)$.
For $y_0 = 1/3$:

(i) Generate x_0 from Binomial$(15, 1/3)$. Suppose the software generated $x_0 = 5$.
(ii) Generate y_1 from $\beta(x_0 + 1, 17 - x_0) = \beta(6, 12)$. Suppose the software generated $y_1 = 0.46$ (approximated to the second digit). Then generate x_1 from Binomial$(15, 0.46)$, resulting in $x_1 = 5$.
(iii) Generate y_2 from $\beta(x_1 + 1, 17 - x_1) = \beta(6, 12)$, resulting in $y_2 = 0.26$. Then generate x_2 from Binomial$(15, 0.26)$, resulting in $x_2 = 5$.

Thus, for $y_0 = 1/3$ a particular realization of Gibbs sampler for the first three iterations are $(5, 0.33)$, $(5, 0.46)$, and $(5, 0.26)$.
For $y_0 = 1/2$:

(i) Generate x_0 from Binomial$(15, 1/2)$. Suppose the software generated $x_0 = 10$.
(ii) Generate y_1 from $\beta(x_0+1, 17-x_0) = \beta(11, 7)$. Suppose the software generated $y_1 = 0.31$ (approximated to the second digit). Then generate x_1 from Binomial$(15, 0.31)$, resulting in $x_1 = 5$.
(iii) Generate y_2 from $\beta(x_1+1, 17-x_1) = \beta(6, 12)$, resulting in $y_2 = 0.34$. Then generate x_2 from Binomial$(15, 0.34)$, resulting in $x_2 = 5$.

Thus, for $y_0 = 1/2$ a particular realization of Gibbs sampler for the first three iterations are $(10, 0.5)$, $(5, 0.31)$, and $(5, 0.34)$.
For $y_0 = 2/3$:

(i) Generate x_0 from Binomial$(15, 2/3)$. Suppose the software generated $x_0 = 11$.
(ii) Generate y_1 from $\beta(x_0+1, 17-x_0) = \beta(12, 6)$. Suppose the software generated $y_1 = 0.61$ (approximated to the second digit). Then generate x_1 from Binomial$(15, 0.61)$, resulting in $x_1 = 9$.
(iii) Generate y_2 from $\beta(x_1+1, 17-x_1) = \beta(10, 8)$, resulting in $y_2 = 0.59$. Then generate x_2 from Binomial$(15, 0.59)$, resulting in $x_2 = 7$.

Thus, for $y_0 = 2/3$ a particular realization of Gibbs sampler for the first three iterations are $(11, 0.66)$, $(9, 0.61)$, and $(7, 0.59)$.

From the three cases with different initial values, we see that when $y_0 = 2/3$ the samples have larger x and y values than the samples with $y_0 = 1/3$. Thus, the choosing of initial values may have influence on the samples. However, this influence would be negligible if the algorithm is run for a large number of times.

13.5.11. If $(X, Y) \sim N\left(\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix}, \begin{pmatrix}\sigma_1^2 & \rho\sigma_1\sigma_2\\ \rho\sigma_1\sigma_2 & \sigma_2^2\end{pmatrix}\right)$, then the conditional distribution of X given $Y = y$ is $(X|Y = y) \sim N(\mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y - \mu_2), \sigma_1^2(1-\rho^2))$.
Apply the above result we have

$$(X|Y = y) \sim N(\rho \cdot y, (1-\rho^2)), \text{ and } (Y|X = x) \sim N(\rho \cdot x, (1-\rho^2)).$$

The Gibbs sampler is given below:
Start with an arbitrary point, $y_{(0)}$. Then obtain $x_{(0)}$ by generating a random value from $N(\rho \cdot y_{(0)}, (1-\rho^2))$.
For $i = 1, \ldots n$, repeat

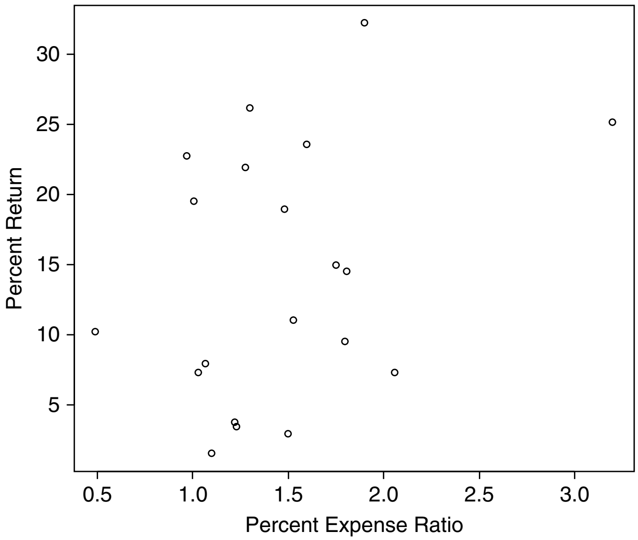
Step 1: Generate $y_{(i)}$ from $N(\rho \cdot x_{(i-1)}, (1-\rho^2))$
Step 2: Generate $x_{(i)}$ from $N(\rho \cdot y_{(i)}, (1-\rho^2))$
Step 3: Obtain the i^{th} sample as $(x_{(i)}, y_{(i)})$. Set $i = i + 1$, go to step 1.

Chapter 14

Some Issues in Statistical Applications: An Overview

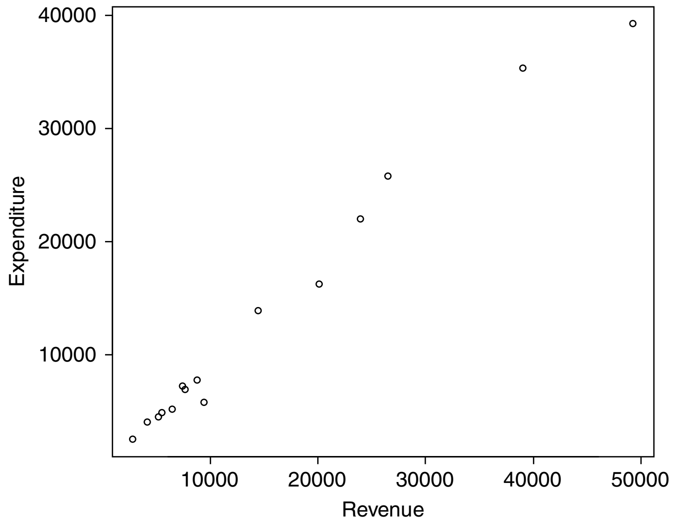
EXERCISES 14.2

14.2.1. The following is a scatter plot of the data.



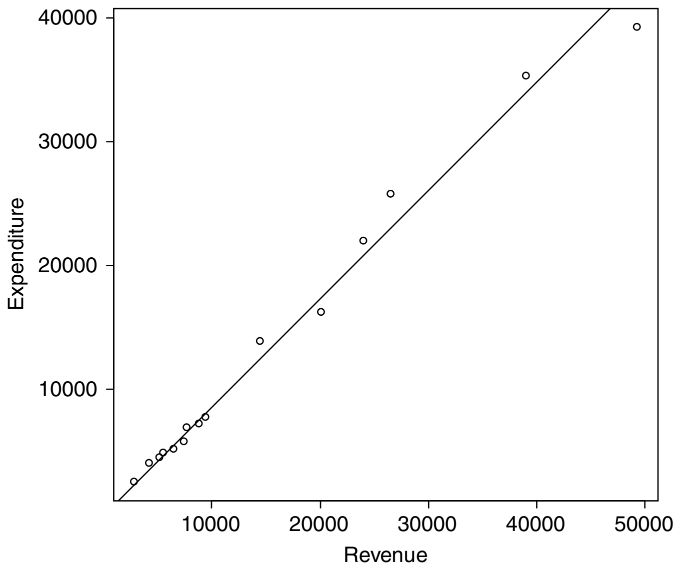
The sample correlation coefficient is 0.3249, indicating a mild positive correlation.

14.2.3. **(a)** The following is a scatter plot of the data.



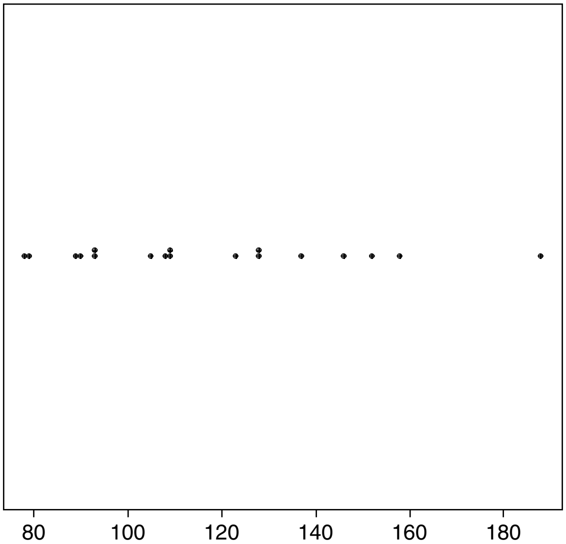
(b) $r = 0.9918$.

(c) The following is a Q-Q plot of revenue versus expenditure.



(d) From the scatter plot being close to a line and $r = 0.9918$ we see that there is a strong positive liner relationship between revenue and expenditure. From the Q-Q plot we see that the quantiles fall nearly along the 45 degree line. Thus, we may conjecture that the revenue and the expenditure have the same probability distribution.

14.2.5. The following is the dot plot for this data.



The above dot plot suggests the distribution of the median house prices is skewed towards the left, because most of the observations are to the left.

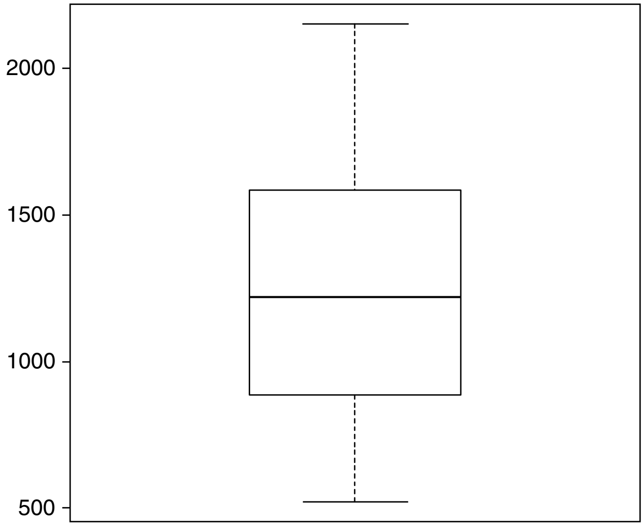
EXERCISES 14.3

14.3.1. **(a)** The following table summarizes the z score, the modified z score, and the distribution free z score.

data	z score	dist–free z	modified z
1215.1	−0.09852	0.011804	−0.12339
1109.9	−0.31406	0.281755	−0.39335
1536.5	0.559969	0.812933	0.701342
1797.8	1.095325	1.483449	1.371858
1630.5	0.752558	1.054144	0.942553
939.7	−0.66277	0.718501	−0.83009
1219.7	−0.0891	0	−0.11159
519.9	−1.52286	1.79574	−1.90733
830	−0.88752	1	−1.11159
780.1	−0.98976	1.128047	−1.23964
1403.3	0.287066	0.471132	0.359541
1869.7	1.242635	1.66795	1.556359
2152.8	1.822656	2.394406	2.282815
1410	0.300793	0.488324	0.376733
532.8	−1.49643	1.762638	−1.87423

Since no z scores and modified z scores have absolute values greater than 3.5, and no distributions free z scores are greater than 5, we then conclude that there is no obvious outliers.

(b) The following is the boxplot.



(c) Outliers in this case may represent an extreme observation which has either very high or very low rate of motor vehicle thefts.

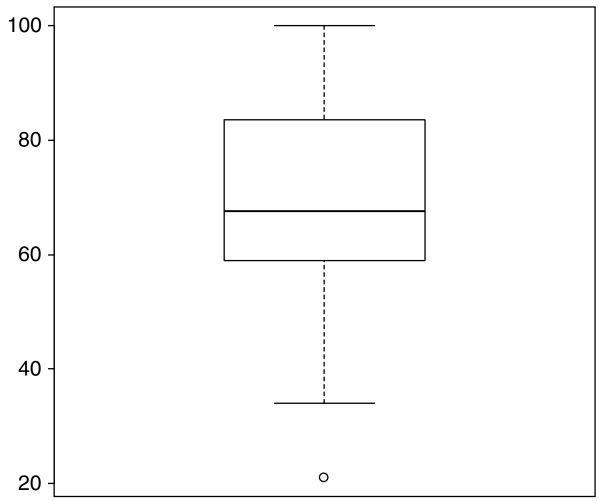
14.3.3. **(a)** The following table summarizes the z score, the modified z score, and the distribution free z score.

data	z score	dist–free z	modified z
67	−0.09135	0.037037	−0.1358
63	−0.29066	0.333333	−0.4321
39	−1.48652	2.111111	−2.20987
80	0.55641	0.925926	0.827163
64	−0.24083	0.259259	−0.35802
95	1.303824	2.037037	1.938274
90	1.054686	1.666667	1.567904
93	1.204169	1.888889	1.790126
21	−2.38342	3.444444	−3.54321
36	−1.636	2.333333	−2.4321
44	−1.23738	1.740741	−1.8395
66	−0.14118	0.111111	−0.20987
100	1.552962	2.407407	2.308644
66	−0.14118	0.111111	−0.20987
72	0.157789	0.333333	0.23457

34	−1.73566	2.481481	−2.58024
78	0.456755	0.777778	0.679015
66	−0.14118	0.111111	−0.20987
68	−0.04152	0.037037	−0.06173
98	1.453307	2.259259	2.160496
74	0.257444	0.481481	0.382719
81	0.606237	1	0.901237
71	0.107961	0.259259	0.160496
100	1.552962	2.407407	2.308644
60	−0.44014	0.555556	−0.65432
50	−0.93842	1.296296	−1.39506
81	0.606237	1	0.901237
66	−0.14118	0.111111	−0.20987
90	1.054686	1.666667	1.567904
89	1.004858	1.592593	1.49383
86	0.855375	1.37037	1.271607
49	−0.98825	1.37037	−1.46913
77	0.406927	0.703704	0.604941
63	−0.29066	0.333333	−0.4321
58	−0.5398	0.703704	−0.80247
43	−1.28721	1.814815	−1.91358

Using z-score test and distribution-free test, there are no outliers. Using modified z-score test the observation 21 is a possible outlier.

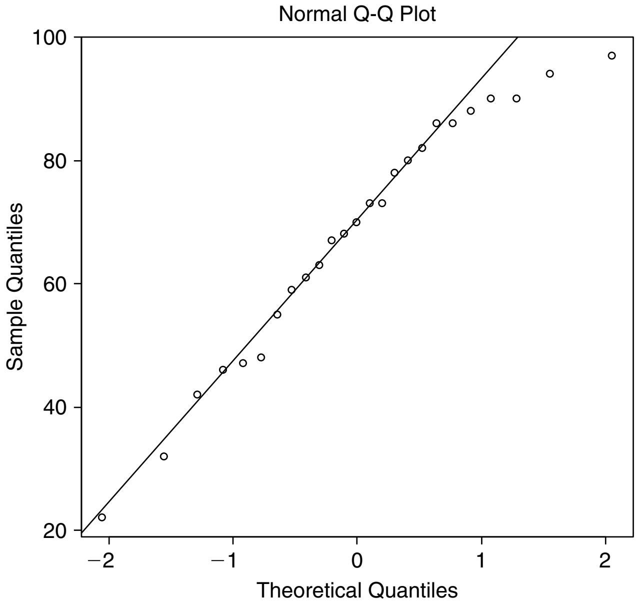
(b) The following is the boxplot.



Hence, the observation 21 is identified as an outlier using the boxplot.

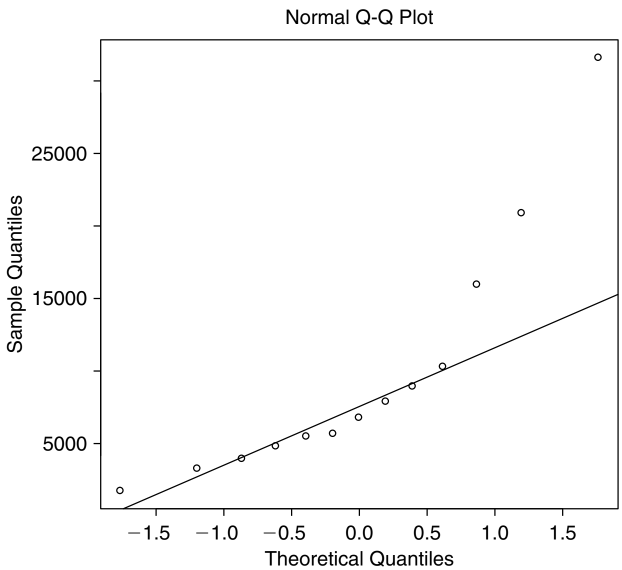
EXERCISES 14.4

14.4.1.

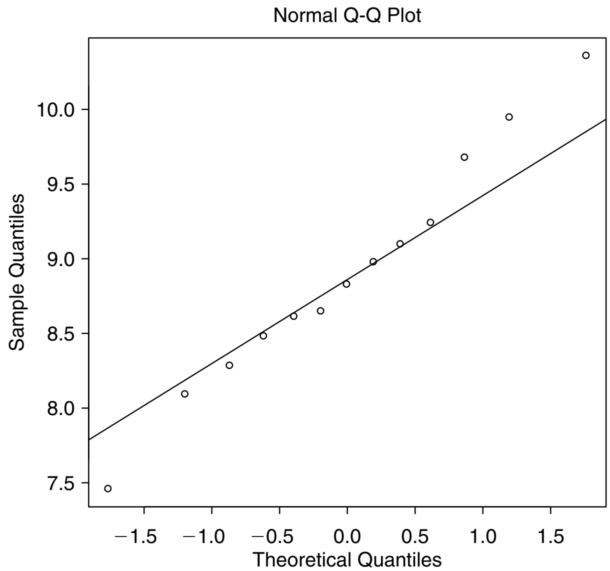


From the above normal probability plot we see that the data follows the straight line fairly well. Hence, the normality of the data is not rejected and no transformation is needed.

14.4.3. **(a)** The following is the normal probability plot of the data. The graph below clearly shows that the data does not follow normal distribution.

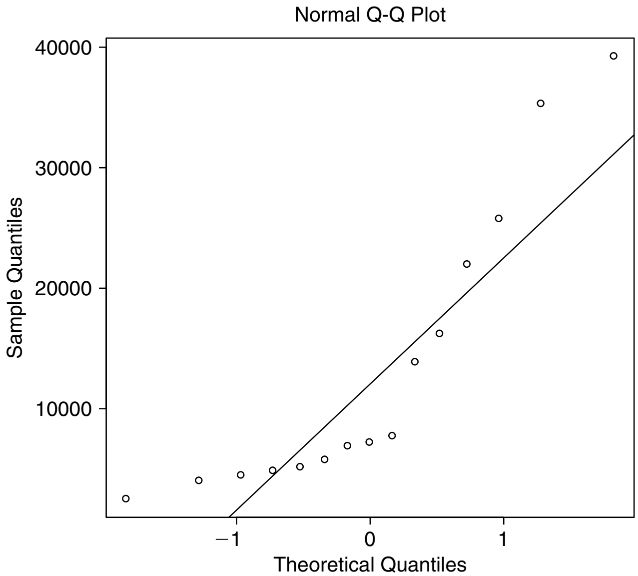


(b) Take the transformation $y = \ln(x)$ and look at the normal probability plot of the transformed data below.

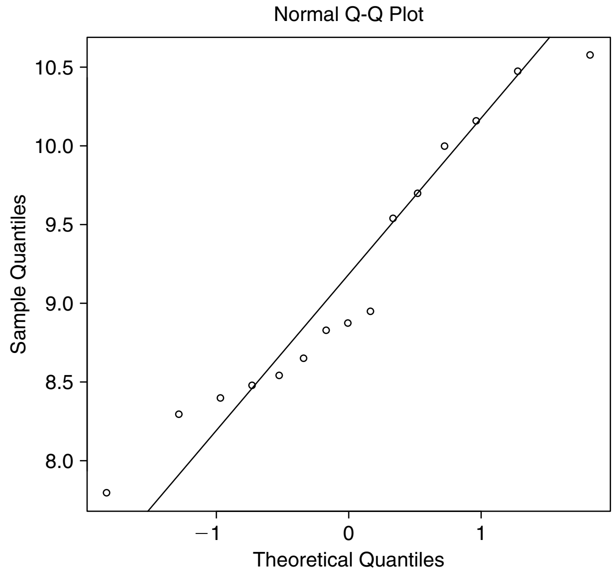


With the transformation, we can see that the transformed data falls much closer to the normal line.

14.4.5. **(a)** The following is the normal probability plot of the data. The graph below clearly shows that the data does not follow normal distribution.

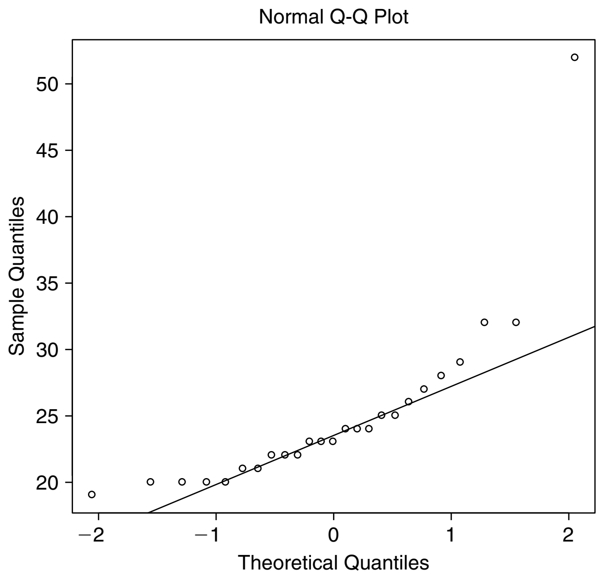


(b) Take the transformation $y = \ln(x)$ and look at the normal probability plot of the transformed data below.

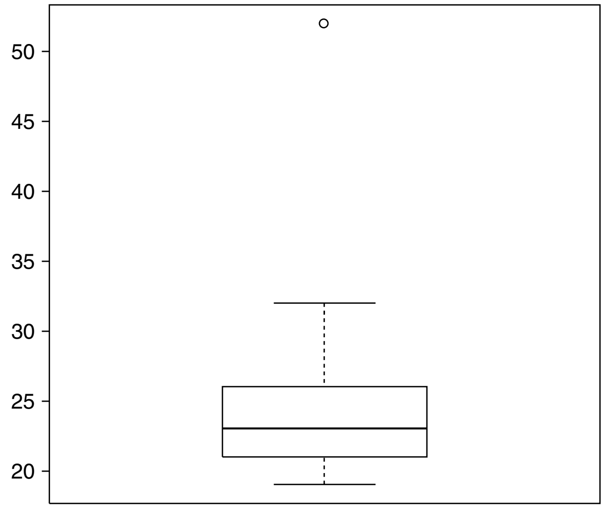


With the transformation, we can see that the transformed data falls much closer to the normal line.

14.4.7. **(a) & (b)** The following is the normal probability plot of the data. We see that the data follows the straight line except for one data points. This suggests that the data may follow normality but with a possible outlier.



(c) The following is the boxplot of the data.



Hence, the observation 52 is identified as a possible outlier using boxplot. Further investigation is needed to check if there was a measurement error about this case. Or this observation may suggest that a particular car is significantly better than the others in terms of mileage per gallon.

14.4.9. Use the data from **Exercise 14.2.1.** Let $X =$ the percent expense ratio and $Y =$ the percent return. Then we can calculate that

$$F = \frac{s_X^2}{s_Y^2} = 0.0038.$$

Since $F < F_{0.025}(19, 19) = 0.3958$, we then reject the null hypothesis at level 0.05. Thus, we suggest that the variances of the two populations are not equal.

14.4.11. Let $X =$ the bonus for female and $Y =$ the bonus for male. The assumption of the test is that the random samples of X and Y are from independent normal distributions. To test the homogeneity of variances of X and Y we calculate the ratio

$$F = \frac{s_X^2}{s_Y^2} = 0.8044$$

Since $F_{0.025}(7, 7) = 0.2002 < F < F_{0.975}(7, 7) = 4.9949$, then we do not reject the null hypothesis at level 0.05. Thus, we suggest that the variances of the two populations are equal.

14.4.13. Let X_1, X_2 and X_3 be the scores of the students taught by the faculty, the teaching assistant and the adjunct, respectively. The assumption of the test is that the random samples of

X_1, X_2 and X_3 are from independent normal distributions. To test homogeneity of variances of X_1, X_2 and X_3 we first compute $\bar{x}_1 = 81.6, \bar{x}_2 = 78.8$ and $\bar{x}_3 = 70.4$. Letting $y_{ij} = |x_{ij} - \bar{x}_i|$, we then obtain the following y_{ij} values.

Deviation		
Faculty	Teaching Assistant	Adjunct
11.4	9.2	15.6
20.6	11.2	14.4
5.4	2.8	2.6
6.6	3.2	19.6
10.4	20.8	23.4

The test statistic is

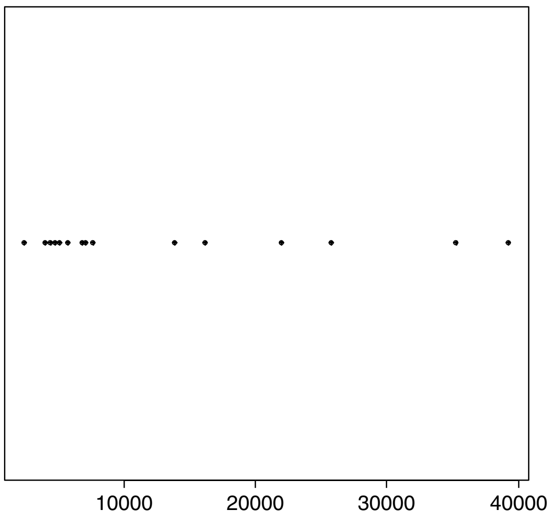
$$z = \frac{\sum_{i=1}^{k} n_i(\bar{y}_{i.} - \bar{y}_{..})^2/(k-1)}{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i.})^2/(n-k)} \quad \text{where } n_1 = 5, n_2 = 5, n_3 = 5, k = 3 \text{ and } n = 15$$

$$= \frac{MST}{MSE} = \frac{43.59}{50.39} = 0.8651$$

Since $z < F_{0.95}(2, 12) = 3.8853$, then at level 0.05 we do not reject the null hypothesis. That is we suggest that the variances of the three populations are equal.

EXERCISES 14.5

14.5.1. **(a)** The following is the dot plot of the data of **Exercise 14.4.5**.



(b) Mean $= \bar{y} = 13373.53$, median $= 7145$, and standard deviation $= s = 11924.47$.

(c) A 95% confidence interval for the mean is

$$\bar{y} \pm t_{0.975}(n-1)\frac{s}{\sqrt{n}} = 13373.53 \pm 2.1448\frac{11924.47}{\sqrt{15}} = (6769.98, 19977.08).$$

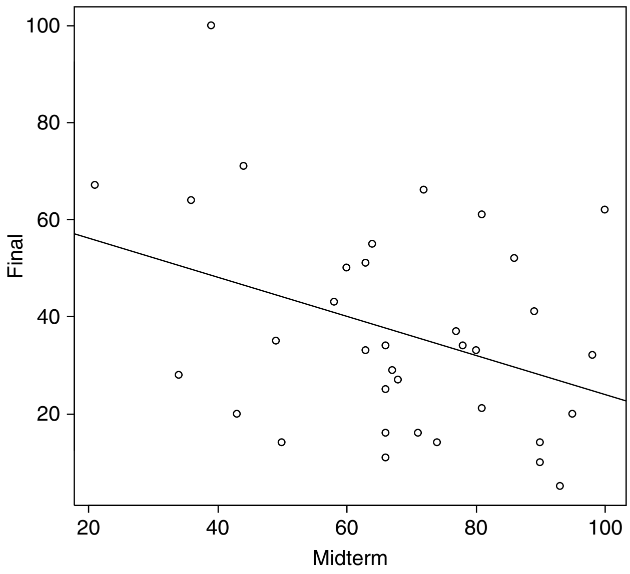
(d) A 95% prediction interval is

$$\begin{aligned}\bar{y} \pm t_{0.975}(n-1) \cdot s\sqrt{1+\frac{1}{n}} &= 13373.53 \pm 2.1448 \cdot 11924.47\sqrt{1+\frac{1}{15}} \\ &= (-13040.67, 39787.74).\end{aligned}$$

Since state expenditure is nonnegative, we can take the 95% prediction interval as $(0, 39787.74)$.

(e) There is a 95% chance that the true mean falls in $(6769.98, 19977.08)$. There is a 95% chance that the next observation falls in $(0, 39787.74)$. The assumption of obtaining the confidence interval and prediction interval is that the data follows normal distribution or has large sample size to employ central limit theorem.

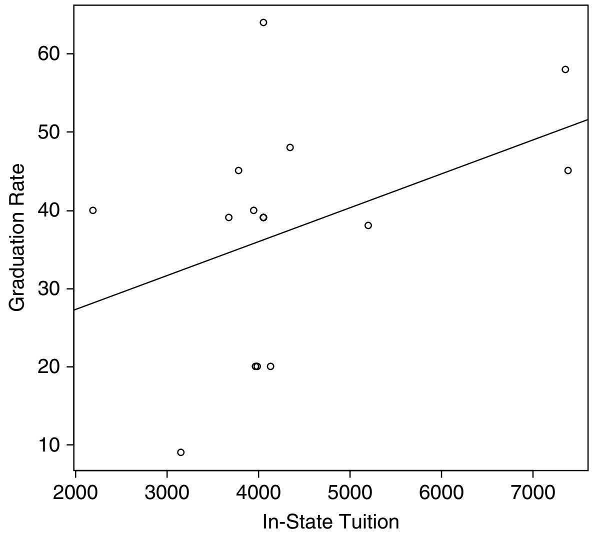
14.5.3. **(a)** Let $X =$ the midterm score and $Y =$ the final score. The following is the scatter plot of the data with the fitted regression line.



(b) The data does not show any particular pattern. No transformation is needed in this case.

(c) Fitting the data we obtain the linear regression model as $\hat{y} = 64.39 - 0.4048x$. However, we have $R^2 = 0.1345$ meaning only 13.45% of the variation in y is explained by the variable x.

14.5.5. **(a)** Let $X =$ the in-state tuition and $Y =$ the graduation rate. The following is the scatter plot of the data with the fitted regression line.

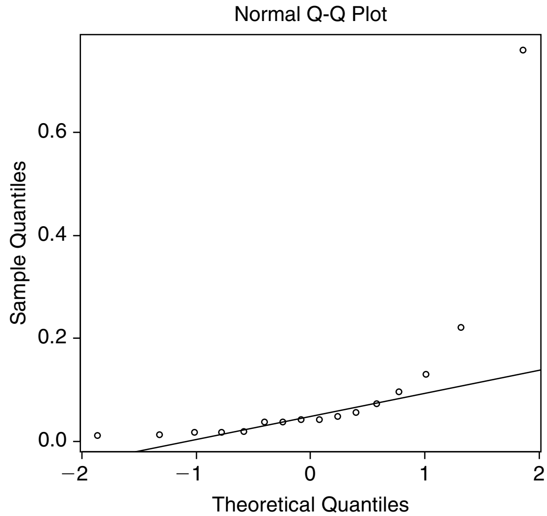


(b) Fitting the data by least squares method we obtain the linear regression model as $\hat{y} = 18.6887 - 0.0043x$.

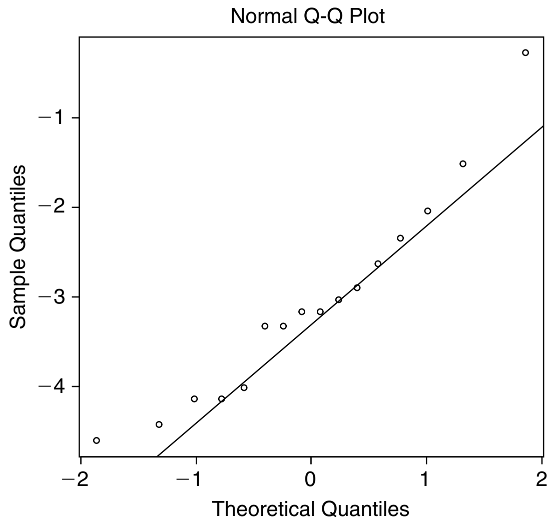
(c) We have $R^2 = 0.1618$ meaning only 16.18% of the variation in y is explained by the variable x. Thus, from the small R^2 and the scatter plot above we suggest that the least squares line is not a good model and must be improved.

EXERCISES 14.6

14.6.1. **(a)** The normal probability plot of the data is given below. From the normal plot we can see that the data significantly deviates from the normal line. Hence, we can not assume the data is normally distributed and nonparametric test is more appropriate.



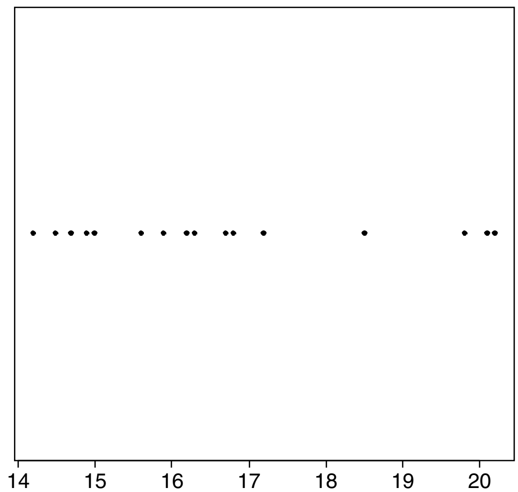
(b) Take the transformation $y = \ln(x)$ and look at the normal probability plot of the transformed data below.



We see that the transform data does not deviate from the normal line a lot. Thus, a parametric test can be used on the logarithmic filtered data.

EXERCISES 14.7

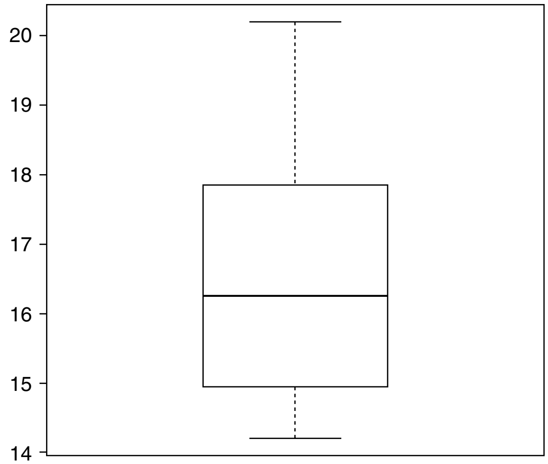
14.7.1. **(a)** Let X = total revenue and Y = pupils per teacher. The following is the dot plot of the data of pupils per teacher.



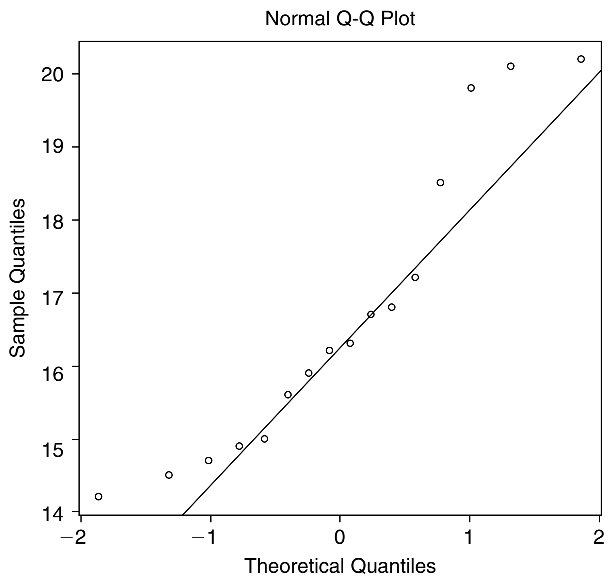
The descriptive statistics of the pupils per teacher data is given below.

n	Mean	Std	Min	Q1	Median	Q3	Max
16	16.6625	2.0063	14.2	14.975	16.25	17.525	20.2

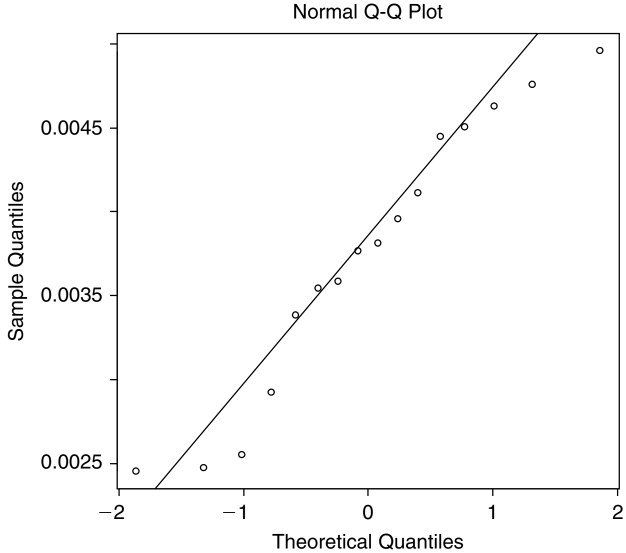
(b) The boxplot of the pupils per teacher data is given below. From the boxplot below we see that no outlier exists.



The following is the normal probability plot of the pupils per teacher data. The data is not normal.



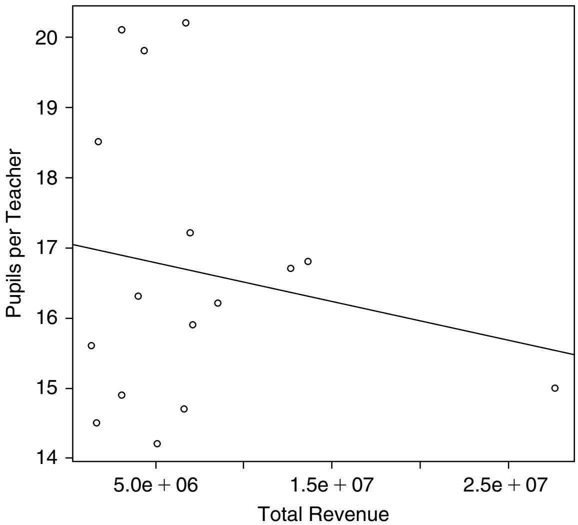
The following gives the normal plot by take the transformation $z = y^{-2}$ which shows that the data becomes approximately normal.



(c) A 95% confidence interval for the mean of the pupils per teacher is

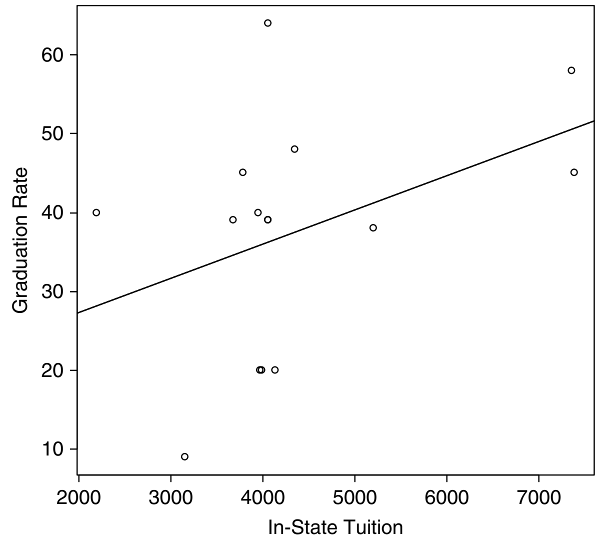
$$\bar{y} \pm t_{0.975}(n-1)\frac{s}{\sqrt{n}} = 16.6625 \pm 2.1315\frac{2.0063}{\sqrt{16}} = (15.59, 17.73).$$

(d) The following is the scatter plot of total revenue vs. pupils per teacher with the fitted regression line.



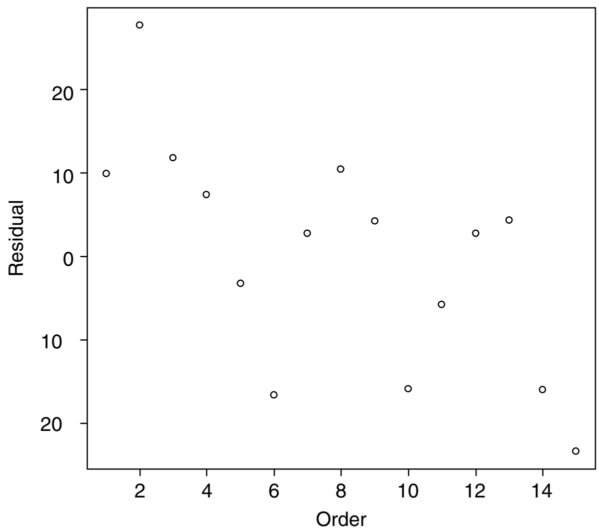
(e) Fitting the data we obtain the linear regression model as $\hat{y} = 17.06 - 5.512 \cdot 10^{-8}x$. However, we have $R^2 = 0.0324$ meaning only 3.24% of the variation in y is explained by the variable x. Thus, the regression model is not a good representation of the relationship between total revenue and pupils per teacher.

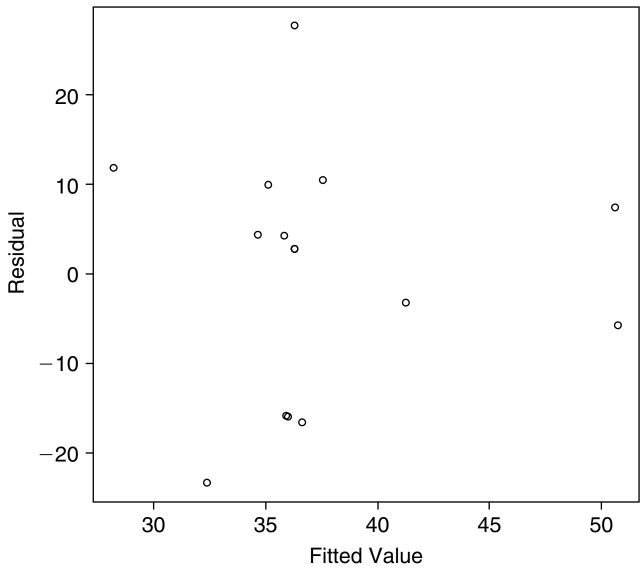
14.7.3. Let X = the in-state tuition and Y = the graduation rate. The following is the scatter plot of graduation rate vs. in-state tuition with the fitted regression line.

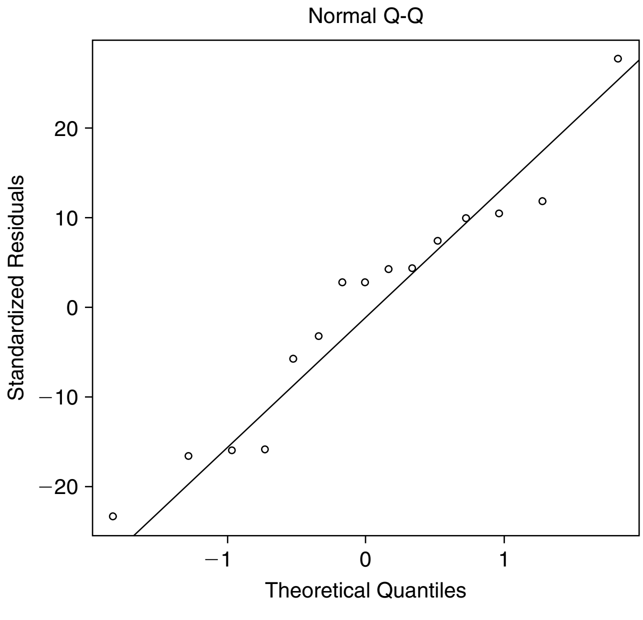


Fitting the data we then obtain the linear regression model as $\hat{y} = 18.6887 - 0.0043x$ with $R^2 = 0.1618$.

To run a residual model diagnostics we look at the following three plots.







There is nothing unusual about the residual plots. Therefore, the basic assumptions in regression analysis for the errors: independency, normality and homogeneity of variances are checked. And it seems to be no reason to reject these assumptions.

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